



Tools – H2OUVE (UEFI Variable Editor)

Ver. 0.17

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SEG Group

Insyde Software

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History

Revision for this material

Rev.	Data	Author	Approver	Summary
0.1	2013/5/27	Chihchin Chuang	Giant Liang	Initial version for SEG
0.2	2013/6/18	Chihchin Chuang	Giant Liang	1.Support modify Boot Type Order and Boot Order 2.Replace some picture for the new execution tips
0.3	2013/7/17	Chihchin Chuang	Giant Liang	1.Support modify Password 2.Support Setup Editor feature on Kernel 5.0
0.4	2013/8/22	Chihchin Chuang	Giant Liang	1.Support Multi-Config 2.Support Image Setup Editor feature on Kernel 5.0
0.5	2013/9/10	Chihchin Chuang	Giant Liang	1.Update picture about Boot Type Order and Boot Order. 2.Add example about how to load Multi-Config.
0.6	2013/11/22	Chihchin Chuang	Giant Liang	1.Replace photo for new name. 2.Support Hotkey setting in Multi-Config. 3.Support replace string in Setup page.
0.7	2014/3/20	Chihchin Chuang	Giant Liang	1.Adjust content of File. 2.Adjust some commands. 3.Support some new features of Multi-Config.
0.8	2014/6/6	Ryan Hsiao	Giant Liang	1. Add Variable Stress Test
0.9	2014/6/17	James SX Lee	Giant Liang	1.Adjust content of SCU file.(Add varstroe ID) 2.Adjust content of Multi-Config file. (Add Name ID, Config ID, and Bios version) 3.Update picture about Multi-Config feature.
0.10	2014/8/7	Chihchin chuang	Giant Liang	Change the commands of Multi-Config.

History

Revision for this material

Rev.	Data	Author	Approver	Summary
0.11	2014/10/29	Kanos Huang	Giant Liang	1. Update usage of all commands 2. Add <code>-d40</code> command set for name-value paired operation 3. Add <code>-gst</code>, <code>-sst</code> command set for name-value paired operation and work like <code>-gs</code> and <code>-ss</code> 4. Change <code>-mc -recv</code> to <code>-mc -format</code> 5. Add <code>-bfirst</code> command to set first boot device 6. Add <code>-uvebf</code> command to enable/disable UVE functionality 7. Update all images
0.12	2015/02/11	Kanos Huang	Giant Liang	1. Add <code>-d40 -b</code>, <code>-d40 -t</code>, <code>-gbd</code>, <code>-sbd</code>, <code>-gbt</code>, <code>-sbt</code> for boot device and boot type process 2. Add <code>-cvl</code> to check if variable locked 3. Add <code>-sc</code> to save customized settings 4. Remove <code>-gst</code>, <code>-sst</code> commands 5. Some commands can't be sure it supports BIOS 3.5 6. Update help message
0.13	2015/03/09	Kanos Huang	Giant Liang	1. Update graphical mode 2. Update the type arguments of multi-config feature from <code>-f</code> to <code>-scu</code> 3. Update links
0.14	2015/04/13	Kanos Huang	Giant Liang	1. Add add/modify/delete for Variable editor 2. Find name of variable in Variable editor 3. Find specific string in String editor 4. Read-only item in Multi-Editor

History

Revision for this material

Rev.	Data	Author	Approver	Summary
0.15	2015/04/14	Kanos Huang	Giant Liang	1. Update some images 2. Update descriptions for read-only and invalid item
0.16	2015/07/13	Kanos Huang	Giant Liang	1. Add -glg command for console 2. Add -gsb command for console 3. Add Logo handler of Multi-Config 4. Add SMBIOS handler of Multi-Config
0.17	2016/03/01	Kanos Huang		1. Update main frame image 2. Update string editor image and description 3. Update support table for kernel 5.1 4. Windows and Linux version support ITB feature after 100.00.16.05 5. Multi-Config was disabled since 100.00.16.06 6. Added -csvx command of ITB feature

Agenda

- [Features](#)
- [Cosole Mode](#)
- [Graphical Mode](#)





Features

Features

- To provide 5 kinds of UEFI Variable Editors
 - SCU(Setup) Editor : To edit "runtime/current setup setting" of platform
 - Image Setup Editor : To edit "setup Default Setting" of BIOS image
 - Variable Editor : To edit "Variable data" at runtime ROM or BIOS image
 - Multi-Config Editor : To edit "Multi-Config Setting" at runtime ROM or BIOS image
 - Variable Stress Test: Support variable access and stress test on variable at runtime or BIOS image.
- Supported OS environment
 - UEFI SHELL
 - DOS
 - WINDOWS
 - LINUX

Features

Status for H2O versions

○ – Support

X – Not Support

kernel Variable Editor		kernel 3.5		kernel 3.7		kernel 5.0	Kernel 5.1
		CRB	OEM	CRB	OEM	CRB	CRB
Common Arguments	-r	X	X	○	○	○	○
	-a	○	○	○	○	○	○
	-l	X	X	○	○	○	○
	-c	X	X	○	○	○	○
	-sc	X	X	○	○	○	○
Setup Editor Arguments	-gs	X	X	○	○	○	○
	-ss	X	X	○	○	○	○
	-ms	X	X	○	○	○	○
Image Setup Editor Arguments	-gd	X	X	○	○	○	○
	-sd	X	X	○	○	○	○
	-gstr	X	X	○	○	○	○
	-sstr	X	X	○	○	○	○

Features

Status for H2O versions

○ – Support ^{update}
X – Not Support

Variable Editor		kernel		kernel 3.5		kernel 3.7		kernel 5.0	kernel 5.1
		CRB	OEM	CRB	OEM	CRB	CRB	CRB	CRB
Setup Editor Name-Value Paired Arguments	-gst	X	X	O	O	O	O	O	O
	-sst	X	X	O	O	O	O	O	O
	-d40	-g	X	X	O	O	O	O	O
		-s	X	X	O	O	O	O	O
		-m	X	X	O	O	O	O	O
		-l	X	X	O	O	O	O	O
		-c	X	X	O	O	O	O	O
		-b	X	X	O	O	O	O	O
		-t	X	X	O	O	O	O	O
	-cdi	X	X	O	O	O	O	O	O

Features

Status for H20 versions

○ – Support ^{update}
 X – Not Support

Variable Editor		kernel		kernel 3.5		kernel 3.7		kernel 5.0	kernel 5.1
		CRB	OEM	CRB	OEM	CRB	OEM	CRB	CRB
Variable Editor Arguments	-gv	O	O	O	O	O	O	O	O
	-sv	O	O	O	O	O	O	O	O
	-re	O	O	O	O	O	O	O	O
	-cvl	X	X	O	O	O	O	O	O

■ -cvl command supported on UEFI mode only

Features

Status for H2O versions

○ – Support update
 X – Not Support

Variable Editor		kernel		kernel 3.5		kernel 3.7		kernel 5.0	kernel 5.1
		CRB	OEM	CRB	OEM	CRB	OEM	CRB	CRB
Multi-Config Editor Arguments	-gk	X	X	X	X	O		X	
	-glg	X	X	X	X	O		X	
	-gsb	X	X	X	X	O		X	
	-mc	-list	X	X	X	X		O	X
		-gen	X	X	X	X		O	X
		-add	X	X	X	X		O	X
		-mod	X	X	X	X		O	X
		-del	X	X	X	X		O	X
		-post	X	X	X	X		O	X
		-act	X	X	X	X		O	X
		-format	X	X	X	X		O	X

- Multi-Config feature supported on Kernel 5.0 only
- Multi-Config feature was disabled since 100.00.16.06

Features

Status for H2O versions

○ – Support

X – Not Support

kernel Variable Editor		kernel 3.5		kernel 3.7		kernel 5.0	kernel 5.1
		CRB	OEM	CRB	OEM	CRB	CRB
Variable Stress Test Arguments	-dbos	O	O	O	O	O	O
	-cnvs	O	O	O	O	O	O
	-cvhc	O	O	O	O	O	O
	-vorv	O	O	O	O	O	O
	-cvur	O	O	O	O	O	O
	-cvus	O	O	O	O	O	O
	-cvrd	O	O	O	O	O	O
	-cvrs	O	O	O	O	O	O
	-cvsf	O	O	O	O	O	O
	-cvs0	O	O	O	O	O	O
	-cgnv	O	O	O	O	O	O
	-cssr	O	O	O	O	O	O
	-cvsd	O	O	O	O	O	O
	-sipi	O	O	O	O	O	O



Console Mode

- 
- **Support Mode**
 - Arguments
 - Common Arguments
 - Setup Editor
 - Image Setup Editor
 - Variable Editor
 - Multi-Config Editor
 - Variable Stress Test

Support mode

- Command (Console) mode
 - Run H2OUIVE with arguments.
(e.g., H2OUIVE -h)
 - **Note** : Here use H2OUIVE instead of execution file name of all version.

```
> sudo H2OUIVE-Lx64 -h
Insyde H2OUIVE (UEFI Variable Editor) Version 100.00.16.00
Copyright (c) 2012 - 2014, Insyde Software Corp. All Rights Reserved.

h2ouive-lx64

Description:
A tool to manage settings, variables, strings, and multi-configurations
for BIOS Setup Utility.

Options:

Common...
-----
-fea                    Report BIOS supported functionalities.
-uevbf <Enabled/Disabled> Enable/Disable H2OUIVE functions in BIOS for
subsequent boot.
-r <File>               Dump the BIOS ROM regions (e.g. Variables, Settings,
Strings, etc.) to raw file.
-l                     Load default/original settings for BIOS setup utility.
i.e. delete setup and associated variables in BIOS (
data section). Same as 'Load Default' option under
BIOS Setup Utility.
-c                     Load custom settings for BIOS setup utility.
i.e. restore setup variables in BIOS (data section).
Same as 'Load Customize Default' option under BIOS
Setup Utility.
-sc <File>              Save custom settings for BIOS setup utility.
i.e. save custom variables in BIOS (data section).
Same as 'Save Customize Default' option under BIOS
Setup Utility.

Setup Editor...
-----
-gs <File> [-all]       Generate setup utility current setting to file.
(-all: optional, include suppress and grayout setting)
-ss <File>              Modify setup utility settings from a configuration
file.
-gd <File> [-i <BiosImageFile>]
Dump default setup utility settings (optionally from
specified Bios Image File) to a configuration file.
-sd <File> [-i <BiosImageFile>] Read saved default configuration file and update
setup utility default settings in specified Bios
File.
-ms [-h] [-fi <Field>] -op <Option> [-fi <Field>] -op <Option>...
'-h' to show more information.
'-fi' to show more information.
-d40 <g <File> | -s <File> | -m <Name> <Value> | -l | -c | -b | -t>
Modify current SCU settings, additional options:
-g <File>               Same as -gs option, but dump to name-value paired
file.
-s <File>              Same as -ss option, but with name-value paired file
content.
-m [-h] <Name> <Value> [-m <Name> <Value>]...
'-h' to show more information
-l                     List all item names.
-c                     Change current settings to default value.
-b <File>              Same as -gs option, but dump boot device list to
name-value paired file
-t <File>              Same as -gs option, but dump boot type list to
name-value paired file
-cdi <File> [-i <Name>] [-i <ImageFile>]
Report duplicate items and their locations in BIOS
Setup Utility.
-gstr <File> <-s>|-b> -i <ImageFile>
Dump configuration and browser strings to
string-formatted file from Bios image file
```

```
-sstr <File> <-s>|-b> -i <ImageFile> [-out <ImageFile>]
Modify setup configuration or setup browser strings
from specified string-formatted file to Bios Image File.
-gbd <File>            Dump current boot device setting to boot-device file.
-sbd <File>            Modify current boot device from specified boot-device
file.
-gbt <File> [-i <BiosImageFile>]
Dump current boot type setting to boot-type file.
-sbt <File> [-i <BiosImageFile>] [-out <ImageFile>]
Modify current boot type from specified boot-type file.
-bfirst <name> [-t]    Make device to first boot.
```

```
Variable Editor...
-----
-gv <File> [-i <BiosImageFile>]
Dump variable information to a variable record file.
-sv <File> [-i <BiosImageFile>] [-out <ImageFile>]
Update variables from specified variable record file.
-re [-vn <VarName>] [-vg <GUID>]
Remove a variable by name or GUID.
-cvl <File> [-vn <Name>] -vg <GUID>
Check variables are locked by input a variable
name and file or by input a specified name and
GUID.
```

```
MultiConfig Editor...
-----
-gk <File> [-i <BiosImageFile>]
Dump BIOS Hotkey settings to a hot-key file.
-mc OPTIONS
The following arguments are the OPTIONS:
-list [-i <ImageFile>]
List all Multi-Config regions.
-gen <File> <-f>|-k>|-sb> <ConfigID> [-i <ImageFile>]
Dump the specified config setting to file.
-add <File> <-f>|-k>|-sb> [-i <ImageFile>] [-out <ImageFile>]
Add a new config setting to multi-config region from
specified file format.
-mod <File> <-f>|-k>|-sb> [-i <ImageFile>] [-out <ImageFile>]
Modify config setting to multi-config region from
specified file format.
-del <-f>|-k>|-sb> <ConfigID> [-i <ImageFile>] [-out <ImageFile>]
Delete specified config setting by ConfigID from
multi-config region.
-act <-f>|-k>|-sb> <ConfigID> [-i <ImageFile>] [-out <ImageFile>]
Activate specified config setting as startup setting.
-act clear <-f>|-k>|-sb> [-i <ImageFile>] [-out <ImageFile>]
Deactivate the specified setting and return to
original startup setting.
-post -f <ConfigID> [-i <ImageFile>] [-out <ImageFile>]
Activate a specified post boot setting by ConfigID
for next boot only.
-post clear -f [-i <ImageFile>] [-out <ImageFile>]
Deactivate current POST Boot setting
-format <ImageFile> [-i <ImageFile>] [-out <ImageFile>]
Format Multi-Config Region based on ImageFile.
```

```
Global...
-----
-a                    Auto Reboot.
```

- 
- Support Mode
 - **Arguments**
 - Common Arguments
 - Setup Editor
 - Image Setup Editor
 - Variable Editor
 - Multi-Config Editor
 - Variable Stress Test

Arguments

Arguments	Description
-? or -h	Display Help information
-fea	Report BIOS supported functionalities.
-uvebf <Enabled Disabled>	Enable/Disable H2OUVE functions in BIOS for subsequent boot.
-r <File>	Dump the BIOS ROM regions (e.g. Variables, Settings, Strings, etc.) to raw file.
-a	Auto Reboot.
-l	Load default/original settings for BIOS setup utility. i.e. delete setup and associated variables in BIOS (data section). Same as "Load Default" option under BIOS Setup Utility.
-c	Load custom settings for BIOS setup utility. i.e. restore setup variables in BIOS (data section). Same as "Load Custom" option under BIOS Setup Utility.
-sc	Save custom settings for BIOS setup utility. i.e. save custom variables in BIOS (data section). Same as "Save Customize default" option under BIOS setup utility.
-gs <File> [-all]	Generate setup utility setting to file. (-all: optional, include suppress and grayout setting)
-ss <File>	Modify setup utility settings from a configuration file.

Arguments

Arguments	Description
<u>-gd <File></u>	Generate setup utility default setting to file.
<u>-sd <File> -i <ImageFile></u>	Read saved default configuration file and update setup utility default settings in specified Bios File.
<u>-d40 <-g [File] -s <File> -m <Name> <Value> -l -c -b -t></u>	Modify current SCU settings with name-value paired format.
<u>-g [File]</u>	Same as -gs option, but dump to name-value paired file.
<u>-s <File></u>	Same as -ss option, but with name-value paired file content.
<u>-m <Name> <Value> [-m <Name> <Value>]...</u>	Modify a setup utility setting by name and value
<u>-l</u>	List all item names.
<u>-c</u>	Change current settings to default value.
<u>-b [File]</u>	Same as -gs option, but dump boot device list to name-value paired file
<u>-t [File]</u>	Same as -gs option, but dump boot type list to name-value paired file
<u>-cdi [File] [-n <Name>] [-i <ImageFile>]</u>	Report duplicate items and their locations in BIOS Setup Utility.

Arguments

Arguments	Description
<u>-gstr <File> -s -b -i <ImageFile></u>	Modify setup configuration or setup browser strings from specified string-formatted file to Bios Image File.
<u>-sstr <File> -s -b -i <ImageFile></u>	Modify setup configuration or setup browser strings from specified string-formatted file to Bios Image File.
<u>-gbd <File> [-i <BiosImageFile>]</u>	Dump current boot device setting to boot-device file.
<u>-sbd <File> [-i <BiosImageFile>]</u>	Modify current boot device from specified boot-device file.
<u>-gbt <File> [-i <BiosImageFile>]</u>	Dump current boot type setting to boot-type file.
<u>-sbt <File> [-i <BiosImageFile>]</u>	Modify current boot type from specified boot-typefile.
<u>-bfirst <name></u>	Make device to first boot.
<u>-gv <File></u>	Dump variable information to a variable record file.
<u>-sv <File></u>	Update variables from specified variable record file.
<u>-re [-vn <VarName>] [-vg <GUID>]</u>	Remove a variable by name or GUID.
<u>-cvl <File> -vn <Name> -vg <GUID></u>	Check variables are locked by input a variable name-guid file or by input a specified name-guid.

Arguments

Arguments	Description
-gk <File> [-I <BIOS image>]	Dump BIOS Hotkey settings to a hot-key file.
-glg <File> [-I <BIOS image>]	Generate BIOS logo setting to file.
-gsb <File> [-type <Type "all">] [-I <BIOS image>]	Generate SMBIOS settings to file.
-mc <MC Arg>	Multi-Config feature
-list	List all Multi-Config regions.
-gen <File> <-f -k -sb -lg> <ConfigID>	Dump the specified config setting to file.
-add <File> <-f -k -sb -lg>	Add a new config setting to multi-config region from specified file format.
-mod <File> <-f -k -sb -lg>	Modify config setting to multi-config region from specified file format.
-del <-f -k -sb -lg> <ConfigID>	Delete specified config setting by ConfigID from multi-config region.
-post -f <ConfigID>	Load the specified config setting from Multi-Config while next BIOS POST.
-post clear -f	Deactivate currernt POST Boot setting.
-act <-f -k -sb -lg> <ConfigID>	Activate specified config setting as startup setting.
-act clear <-f -k -sb -lg>	Deactivate the specified setting and return to original startup setting.
-format <ImageFile>	Format Multi-Config Region based on ImageFile.

Arguments

Arguments	Description
-dbos	Display Boot Order Sequence.
-cnvs	Clear NV Spare data.
-cvhc	Check Variable Header Continuity.
-vorv [-count]	Variable store Out of Region Verification.
-cvur [-count]	Check Variable Usage situation by Reboot.
-cvus [-count]	Check Variable Usage situation by Shutdown.
-cvrd [-count]	Check Variable Reclamation on Dxe.
-cvrs [-count]	Check Variable Reclamation on Smm.
-cvsf [-count] [-bc]	Check Variable fill Setting (fill 0xFF).
-cvs0 [-count] [-bc]	Check Variable fill Setting (fill 0x00).
-cgnv	Check GetNextVariableName on RT.
-cssr	Check Synchronization between Smm and RT SetVariable.
-cvsd	Check Variable is existing while changing variable State to Deleted_transition in Smm variable services.
-sipi	Send IPI.
-count <number>	Specify the count number.
-bc <number>	Partial erase – Number * block size (0x1000)

- 
- Support Mode
 - Arguments
 - **Common Arguments**
 - Setup Editor
 - Image Setup Editor
 - Variable Editor
 - Multi-Config Editor
 - Variable Stress Test

Common Arguments

Argument “-fea” : Report BIOS supported functionalities.

- Type command “H2OUVE -fea”

```
> H2OUVE-Sx64 -fea
```

```
Common Feature:
```

```
-r : Supported!
```

```
SCU Editor Feature:
```

```
-gs <File> [-all] : Supported!
```

```
-ss <File> : Supported!
```

```
Variable Editor Feature:
```

```
-gv <File> : Supported!
```

```
-sv <File> : Supported!
```

```
-re -vn <VarName> -vg <GUID> : Supported!
```

```
Multi-Config Feature:
```

```
Read flash part data for searching Multi-Config, please wait a minute.
```

```
-gk <File> : Unsupported!
```

```
-mc OPTIONS : Unsupported!
```

Common Arguments

Argument "-r" : Dump the BIOS ROM regions (e.g. Variables, Settings, Strings, etc.) to raw file

- Type command "H2OUE -r <file>"

```
> H2OUE-Sx64 -r img.fd
```

```
Now Dumping BIOS Image File to img.fd.  
Dump BIOS image successfully!
```

Common Arguments

Argument "-a", "-l" and "-c"

- Type command "H2OUVE -a"
 - System will be Reboot
- Type command "H2OUVE -l"
 - System will load Optimal default while next system reboot.
 - If this command execute without command "-a", then it'll show warning message.
- Type command "H2OUVE -c"
 - System will load Custom default.

Common Arguments

Argument “-sc” : Save custom setup default settings.

- Type command “H2OUVE -sc <File>”
 - Save custom setup default settings by input file which could be dumped by -gs
 - It will also show which item is changed

```
> sudo H2OUVE-L -sc scu.txt
Succeed to read scu.txt file.
1: Modify Boot Type to Dual Boot Type
2: Modify Quick Boot to Disabled
3: Modify Wake on PME to Disabled
Write custom default successfully!
```


- 
- Support Mode
 - Arguments
 - Common Arguments
 - **Setup Editor**
 - Image Setup Editor
 - Variable Editor
 - Multi-Config Editor
 - Variable Stress Test

SCU(Setup) Editor

Argument "-gs" : Generate **Current Setup Setting** to a file

- Type command "H2OUVE -gs <file>"
 - The settings of setup menu just like SCU that will be listed in the output file.
- Type command "H2OUVE -gs <file> -all"
 - It'll show all settings of setup menu.

```
> H2OUVE-$ -gs file.txt
```

```
Now parsing Hii data, please wait a minute...  
Complete parsing Hii data!!!  
Saving SCU setting file: file.txt
```

```
>
```

SCU(Setup) Editor

Content of output file about SCU

- Put the "*" between "[" and "]" to change the setting
 - This is for "Oneof" Question in VFR
- Set the numbers between "[" and "]" to change the setting. The value must be in between maximum and minimum.
 - This is for "Numeric " Question in VFR

```
(0x75, 1, 0x1234) EFI Device First  
Callback ID = 0x105c  
(1)[ ] Disabled  
(0)[*] Enabled (Default)
```

```
(0x69, 2, 0x1234) Timeout  
Callback ID = 0xb  
Minimum: 0  
Maximum: 65535  
Step: 1  
Default: None  
Setting: [0]
```



```
(0x75, 1, 0x1234) EFI Device First  
Callback ID = 0x105c  
(1)[*] Disabled  
(0)[ ] Enabled (Default)
```

```
(0x69, 2, 0x1234) Timeout  
Callback ID = 0xb  
Minimum: 0  
Maximum: 65535  
Step: 1  
Default: None  
Setting: [10]
```

SCU(Setup) Editor

Content of output file about SCU

- About "Text" Question
 - User Only CAN modify the content of brackets[] to assign data.

```
IPv4 IP Address  
  Callback ID = 0x2D01  
  [0.0.0.0]
```

```
IPv4 Subnet Mask  
  Callback ID = 0x2D02  
  [0.0.0.0]
```

```
IPv4 Gateway Address  
  Callback ID = 0x2D03  
  [0.0.0.0]
```



```
IPv4 IP Address  
  Callback ID = 0x2D01  
  [192.192.150.133]
```

```
IPv4 Subnet Mask  
  Callback ID = 0x2D02  
  [255.255.255.0]
```

```
IPv4 Gateway Address  
  Callback ID = 0x2D03  
  [192.192.150.254]
```

SCU(Setup) Editor

Content of output file about SCU

- Modify Boot Type Order
 - Modify the number of brackets[] to change the order.
 - Do NOT Modify the numbers of BootTypeOrderSize and BootTypeOrderOffset.

```
===== Boot Type Order =====  
BootTypeOrderSize  = 0x8  
BootTypeOrderOffset = 0x95  
(238)[00] BOOT_POLICY_OTHER_DEVICE  
(002)[01] BBS_HARDDISK  
(001)[02] BBS_FLOPPY  
(005)[03] BBS_USB  
(003)[04] BBS_CDROM
```



```
===== Boot Type Order =====  
BootTypeOrderSize  = 0x8  
BootTypeOrderOffset = 0x95  
(238)[02] BOOT_POLICY_OTHER_DEVICE  
(002)[01] BBS_HARDDISK  
(001)[03] BBS_FLOPPY  
(005)[00] BBS_USB  
(003)[04] BBS_CDROM
```


SCU(Setup) Editor

Content of output file about SCU

- Modify Boot Order
 - Modify the number of brackets[] to change the order.
 - Legacy and EFI are two group, and there is "<Legacy>" or "<EFI>" before the device name.
 - User only can swap the number of device which in the same group.
 - If the boot device is not exist just like EFI DVD/CDROM or EFI Network , and they will keep at the end of list.

```
===== Boot Order =====
(08193)[0] <EFI> EFI USB Device
(00000)[1] <EFI> Internal EFI Shell
(00002)[2] <Legacy> JetFlashTranscend 8GB
(00001)[3] <Legacy> JetFlashTranscend 4GB
(08194)[4] <EFI> EFI DVD/CDROM
(08195)[5] <EFI> EFI Network
=====
```



```
===== Boot Order =====
(08193)[1] <EFI> EFI USB Device
(00000)[0] <EFI> Internal EFI Shell
(00002)[3] <Legacy> JetFlashTranscend 8GB
(00001)[2] <Legacy> JetFlashTranscend 4GB
(08194)[4] <EFI> EFI DVD/CDROM
(08195)[5] <EFI> EFI Network
=====
```

SCU(Setup) Editor

Content of output file about SCU

- Add Password
 - Keep the brackets[] of Current Password clear.
 - Modify the content of brackets[] of New Password and Repeat Password to set the password.

```
===== Set Supervisor Password =====  
Callback ID = 0x1037  
Minimum Size: 1  
Maximum Size: 10  
Current Password:[]  
New Password:  []  
Repeat Password: []  
=====
```



```
===== Set Supervisor Password =====  
Callback ID = 0x1037  
Minimum Size: 1  
Maximum Size: 10  
Current Password:[]  
New Password:  [ABCD]  
Repeat Password: [ABCD]  
=====
```

- Modify Password
 - Modify the content of brackets[] of Passwords to modify the password.

```
===== Set Supervisor Password =====  
Callback ID = 0x1037  
Minimum Size: 1  
Maximum Size: 10  
Current Password:[]  
New Password:  []  
Repeat Password: []  
=====
```



```
===== Set Supervisor Password =====  
Callback ID = 0x1037  
Minimum Size: 1  
Maximum Size: 10  
Current Password:[ABCD]  
New Password:  [1234]  
Repeat Password: [1234]  
=====
```

SCU(Setup) Editor

Content of output file about SCU

- Clear Password
 - Fill the content of brackets[] of Current Password.
 - Keep the brackets[] of New Password and Repeat Password clear.

```
===== Set Supervisor Password =====  
Callback ID = 0x1037  
Minimum Size: 1  
Maximum Size: 10  
Current Password:[ ]  
New Password:    [ ]  
Repeat Password: [ ]
```



```
===== Set Supervisor Password =====  
Callback ID = 0x1037  
Minimum Size: 1  
Maximum Size: 10  
Current Password:[1234]  
New Password:    [ ]  
Repeat Password: [ ]
```

SCU(Setup) Editor

Argument "-ss" : Modify **Current Setup Setting** with the data in the (input) file

- Type command "H2OUE -ss <file>"

```
> H2OUE-Sx64.efi -ss file.txt
```

```
Now Parsing File From file.txt.
```

```
Succeed to read file.txt file.
```

```
Now Parsing File and save settings, please wait a minute...
```

```
Succeed to write SCU settings!
```

SCU(Setup) Editor

Argument "-gbd" : Dump **Current Boot device** to file

- Type command "H2OUVE -gbd <file>"

```
> sudo h2ouve-lx64 -gbd bd.txt  
Saving Boot Device setting file: bd.txt
```

```
===== --1-- Boot Order =====  
(00000)[0] <EFI> "EFI Hard Disk 1 (Windows Boot Manager)"  
(00001)[1] <EFI> "EFI Hard Disk 2 (ubuntu)"  
=====
```

SCU(Setup) Editor

Argument "-sbd" : Save **Current Boot device** by input file

- Type command "H2OUVE -sbd <file>"

```
===== 1-- Boot Order =====  
(00000)[0] <EFI> "EFI Hard Disk 1 (Windows Boot Manager)"  
(00001)[1] <EFI> "EFI Hard Disk 2 (ubuntu)"  
=====
```



```
===== 1-- Boot Order =====  
(00000)[1] <EFI> "EFI Hard Disk 1 (Windows Boot Manager)"  
(00001)[0] <EFI> "EFI Hard Disk 2 (ubuntu)"  
=====
```

```
> sudo H2OUVE-Lx64 -sbd bd.txt  
Now Parsing File From bd.txt.  
Bios Checking...OK!  
Modify order 1 to "EFI Hard Disk 2 (ubuntu)"  
Modify order 2 to "EFI Hard Disk 1 (Windows Boot Manager)"  
Succeed to save Boot Device setting.
```


SCU(Setup) Editor

Argument "-gbt" : Dump **Current Boot type** to file

- Type command "H2OUVE -gbt <file>"

```
> sudo H2OUVE-Lx64 -gbt bt.txt  
Saving Boot Type setting file: bt.txt
```

```
===== Boot Type Order =====  
(128)[0] "Network"  
(002)[1] "Hard Disk"  
(238)[2] "RAID"  
(005)[3] "USB Storage"  
(003)[4] "CD/DVD ROM"  
=====
```

SCU(Setup) Editor

Argument "-sbt" : Save **Current Boot type** by input file

- Type command "H2OUVE -sbt <file>"

```
===== Boot Type Order =====  
(128)[0] "Network"  
(002)[1] "Hard Disk"  
(238)[2] "RAID"  
(005)[3] "USB Storage"  
(003)[4] "CD/DVD ROM"  
=====
```



```
===== Boot Type Order =====  
(128)[1] "Network"  
(002)[0] "Hard Disk"  
(238)[2] "RAID"  
(005)[3] "USB Storage"  
(003)[4] "CD/DVD ROM"  
=====
```

```
> sudo H2OUVE-Lx64 -sbt bt.txt  
Now Parsing File From bt.txt.  
Bios Checking...OK!  
Succeed to Save Boot Type setting.
```

SCU(Setup) Editor

Argument "-ms" : Modify specific item of **Current Setup Setting**

- Type command "H2OUVE -ms -fi <Item Name> -op <Option> "
 - Item Name is case sensitive.

```
> H2OUVE-$ -ms -fi Quick Boot -op Enabled
```

```
Now parsing Hii data, please wait a minute...
```

```
Successful update SCU "Quick Boot" = "Enabled"!
```

```
>
```

- If it do not need to modify the setting of item, it will show this message

```
> H2OUVE-$ -ms -fi Quick Boot -op Enabled
```

```
Now parsing Hii data, please wait a minute...
```

```
"Quick Boot" has been set "Enabled".
```

```
>
```

SCU(Setup) Editor

Argument "-ms" : Modify specific item of **Current Setup Setting**

- Modify Boot Type Order
 - According the name strings to modify the Boot Type Order.
 - **MUST** use "\", " to separate the name strings **WITHOUT** redundant space.

```
===== Boot Type Order =====  
BootTypeOrderSize   = 0x8  
BootTypeOrderOffset = 0x95  
(238)[00] BOOT_POLICY_OTHER_DEVICE  
(002)[01] BBS_HARDDISK  
(001)[02] BBS_FLOPPY  
(005)[03] BBS_USB  
(003)[04] BBS_CDROM  
=====
```

```
> H2OUVE-$ -ms -fi Boot Type Order -op BBS_USB,BBS_HARDDISK,BBS_CDROM,BBS_FLOPPY,BOOT_POLICY_OTHER_DEVICE
```

```
Now parsing Hii data, please wait a minute...
```

```
Successful update SCU "Boot Type Order" = "BBS_USB,BBS_HARDDISK,BBS_CDROM,BBS_FLOPPY,BOOT_POLICY_OTHER_DEVICE"!
```

```
>
```

SCU(Setup) Editor

Argument "-ms" : Modify specific item of **Current Setup Setting**

- Modify Boot Type Order
 - If user entered an invalid input, and utility will show the current setting.

```
Now parsing Hii data, please wait a minute...
The size is incorrect, the number of BootTypeOrder is 5!!
Please refer to the following for modifying Boot Type Order:
===== Boot Type Order =====
(002)[00] BBS_HARDDISK
(238)[01] BOOT_POLICY_OTHER_DEVICE
(005)[02] BBS_USB
(003)[03] BBS_CDROM
(001)[04] BBS_FLOPPY
```

SCU(Setup) Editor

Argument "-ms" : Modify specific item of **Current Setup Setting**

- Modify Boot Order

- According the number in the brackets() to modify the Boot Order.
- The rule for modify Boot Order has been explained at [here](#).

Boot Order		
08193	[0]	<EFI> EFI USB Device
00000	[1]	<EFI> Internal EFI Shell
00002	[2]	<Legacy> JetFlashTranscend 8GB
00001	[3]	<Legacy> JetFlashTranscend 4GB
08194	[4]	<EFI> EFI DVD/CDROM
08195	[5]	<EFI> EFI Network

```
> H2OUVE-$ -ms -fi Boot Order -op 2 1 8193 0 8194 8195
```

```
Now parsing Hii data, please wait a minute...
```

```
Successful update SCU "Boot Order" = "2 1 8193 0 8194 8195"!
```

```
>
```


SCU(Setup) Editor

Argument "-ms" : Modify specific item of **Current Setup Setting**

- Modify Boot Order
 - If user entered an invalid input, and utility will show the current setting.

```
Now parsing Hii data, please wait a minute...
The size is incorrect, the number of BootTypeOrder is 6!!
Please refer to the following for modifying Boot Order:
===== Boot Order =====
(00001)[0] <Legacy> JetFlashTranscend 8GB
(00002)[1] <Legacy> UDISK    PDM06_8G B312.0
(00000)[2] Internal EFI Shell
(08193)[3] EFI USB Device
(08194)[4] EFI DVD/CDROM
(08195)[5] EFI Network
```

SCU(Setup) Editor

Argument "-ms" : Modify specific item of **Current Setup Setting**

- Add Password

- Command format after "-op" : "" "Password"

```
> H2OUVE-$ -ms -fi Set Supervisor Password -op "" "123"  
Now parsing Hii data, please wait a minute...  
Successful update SCU "Set Supervisor Password" = "123"!
```

- Change Password

- Command format after "-op" : "Old Password" "New Password"

```
> H2OUVE-$ -ms -fi Set Supervisor Password -op "123" "456"  
Now parsing Hii data, please wait a minute...  
Successful update SCU "Set Supervisor Password" = "456"!
```

- Clear Password

- Command format after "-op" : "Old Password" ""

```
> H2OUVE-$ -ms -fi Set Supervisor Password -op "456" ""  
Now parsing Hii data, please wait a minute...  
Successful update SCU "Set Supervisor Password" = ""!
```

SCU(Setup) Editor

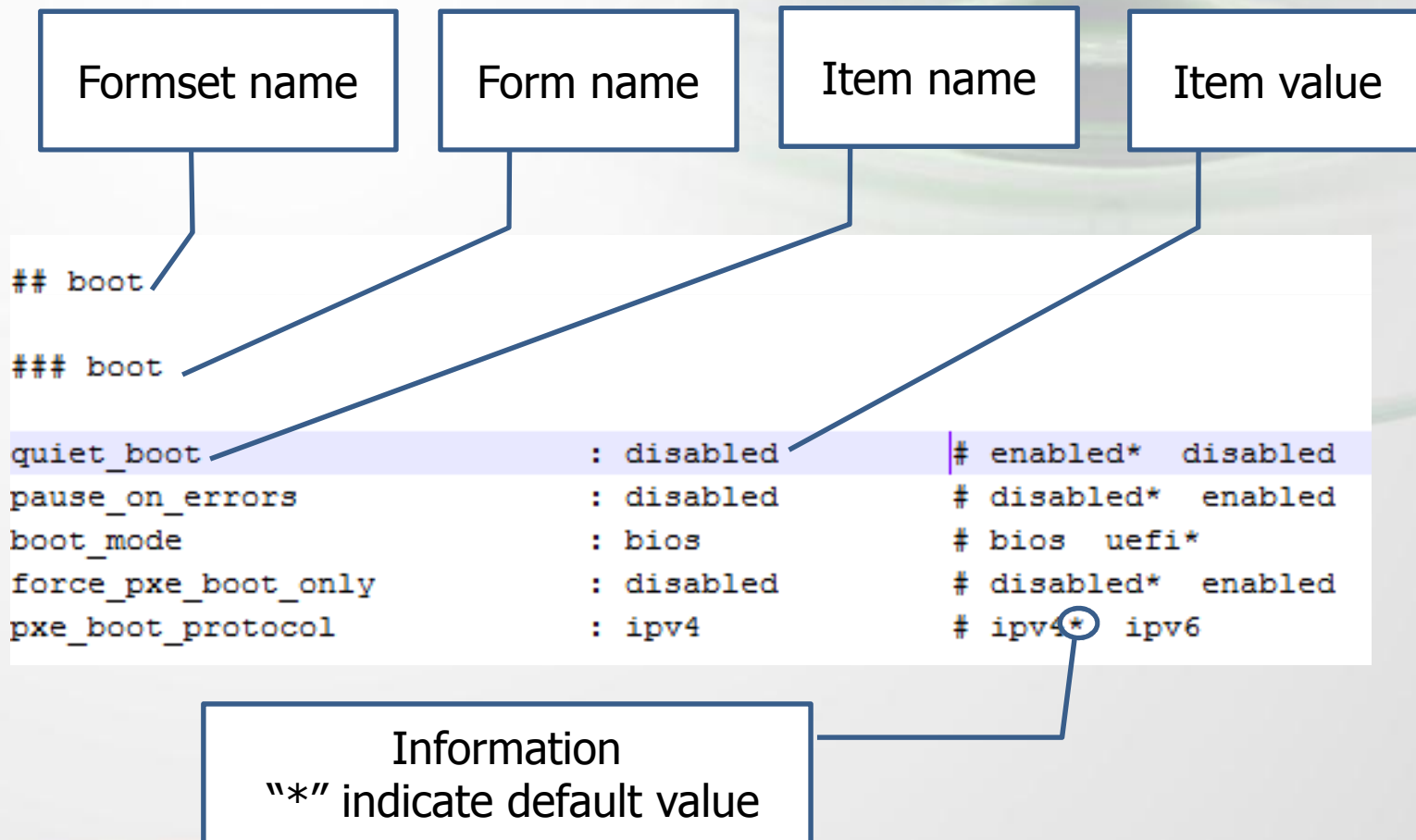
Argument “-d40” : Modify SCU settings with name-value paired format

- Type command “H2OUVE -d40 <-g [File] | -s <File> | -m <Name> <Value> | -l | -c | -b | -t>”
- Name-value paired format list below:
 - Concatenate name string without spaces between each words.
 - Maintain lower case.
 - Common name/value delimitation by colon (:).
 - List relevant information on same line as name-value pair.
 - Everything past the “#” are the available settings.
 - The “*” indicates the default setting, but is just informational.
- This command set doesn't support setup settings with duplicate name, user can use “-cdi” command to check any duplicate name exists.

SCU(Setup) Editor

Name-value paired content of output file

- Descriptions about name-value paired file format



SCU(Setup) Editor

Name-value paired content of output file

- Boot device doesn't have default option
- Boot type may not has default option
- Every boot type should be different
- When change sequence of boot type, modify at least 2 boot types

```
## boot_type
boot_type_1      : hard_disk      # hard_disk network raid usb_storage cd/dvd_rom
boot_type_2      : usb_storage    # hard_disk network raid usb_storage cd/dvd_rom
boot_type_3      : raid           # hard_disk network raid usb_storage cd/dvd_rom
boot_type_4      : network        # hard_disk network raid usb_storage cd/dvd_rom
boot_type_5      : cd/dvd_rom     # hard_disk network raid usb_storage cd/dvd_rom
```

SCU(Setup) Editor

Name-value paired content of output file

- Modify normal item

```
quiet_boot      : disabled
pause_on_errors : disabled
boot_mode       : bios
force_pxe_boot_only : disabled
pxe_boot_protocol : ipv4
```



```
quiet_boot      : enabled
pause_on_errors : disabled
boot_mode       : bios
force_pxe_boot_only : disabled
pxe_boot_protocol : ipv4
```

- Modify boot type

```
## boot type
boot_type_1      : hard_disk
boot_type_2      : usb_storage
boot_type_3      : raid
boot_type_4      : network
boot_type_5      : cd/dvd_rom
```



```
## boot type
boot_type_1      : usb_storage
boot_type_2      : hard_disk
boot_type_3      : raid
boot_type_4      : network
boot_type_5      : cd/dvd_rom
```


SCU(Setup) Editor

Argument “-d40 -g” : Same as -gs option, but dump to name-value paired file

- Type command “H2OUVE -d40 -g”
 - List all name-value paired setup settings.

```
> sudo H2OUVE-L -d40 -g
- Config Name ID: []
- Config ID: [](In hex)
- BIOS Version(From BVDt): "Metallica.0.0.3"

## boot

### boot

quiet_boot           : enabled          # enabled* disabled
pause_on_errors      : disabled         # disabled* enabled
boot_mode             : bios             # bios* uefi
force_pxe_boot_only  : disabled         # disabled* enabled
```

- Type command “H2OUVE -d40 -g <File>”
 - Dump all name-value paired setup settings in <File>

```
> sudo H2OUVE-L -d40 -g file.txt
Now parsing Hii data, please wait a minute...
Complete parsing Hii data!!!
Saving SCU setting file: file.txt
```

SCU(Setup) Editor

Argument “-d40 -s” : Same as -ss option, but with name-value paired file content

- Type command “H2OUVE -d40 -s <File>”
 - Update current setup settings by name-value paired format file

```
> sudo H2OUVE-L -d40 -s file.txt  
Now Parsing File From file.txt.  
Succeed to read file.txt file.  
Nothing changed.
```

- All settings file dumped by -d40 can be saves by -d40 -s, includes...
 - -d40 -g
 - -d40 -b
 - -d40 -t

SCU(Setup) Editor

Argument "-d40 -m" : Modify a setup utility setting by name and value

- Type command "H2OUE -d40 -m <Name> <Value>"
 - Update one current setup item by <Name> <Value>

```
> sudo H2OUE-L -d40 -m quiet_boot disabled
1: Modify quiet_boot to disabled
Succeed to write SCU settings!
1 items changed.
```

- Type command "H2OUE -m <Name>"
 - Get current value by input <Name>

```
> sudo H2OUE-Lx64 -d40 -m quiet_boot
enabled          # enabled*  disabled
```

Current Value

SCU(Setup) Editor

Argument “-d40 -m” : Modify a setup utility setting by name and value

- Type command “H2OUVE -d40 -m <Name> <Value> [-m <Name> <Value>]...”
 - Update multiple current setup items by [-m <Name> <Value>]

```
> sudo H2OUVE-L -d40 -m quiet_boot enabled -m boot_mode uefi
1: Modify quiet_boot to enabled
2: Modify boot_mode to uefi
Succeed to write SCU settings!
2 items changed.
```

- Please run more than 2 items when changing boot devices and boot types

SCU(Setup) Editor

Argument “-d40 -l” : List all item names

- Type command “H2OUVE -d40 -l”
 - List all names of item in current setup settings

```
> sudo H2OUVE-L -d40 -l
quiet_boot
pause_on_errors
boot_mode
force_pxe_boot_only
restore_on_ac_power_loss
power_staggering_ac_recovery
power_button
nmi_on_error
wake_on_lan
serial_communication
serial_port_address
serial_connector
serial_port_mode
remote_terminal_type
redirection_after_boot
set_supervisor_password
power_management
```

SCU(Setup) Editor

Argument “-d40 -c” : Change current settings to default value

- Type command “H2OUVE -d40 -c”
 - Update all current setup settings with their default value

```
> sudo H2OUVE-L -d40 -c
1: Modify boot_mode to bios
2: Modify wake_on_pme to disabled
3: Clear set_supervisor_password
4: Clear set_user_password
5: Clear set_all_hdd_password
6: Clear set_all_master_hdd_password
6 items changed.
Succeed to write SCU settings!
```


SCU(Setup) Editor

Argument “-d40 -b” : Dump boot device to name-value file

- Type command “H2OUVE -d40 -b [File]”
 - Dump all boot devices in name-value file

```
> sudo H2OUVE-Lx64 -d40 -b b40.txt
Complete parsing Hii data!!!
Saving SCU setting file: b40.txt
```

- Save boot devices

```
## boot_device
boot_device_1 : efi_hard_disk_2_(ubuntu) # (efi) efi_hard_disk_2_(ubuntu)
boot_device_2 : efi_hard_disk_1_(windows_boot_manager) # (efi) efi_hard_disk_1_(windows_boot_manager)
```

```
## boot_device
boot_device_1 : efi_hard_disk_1_(windows_boot_manager) # (efi) efi_hard_disk_1_(windows_boot_manager)
boot_device_2 : efi_hard_disk_2_(ubuntu) # (efi) efi_hard_disk_2_(ubuntu)
```

```
> sudo H2OUVE-Lx64 -d40 -s b40.txt
Now Parsing File From b40.txt.
Succeed to read b40.txt file.
Modify boot_device_1 to "efi_hard_disk_1_(windows_boot_manager)"
Modify boot_device_2 to "efi_hard_disk_2_(ubuntu)"
2 items changed.
```

SCU(Setup) Editor

Argument “-d40 -t” : Dump boot type to name-value file

- Type command “H2OUIVE -d40 -t [File]”
 - Dump all boot types in name-value file

```
> sudo H2OUIVE-Lx64 -d40 -t t40.txt
Saving SCU setting file: t40.txt
```

- Save boot types

```
## boot_type
boot_type_1      : hard_disk
boot_type_2      : network
boot_type_3      : raid
boot_type_4      : usb_storage
boot_type_5      : cd/dvd_rom
```



```
## boot_type
boot_type_1      : hard_disk
boot_type_2      : network
boot_type_3      : raid
boot_type_4      : cd/dvd_rom
boot_type_5      : usb_storage
```

```
> sudo H2OUIVE-Lx64 -d40 -s t40.txt
Now Parsing File From t40.txt.
Succeed to read t40.txt file.
Modify boot_type_4 to "cd/dvd_rom"
Modify boot_type_5 to "usb_storage"
2 items changed.
```

SCU(Setup) Editor

Argument “-cdi” : Report duplicate items and their locations in BIOS Setup Utility

- Type command “H2OUVE -cdi”
 - List all duplicate items in setup settings on screen

```
> sudo H2OUVE-L -cdi  
  
## FormsetID: [Formset name] -> FormID: [Form name]  
  
There are 0 duplicate items.
```

- Type command “H2OUVE -cdi [File]”
 - Dump all duplicate items in setup setting to specified file

```
> sudo H2OUVE-L -cdi file.txt  
Now parsing Hii data, please wait a minute...  
Complete parsing Hii data!!!  
Saving SCU setting file: file.txt
```

SCU(Setup) Editor

Argument "-cdi" : Report duplicate items and their locations in BIOS Setup Utility

- Type command "H2OUVE -cdi -n <Name>"
 - List locations if any item in setup setting has same name

```
> sudo H2OUVE-L -cdi -n "Quiet Boot"
0x02: [Boot] -> 0x0001: [Boot]
```

- Type command "H2OUVE -cdi -i <ImageFile>"
 - List all duplicate items in setup setting of Bios Image File

```
> sudo H2OUVE-L -cdi -i bios image file.fd
## FormsetID: [Formset name] -> FormID: [Form name]

1. Spin-Up Device
  0x1: [Advanced] -> 0x23: [SATA Configuration]
  0x1: [Advanced] -> 0x23: [SATA Configuration]
  0x1: [Advanced] -> 0x23: [SATA Configuration]
  0x1: [Advanced] -> 0x23: [SATA Configuration]
  0x1: [Advanced] -> 0x23: [SATA Configuration]
  0x1: [Advanced] -> 0x23: [SATA Configuration]

2. SATA Device Type
  0x1: [Advanced] -> 0x23: [SATA Configuration]
  0x1: [Advanced] -> 0x23: [SATA Configuration]
  0x1: [Advanced] -> 0x23: [SATA Configuration]
```

SCU(Setup) Editor

Argument “-cdi” : Report duplicate items and their locations in BIOS Setup Utility

- Type command “H2OUVE -cdi [-n <Name>] [-i <ImageFile>]”
 - List locations if any item in setup setting of bios image file has same name

```
> sudo H2OUVE-L -cdi -i bios_image_file.fd -n "Quiet Boot"
0x05: [Boot] -> 0x0001: [Boot]
```

- List all duplicate items

```
> sudo H2OUVE-Lx64 -cdi

## FormsetID: [Formset name] -> FormID: [Form name]

There are 0 duplicate items.
```

- List all duplicate items in image file

```
> sudo H2OUVE-Lx64 -cdi -i this.fd

## FormsetID: [Formset name] -> FormID: [Form name]

There are 0 duplicate items.
```

SCU(Setup) Editor

Argument "-bfirst" : Make device to first boot.

- Type command "H2OUE -bfirst <name>"
 - Set input name of boot device to be first boot device

```
> sudo H2OUE-L -bfirst "USB1:      USB FLASH DRIVE "  
Boot device: USB1:      USB FLASH DRIVE  
[0] "HDD1:Hitachi HDS721010CLA332  "  
[1] "USB1:      USB FLASH DRIVE "  
[2] "USB2:Generic Flash Disk      "  
  
Make "USB1:      USB FLASH DRIVE " to be first boot device  
Success!
```

SCU(Setup) Editor

Argument "-uvebf" : Enable/Disable H2OUVE functions in BIOS for subsequent boot

- Type command "H2OUVE -uvebf <Enabled|Disabled>"
 - Enable H2OUVE functionalities

```
> sudo H2OUVE-L -uvebf Enabled
Succeed to set!
Please reboot for working.
If it doesn't work, please ask BIOS to support it feature.
```

- Disable H2OUVE functionalities

```
> sudo H2OUVE-L -uvebf Disabled
Succeed to set!
Please reboot for working.
If it doesn't work, please ask BIOS to support it feature.
```


- 
- Support Mode
 - Arguments
 - Common Arguments
 - Setup Editor
 - **Image Setup Editor**
 - Variable Editor
 - Multi-Config Editor
 - Variable Stress Test

Image Setup Editor

Argument “-gd” : Generate **Setup Default Setting** to a file

- Type command “H2OUVE -gd <file> [-i <ImageFile>]”
 - All the default settings of setup menu will be listed in the output file,
And the [contents format](#) of file is the same with SCU Editor.
 - Runtime

```
> H2OUVE-Sx64 -gd file.txt  
  
Now parsing data, please wait a minute...  
Now parsing Hii data, please wait a minute...  
  
Complete parsing image file!!!  
Saving SCU default setting file: file.txt
```

- Image file

```
> H2OUVE-Sx64 -gd file.txt -i img.fd  
  
Now parsing data, please wait a minute...  
Succeed to read img.fd file.  
Succeed to read file.  
Now parsing Hii data, please wait a minute...  
Complete parsing image file!!!  
Saving SCU default setting file: file.txt
```

Image Setup Editor

Argument "-sd" : Modify **Setup Default Setting** with the data in the file

- Type command "H2OUVE -sd <file> -i <ImageFile> -out <NewImageFile>"
 - Argument "-out" is optional to save as the other new image name.

```
> H2OUVE-Sx64 -sd file.txt -i img.fd -out out.fd
```

```
Now Parsing File From file.txt.  
Succeed to read file.txt file.  
Succeed to read img.fd file.  
Now parsing Hii data, please wait a minute...  
Compress Firmware Volume....  
Write data to out.fd file, please wait a minute...  
Succeed to write setup default setting to image file "out.fd".
```

Image Setup Editor

Argument "-gstr" : **Generate strings** which in Setup page to a file.

- Strings with two type in setup page.
 - Setup utility
 - > H2OUVE -gstr <ImageFile> <file> -s

```
> H2OUVE-Sx64 -gstr file_s.txt -s -i img.fd
```

```
Now Parsing File From img. fd.  
Succeed to read img. fd file.  
Now parsing Hii data, please wait a minute...  
Complete parsing image file!!!  
Saving setting to file: file_s.txt
```

- Setup browser
 - > H2OUVE -gstr <ImageFile> <file> -b

```
> H2OUVE-Sx64 -gstr file_b.txt -b -i img.fd
```

```
Now Parsing File From img. fd.  
Succeed to read img. fd file.  
Now parsing Hii data, please wait a minute...  
Complete parsing image file!!!  
Saving setting to file: file_b.txt
```

Image Setup Editor

Content of output file with "-gstr" argument

- Modify the string in the " ", and then "-sstr" can apply the string base on this file.
- It has word limit at the top of whole strings
- The first number in brackets means which package it belong to (For string of setup utility).

Remaining character (Package0): 2048
Remaining character (Package1): 2048

[0,682) Prompt
en-US "Event Log Viewer"

[0,683) Help
en-US "The utility is for user to view event log of all storages."

[1,2B) Prompt
en-US "Dual Boot Type"

[1,2C) Prompt
en-US "Legacy Boot Type"

[1,2D) Prompt
en-US "UEFI Boot Type"

[1,5A) Prompt
en-US "Quick Boot"

The remaining character that can be added.

Remaining character: 2048

[1]
en-US "English"

[2]
en-US "F9=Reset to Defaults"

Strings for setup browser

Image Setup Editor

Argument "-sstr" : **Save strings** to Setup page with the data in the file.

- Argument "-out" is supported for this argument. (optional)

- Setup utility

- > H2OUVE -sstr <file> <ImageFile> -s

```
> H2OUVE-$x64 -sstr file_s.txt -s -i img.fd
```

```
Now Parsing File From file_s.txt.  
Succeed to read file_s.txt file.  
Succeed to read img.fd file.  
Compress Firmware Volume....  
Write data to img.fd file, please wait a minute...  
Succeed to write setting to image file "img.fd".
```

- Setup browser

- > H2OUVE -sstr <file> <ImageFile> -b

```
> H2OUVE-$x64 -sstr file_b.txt -b -i img.fd
```

```
Now Parsing File From file_b.txt.  
Succeed to read file_b.txt file.  
Compress Firmware Volume....  
Write data to img.fd file, please wait a minute...  
Succeed to write setting to image file "img.fd".
```

Image Setup Editor

Example for Argument "-sstr".

- 1. Modify content of output file with "-gstr" argument.
 - This example is for setup utility type(-s)

(1,5A) Prompt
en-US "Quick Boot"



(1,5A) Prompt
en-US "Fast Boot"

- 2. Save string to setup page in image file.
 - >H2OUVE -sstr <file> <ImageFile> -s
- 3. Use the image file to update BIOS.
- 4. The string has changed.



Boot Type	<Dual Boot Type>
Quick Boot	<Enabled>
Quiet Boot	<Enabled>
Network Stack	<Disabled>

Boot Type	<Dual Boot Type>
Fast Boot	<Enabled>
Quiet Boot	<Enabled>
Network Stack	<Disabled>

- 
- Support Mode
 - Arguments
 - Common Arguments
 - Setup Editor
 - Image Setup Editor
 - **Variable Editor**
 - Multi-Config Editor
 - Variable Stress Test

Variable Editor

Argument “-gv” : Generate **all the UEFI Variable** to a file

- Type command “H2OUVE -gv <file> [-i <ImageFile>]”

- Runtime

```
> H2OUVE-Sx64 -gv var.txt
```

```
Now parsing Variable Information.
```

```
Succeed to read img.fd file.
```

```
Complete parsing Variable Information!!!
```

```
Saving Variable Information file: var.txt
```

- Image

```
> H2OUVE-Sx64 -gv var_i.txt -i img.fd
```

```
Now parsing Variable Information.
```

```
Succeed to read img.fd file.
```

```
Complete parsing Variable Information!!!
```

```
Saving Variable Information file: var_i.txt
```

Variable Editor

Content of output file with "-gv" argument

- File format:

[Index] Variable Name

GUID:XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXXXX

Attributes: 0xXX

DataSize: 0xXX

Data:

00000000: XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX

00000010: XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX

.....

- Index must start from 001 and be in order.
- The number of data bytes must be equal to DataSize. (Except data size = 0)
- "X" represent a hex number.
- It only supports for modifying the data region of variable on image.

```
[001] CurrentFD
      GUID: 59D1C24F-50F1-401A-B101-F33E0DAED443
      Attributes: 0x3
      DataSize: 0x1
      Data:
          00000000: 26

[002] BootState
      GUID: 60B5E939-0FCF-4227-BA83-6BBED45BC0E3
      Attributes: 0x3
      DataSize: 0x1
      Data:
          00000000: 01

[003] GigNVDData
      GUID: A31ABB16-C627-475B-988E-7EE0776740F3
      Attributes: 0x3
      DataSize: 0x3
      Data:
          00000000: 00 00 00
```

Variable Editor

Content of output file with "-gv" argument

- Modify Variable data value:

```
[012] Timeout
      GUID: 8BE4DF61-93CA-11D2-AA0D-00E098032B8C
      Attributes: 0x7
      DataSize: 0x2
      Data:
          00000000: 00 00
```



```
[012] Timeout
      GUID: 8BE4DF61-93CA-11D2-AA0D-00E098032B8C
      Attributes: 0x7
      DataSize: 0x2
      Data:
          00000000: 10 00
```

- Delete Variable: (Runtime Only)
 - Modify **DataSize** or **Attributes** value to zero.

```
[012] Timeout
      GUID: 8BE4DF61-93CA-11D2-AA0D-00E098032B8C
      Attributes: 0x7
      DataSize: 0x0
      Data:
          00000000: 10 00
```



```
[012] Timeout
      GUID: 8BE4DF61-93CA-11D2-AA0D-00E098032B8C
      Attributes: 0x0
      DataSize: 0x2
      Data:
          00000000: 10 00
```

Variable Editor

Content of output file with "-gv" argument

- Create/Add some Variables : (Runtime Only)
 - Follow the format to add the description and ensure index number must be in order.

```
[001] Test Variable1
      GUID: 11112222-3333-4444-5555-666677778888
      Attributes: 0x6
      DataSize: 0x1
      Data:
          00000000: 00

[002] Test Variable2
      GUID: 11112222-3333-4444-5555-666677778888
      Attributes: 0x7
      DataSize: 0x8
      Data:
          00000000: 00 70 EB 79 00 00 00 00

[003] Test Variable3
      GUID: 99998888-7777-6666-5555-444433332222
      Attributes: 0x7
      DataSize: 0x1
      Data:
          00000000: 01
```

Variable Editor

Argument “-sv” : Modify **Variable** with the data in the file

- Type command “H2OUVE -sv <file> [-i <ImageFile>]”
 - Runtime

```
> H2OUVE-Sx64 -sv gv.txt
```

```
Now Parsing File From gv.txt.  
Succeed to read gv.txt file.  
Succeed to write Variable.
```

- Image

```
> H2OUVE-Sx64 -sv var_i.txt -i img.fd
```

```
Now Parsing File From var_i.txt.  
Succeed to read var_i.txt file.  
Succeed to read img.fd file.  
Write data to img.fd file, please wait a minute...  
Succeed to write Variable to Image file "img.fd".
```

Variable Editor

Argument "-re" : Delete specific **Variable** by Name or GUID

- Type command "H2OUVE -re -vn <VarName> -vg <GUID>"

```
> H2OUVE-S -re -vn TestVar -vg 11112222-3333-4444-5555-666677778888
```

```
Delete Variable.....OK
```

```
>
```


Variable Editor

Argument "-re" : Delete specific **Variable** by Name or GUID

- If only type command as "H2OUIVE -re -vn <VarName>" or "H2OUIVE -re -vg <GUID>", the tool will list all matched Variables one by one and delete it by asking user.

```
> H2OUIVE-S -re -vn TestVar
```

```
Variable Name: TestVar
  GUID = 11112222-3333-4444-5555-666677778888
  Attributes: 0x6
  Variable Data Size = 01
  Data:
  00000000: 01                                     *. *
Are you sure you want to Delete TestVar Variable?(y/n):
```

```
> H2OUIVE-S -re -vg 11112222-3333-4444-5555-666677778888
```

```
Variable Name: TestVar
  GUID = 11112222-3333-4444-5555-666677778888
  Attributes: 0x6
  Variable Data Size = 01
  Data:
  00000000: 01                                     *. *
Are you sure you want to Delete TestVar Variable?(y/n):
```

Variable Editor

Argument “-cvl” : Check if **Variable** is locked

- Type command “H2OUVE -cvl [<File> | -vn <Name> -vg <GUID>]”
 - Supported on UEFI mode
 - Check if variables are locked list in file which is a INI file composed by variable name and GUID

```
[Variable]
VariableName01="Setup"
GUID01="A04A27F4-DF00-4D42-B552-39511302113D"
VariableName02="SetupMode"
GUID02="8BE4DF61-93CA-11D2-AA0D-00E098032B8C"
```

One variable

```
> sudo H2OUVE-Lx64 -cvl VariableTestTool.ini
Name                               GUID
-----
Setup                               A04A27F4-DF00-4D42-B552-39511302113D ... not locked.
SetupMode                           8BE4DF61-93CA-11D2-AA0D-00E098032B8C ... locked.
Finish!
```

- Check if a specific variable is locked or not

```
> sudo H2OUVE-Lx64 -cvl -vn Setup -vg A04A27F4-DF00-4D42-B552-39511302113D
Variable name: Setup
Variable GUID: A04A27F4-DF00-4D42-B552-39511302113D
Variable is not locked.
```

- 
- Support Mode
 - Arguments
 - Common Arguments
 - Setup Editor
 - Image Setup Editor
 - Variable Editor
 - **Multi-Config Editor**
 - Variable Stress Test

Multi-Config Editor

Important information

- **This feature was not available since 100.00.16.06.**
- **Kernel 5.1 doesn't support Multi-Config.**

Multi-Config Editor

About input file

- Set **Config Name ID**, **Config ID** for adding or modifying Config data.

- Config Name ID: []
- Config ID: [](In hex)
- BIOS Version(From BVDT): "SharkBay.05.04.22.0000"

- Set Config Name ID and Config ID between the brackets [].
- Maximum Config ID is 0xFE, 0xFF can't be a Config ID.
- The field of Bios Version is for checking the context of file is compatible with the BIOS. (The field is filled automatically by H2OUVE when the file is generating.)

Multi-Config Editor

About input file of `-scu` type and `-k` type

- **For `"-scu"` type**

- The format is same as the content of output file with `"-gd"` argument

- **For `"-k"` type**

- The format is same as the content of output file with `"-gk"` argument
- First, use `"-gk"` argument to get the current hotkey setting to file, then you modify this file to do other operation about Multi-Config.

```
===== HotKey0 =====  
Key: (0x3B,0x0,0x0,0x0)  
Operation: 6  
  
===== HotKey1 =====  
Key: (0x3C,0x0,0x0,0x0)  
Operation: 5  
  
===== HotKey2 =====  
Key: (0x53,0x0,0x0,0x0)  
Operation: 7
```


Multi-Config Editor

About input file for -lg type

- The format is same as the content of output file with "-glg" argument.
- Firstly, using "-gk" option to get the current logo setting to file, then modifying this file.

```
Logo Type: BGRT

##### Image File Setting0(BIOS) #####

Status: [Disabled]
Image Name: []
Attribute: [8]
Coordinate X: [0]
Coordinate Y: [0]

#####

##### Image File Setting1(BIOS) #####

Status: [Default]
Image Name: []
Attribute: [9]
Coordinate X: [0.0%]
Coordinate Y: [0.0%]

#####
```


Multi-Config Editor

About input file for -sb type

- The format is same as the content of output file with "-gsb" argument.
- Firstly, using "-gsb" option to get the current SMBIOS setting to file, then modifying this file.

```
- Config Name ID: []  
- Config ID: [] (In hex)  
- BIOS Version(From BVDt): "SharkBay.05.05.24.0000"  
  
----- DMI Setting Block0 -----  
Type: []  
- Setting: ([], [], [], "")  
----- End Setting Block0 -----
```

Multi-Config Editor

Argument "-gk" :Get **Hotkey setting** to file.

- Dump dynamic hotkey setting to file.
 - Hotkey
 - String
 - System Information

```
===== String =====  
Key: N/A  
Text: "CPUID : " (min,max)
```

```
===== HotKey0 =====  
Key: (0x3B,0x0,0x0,0x0)  
Operation: 6  
  
===== HotKey1 =====  
Key: (0x3C,0x0,0x0,0x0)  
Operation: 5  
  
===== HotKey2 =====  
Key: (0x53,0x0,0x0,0x0)  
Operation: 7
```

```
===== System Info0 =====  
Type: 0  
String: "InsydeH2O Version : %BIOSVER%" (min,min)  
  
===== System Info1 =====  
Type: 1  
String: "BIOS Build Date : %BUILDDATE%" (min,min)  
  
===== System Info2 =====  
Type: 2  
String: "Processor Type : %CPUTYPE%" (min,min)  
  
===== System Info3 =====  
Type: 3  
String: "System Memory Speed : %MEMSPEED%" (min,min)
```

Multi-Config Editor

Argument "-glg" :Get **BIOS Logo setting** to file.

- Using this setting to decide how to present logo during post.

```
Logo Type: BGRT

##### Image File Setting0(BIOS) #####

Status: [Disabled]
Image Name: []
Attribute: [8]
Coordinate X: [0]
Coordinate Y: [0]

#####

##### Image File Setting1(BIOS) #####

Status: [Default]
Image Name: []
Attribute: [9]
Coordinate X: [0.0%]
Coordinate Y: [0.0%]

#####
```

- BGRT: It only can show one logo image at most, and the position is decided by BIOS.
- OEM badge: It can show many logo images, and the position can be decided by user

- Disabled: Disalbed this Logo setting.
- Enabled: Use this Logo setting.
- Default: Use BIOS Logo setting

- Give it a image with absolute path

Give Attribute number listed below:

- (0) Display at LeftTop
- (1) Display at CenterTop
- (2) Display at RightTop
- (3) Display at CenterRight
- (4) Display at RightBottom
- (5) Display at CenterBottom
- (6) Display at LeftBottom
- (7) Display at CenterLeft
- (8) Display at Center
- (9) Display with Customized

- Coordinate X, Y are offset when Attribute is **0~8**.
- Coordiante X, Y are present in percentage when Attribute is **9**.

Multi-Config Editor

Argument "-gsb" :Get **SMBIOS setting** to file.

- Using argument to decide which type of SMBIOS should be dumped.
- The -gsb command can list all types and elements based on the arguments.
- The first 3 elements (Type, Length, Handle) are read-only, please do not modify them.
- Modifying the type, offset, attribute and value between "DMI Setting block" and "End Setting block" block.

```
#
# For Example:
# ----- DMI Setting Block -----
# Type: [0]
# - Setting: ([0], [05h] [4] "SharkBay.05.04.11.0000")
# - Setting: ([0], [12h] [5] "03h 0dh")
# - Setting: ([0], [14h] [1] "1ch")
# ----- End Setting Block -----
#
```

Instance

Attribute

Value based on
Attribute

SMBIOS Type

Offset

```
===== System Information (Type 1) =====
Offset Attribute - Title
(00h) 4 - Type
(01h) 4 - Length
(02h) 4 - Handle
(04h) 4 - Manufacturer
(05h) 4 - Product Name
(06h) 4 - Version
(07h) 4 - Serial Number
(08h) 6 - UUID
(18h) 1 - Wake-up Type
(19h) 4 - SKU Number
(1Ah) 4 - Family
----- DMI Setting Block1 -----
Type: [1]
- Setting: ([], [], [], "")
----- End Setting Block1 -----
```

Multi-Config Editor

About hotkey setting format

Key(ScanCode, Shift, Alt, Ctrl)

Relative operation

===== HotKey1 =====

Key: (0x1,0x0,0x0,0x0)

Operation: 1

Text_before: "Press Esc for boot options" (min,max)

Text_after: "Esc is pressed. Go to boot options." (min,max)

Graphic_after: "Esc is pressed. Go to boot options." (center,center)

Foreground: (0xFF, 0xFF, 0xFF)

Background: (0x00, 0x00, 0x00)

- (0) No Operation
- (1) Front page
- (2) Setup
- (3) Device manager
- (4) Boot manager
- (5) Boot from file
- (6) Secure boot option

Strings with different modes and location.(e.g., **Text_before** means string show at **Text** mode **before** pressing relative hotkey)

Location can input any valid number or input "min", "center", and "max" to set relative location with difference mode.

(**min**, **min**) is the **upper left** corner

(**max**, **max**) is the **lower right** corner

Set **Background** and **Foreground** color in **Hex**, and the order is "(Red, Green, Blue)".
(Only support on Graphic mode)

Multi-Config Editor

About string setting format

- It just like hotkey setting format, but without relative key.
 - It's always displayed with different modes.

```
===== String =====  
Key: N/A  
Text: "CPUID : " (min,max)
```



If you want to add another string setting, just copy this format and paste it beneath.

You can input any string in the “ ”, and set location in the ().

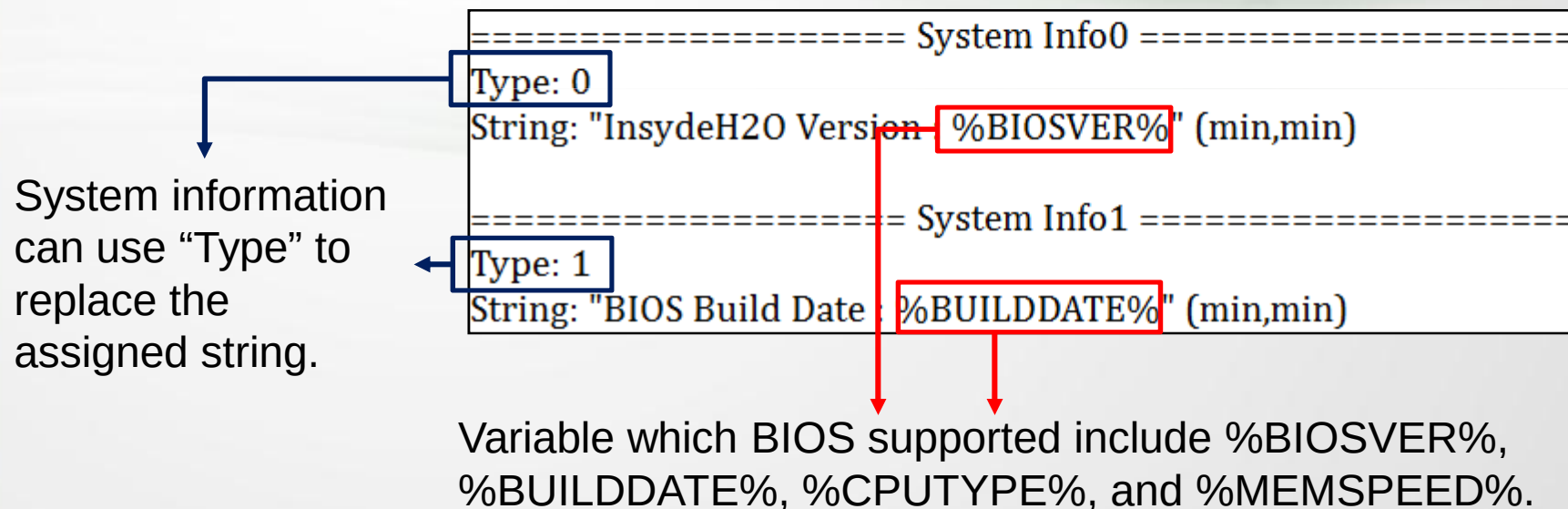
If many strings show at same location which use **identifiers**, and they will be shift to suitable location.

Note : String setting can **ADD ONLY**, and if you want to hide some strings from BIOS, you can add the string with many space at same location (location must input number, **CAN'T** input identifiers just like “min”)

Multi-Config Editor

About system information setting format

- It just like string setting format, but with some system information variable.
 - It will replace the contents according to the different platforms.



Multi-Config Editor

Argument “-list” :Multi-Config, **List all Config settings**, and dump specified data.

- List Multi-Config information from Runtime ROM or BIOS Image.
 - Runtime ROM

```
> sudo h2ouve-lx64 -mc -list
```

No.	Type	ConfigID	Valid	Mode	Active(POST/Data)	Name
0h	Setup	1h	True	R/W	(No/ No)	
1h	Setup	3h	True	R/W	(No/ No)	
2h	HotKey	2h	True	R/W	(No/ No)	
3h	HotKey	3h	True	R/W	(No/ No)	

```
The Number Of Current Config Data: 4  
Load setup default data from: BIOS  
Remaining size of Multi-Config: 0x3c46
```

- BIOS Image

```
> sudo h2ouve-lx64 -mc -list -i thisdcs.fd
```

No.	Type	ConfigID	Valid	Mode	Active(POST/Data)	Name
0h	Setup	0h	True	R/W	(No/ No)	
1h	Setup	1h	True	R/W	(No/Yes)	
2h	HotKey	0h	True	R/W	(No/ No)	
3h	HotKey	1h	True	R/W	(No/ No)	

```
The Number Of Current Config Data: 4  
Load setup default data from: Multi-Config  
Remaining size of Multi-Config: 0x3c46
```

Multi-Config Editor

Argument “-list” :Multi-Config, **List all Config settings**, and dump specified data.

- About the meaning of each field.

Multi-Config						
No.	Type	ConfigID	Valid	Mode	Active(POST/Data)	Name
0h	Setup	1h	True	R/W	(No/ No)	
1h	Setup	3h	True	R/W	(No/ No)	
2h	HotKey	2h	True	R/W	(No/ No)	
3h	HotKey	3h	True	R/W	(No/ No)	

- No. : The number in the Multi-Config list
- Type : The type of Config setting (Full is Full setup setting, Hotkey is dynamic hotkey)
- ConfigID : The number in each type (**Type and ConfigID will combine as a unique value**)
- Data : The attribute of Data which include RO and R/W.
- Active : Which setting will be active.
 - POST means that it is effectual once on next POST time.
 - Data means that it is always effectual.
- Name : Identity name for each Config data, it can help customers to recognize the purpose of the Config data. It is a ASCII, and at most **8 characters**.

Multi-Config Editor

Argument "-gen" :Multi-Config, **Generate Config settings to file.**

- Dump the specified Config setting by combining Type and ConfigID.

- Runtime ROM

```
> sudo h2ouve-lx64 -mc -gen gd.txt -scu 1
Now parsing data, please wait a minute...
Now parsing Hii data, please wait a minute...
Complete parsing!!!
Saving setting to file: gd.txt
```

- BIOS Image

```
> sudo h2ouve-lx64 -mc -gen gd.txt -scu 0 -i thisdcs.fd
Now parsing data, please wait a minute...
Now parsing Hii data, please wait a minute...
Complete parsing!!!
Saving setting to file: gd.txt
```

Multi-Config Editor

Argument “-add” :Multi-Config, **Add an new Config setting** with specified file.

- Add an new Config setting with specified file and specified type of data, and you can assign the ConfigID of Config setting(**optional**).

- Runtime ROM

```
> sudo h2ouve-lx64 -mc -add config.txt -scu
Bios Checking...OK!
Succeed to read config.txt file.
Now Parsing File and save settings, please wait a minute...
Succeed to save config data to multi config region. ConfigID: 0h
Succeed to save config data!
```

- BIOS Image

```
> sudo h2ouve-lx64 -mc -add config.txt -scu -i thisdcs.fd
Bios Checking...OK!
Succeed to read config.txt file.
Now Parsing File and save settings, please wait a minute...
Succeed to save config data to multi config region. ConfigID: 1h
Succeed to save config data!
```

Multi-Config Editor

Argument "-mod" : Multi-Config, **Modify Config setting** with specified file.

- Modify the specified Config setting with specified file.

- Runtime ROM

```
> sudo h2ouve-lx64 -mc -mod gd.txt -scu
Bios Checking...OK!
Succeed to read gd.txt file.
Now Parsing File and save settings, please wait a minute...
Succeed to save config data to multi config region. ConfigID: 1h
Succeed to save config data!
```

- BIOS Image

```
> sudo h2ouve-lx64 -mc -mod gd.txt -scu -i thisdcs.fd
Bios Checking...OK!
Succeed to read gd.txt file.
Now Parsing File and save settings, please wait a minute...
Succeed to save config data to multi config region. ConfigID: 0h
Succeed to save config data!
```

Multi-Config Editor

Argument “-mod” : Multi-Config, **Modify Config setting** with specified file.

- Modify the status and attribute of Config setting.
 - The result with argument “-mc -mod [invalid|valid] -f 1”.

Multi-Config					
No.	Type	ConfigID	Valid	Mode	Active(POST/Data)
0h	Setup	1h	True	R/W	(No/ No)
1h	Setup	3h	False	R/W	(No/ No)
2h	HotKey	2h	True	R/W	(No/ No)
3h	HotKey	3h	True	R/W	(No/ No)



Multi-Config					
No.	Type	ConfigID	Valid	Mode	Active(POST/Data)
0h	Setup	1h	True	R/W	(No/ No)
1h	Setup	3h	True	R/W	(No/ No)
2h	HotKey	2h	True	R/W	(No/ No)
3h	HotKey	3h	True	R/W	(No/ No)

- The result with argument “-mc -mod [RO|R/W] -f 0”.

Multi-Config					
No.	Type	ConfigID	Valid	Mode	Active(POST/Data)
0h	Setup	1h	True	R/W	(No/ No)
1h	Setup	3h	True	RO	(No/ No)
2h	HotKey	2h	True	R/W	(No/ No)
3h	HotKey	3h	True	R/W	(No/ No)



Multi-Config					
No.	Type	ConfigID	Valid	Mode	Active(POST/Data)
0h	Setup	1h	True	R/W	(No/ No)
1h	Setup	3h	True	R/W	(No/ No)
2h	HotKey	2h	True	R/W	(No/ No)
3h	HotKey	3h	True	R/W	(No/ No)

Multi-Config Editor

Argument "-del" : Multi-Config, **Delete specified Config setting.**

- Delete the specified Config setting by combining Type and ConfigID.

- Runtime ROM

```
> sudo h2ouve-lx64 -mc -del -scu 0  
Succeed to delete config data!
```

- BIOS Image

```
> sudo h2ouve-lx64 -mc -del -scu 0 -i thisdcs.fd  
Succeed to delete config data!
```


Multi-Config Editor

Argument "-format" : Multi-Config, **Format Multi-Config Region** based on ImageFile.

- Base on image file to format the Multi-Config region, and it can't accept the image file that is from other platforms.
 - Runtime ROM: Load Original.fd to format runtime Multi-Config region

```
> H20UVE-L -mc -format Original.fd  
Succeed to recovery Multi-Config region!
```

- BIOS Image: Load Original.fd to format Multi-Config region of img.fd

```
> H20UVE-L -mc -format Original.fd -i img.fd  
Succeed to recovery Multi-Config region!
```

Multi-Config Editor

Argument "-act" : **Active Multi-Config setting.**

- Active the specified Data setting

- Runtime ROM

```
> sudo h2ouve-lx64 -mc -act -scu 1  
Succeed to set!
```



Multi-Config					
No.	Type	ConfigID	Valid	Mode	Active(POST/Data)
0h	Setup	0h	True	R/W	(No/ No)
1h	Setup	1h	True	R/W	(No/Yes)
2h	HotKey	0h	True	R/W	(No/ No)
3h	HotKey	1h	True	R/W	(No/ No)

The Number Of Current Config Data: 4
Load setup default data from: Multi-Config
Remaining size of Multi-Config: 0x3c4b

- BIOS Image

```
> sudo h2ouve-lx64 -mc -act -scu 1 -i thisdcs.fd  
Succeed to set!
```



Multi-Config					
No.	Type	ConfigID	Valid	Mode	Active(POST/Data)
0h	Setup	0h	True	R/W	(No/ No)
1h	Setup	1h	True	R/W	(No/Yes)
2h	HotKey	0h	True	R/W	(No/ No)
3h	HotKey	1h	True	R/W	(No/ No)

The Number Of Current Config Data: 4
Load setup default data from: Multi-Config
Remaining size of Multi-Config: 0x3c4b

- Clear setting

```
> sudo h2ouve-lx64 -mc -act clear -scu  
Succeed to clear!
```

Multi-Config Editor

Example for Argument "-mc".

- 1. SCU setting from BIOS.

Boot Type	<Dual Boot Type>
Quick Boot	<Enabled>
Quiet Boot	<Enabled>
Network Stack	<Disabled>

- 2. Load the first setting of Multi-Config.

- >H2OUVE -mc -act -f 1

The First setting of Multi-Config

(0x79, 1, 0x1234) Boot Type	(0x6D, 1, 0x1234) Quick Boot
Callback ID = 0x105e	Callback ID = 0x1
(0)[] Dual Boot Type	(1)[] Enabled (Default)
(1)[] Legacy Boot Type	(0)[*] Disabled
(2)[*] UEFI Boot Type (Default)	

- 3. Restart the platform.

Boot Type	<Dual Boot Type>
Quick Boot	<Enabled>
Quiet Boot	<Enabled>
Network Stack	<Disabled>

- 4. Press "F9" to Load Optimal Defaults in SCU.

Boot Type	<UEFI Boot Type>
Quick Boot	<Disabled>
Quiet Boot	<Enabled>
Network Stack	<Disabled>

Multi-Config Editor

Argument "-post" : Multi-Config, **POST setting**.

- Active the specified Data setting once at first POST time.(Only for Full setup setting type)

- Runtime ROM

```
> sudo h2ouve-lx64 -mc -post -scu 1  
Succeed to set!
```



Multi-Config					
No.	Type	ConfigID	Valid	Mode	Active(POST/Data)
0h	Setup	1h	True	R/W	(Yes/ No)
1h	Setup	3h	True	R/W	(No/ No)
2h	HotKey	2h	True	R/W	(No/ No)
3h	HotKey	3h	True	R/W	(No/ No)

- BIOS Image

```
> sudo h2ouve-lx64 -mc -post -scu 0 -i thisdcs.fd  
Succeed to set!
```



Multi-Config					
No.	Type	ConfigID	Valid	Mode	Active(POST/Data)
0h	Setup	0h	True	R/W	(Yes/ No)
1h	Setup	1h	True	R/W	(No/ No)
2h	HotKey	0h	True	R/W	(No/ No)
3h	HotKey	1h	True	R/W	(No/ No)

- Clear setting

```
> sudo h2ouve-lx64 -mc -post clear -scu  
The setting has been cleared!
```

- 
- Support Mode
 - Arguments
 - Common Arguments
 - Setup Editor
 - Image Setup Editor
 - Variable Editor
 - Multi-Config Editor
 - **Variable Stress Test**

Variable Stress Test

Argument "-dbos" : **Display Boot Order Sequence.**

- Type command "H2OUE -dbos"
- Display the Boot Order sequence

```
fs0:\Insyde\Tool\H2OUE\Demo> H2OUE-Sx64.efi -dbos
```

```
-----Boot order-----  
1  EFI USB Device  
2  Internal EFI Shell  
3  ADATA   USB Flash Drive  
4  EFI DVD/CDROM  
5  EFI Network
```


Variable Stress Test

Argument "-cnvs" : **Clear NV Spare data.**

- Type command "H2OUE -cnvs"
- Clear NV Spare with fill in 0xFF.
- This function is not supported in secure flash mode.

```
fs0:\Insyde\Tool\H2OUE\Demo> H2OUE-Sx64.efi -cnvs
```

```
Read BIOS data...
```

```
Start testing.....
```

```
Check NV_FTW_SPARE
```

```
Tool already clean NV_FTW_SPARE's data
```

```
System reboot !
```

```
End testing.
```


Variable Stress Test

Argument "-cvhc" : **Check Variable Header Continuity.**

- Type command "H2OUE -cvhc"
- Auto detect the platform is Secure Boot or not, then check variable header continuity.
- If the variable header has checksum or variable sequence is non-contiguous, show the error message.

```
fs0:\Insyde\Tool\H2OUE\Demo> H2OUE-Sx64.efi -cvhc
```

```
Read BIOS data...
```

```
Secure Boot Supported... Check for Secure Variable Header.
```

```
Start testing.....
```

```
Variable header check.....PASS
```

```
End testing.
```

Variable Stress Test

Argument “-vorv” : **Variable store Out of Region Verification.**

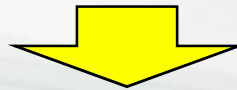
- Type command “H2OUVE -vorv [-count]”
- The argument -count used to specified the number of times.
- This function is not supported in secure flash mode.
- There are 2 case to testing:
 - The data size of last one variable is out of variable storage region.
 - The last one variable has only 8-byte variable header.
(StartID, State, and Attributes)
 - The system should do reclaim and remove those incomplete variables.

Variable Stress Test

Argument "-vorv" : **Variable store Out of Region Verification.**

```
fs0:\Insyde\Tool\H20UVE\Demo> H20UVE-Sx64.efi -vorv
```

```
Read BIOS data...
Start testing.....
1.
This is 1 times to run longrun
Do variable out of region check Case0 - Begin
```



```
Read BIOS data...
Start testing.....
Do variable out of region check Case1 - Begin
Write variable store region to ROM
Do variable out of region check - End

Elapsed time: 24540 (ms)
2013/01/01 00:18:39
System reboot !
```

Variable Stress Test

Argument “-cvur” : **Check Variable Usage situation by Reboot.**

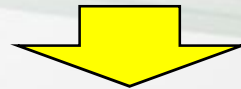
- Type command “H2OUVE -cvur [-count]”
- The argument -count used to specified the number of times.
- To detect that BIOS updates the variables during the POST time.
- There are 3 situations:
 - Variables are added or modified : list those variables
 - System do reclaim : show system do reclaim message
 - No change : pass the test

Variable Stress Test

Argument "-cvur" : **Check Variable Usage situation by Reboot.**

```
fs0:\Insyde\Tool\H2OUVE\Demo> H2OUVE-Sx64.efi -cvur
```

```
Read BIOS data...
Start testing.....
1.
This is 1 times to run longrun
Variable usage condition check #0
NV used bytes: 0x2DF5
Elapsed time: 44784 (ms)
2013/01/01 00:23:08
System reboot !
```



```
Read BIOS data...
Start testing.....
Variable usage condition check #1
NV used bytes: 0x2E3D
Test Fail - variables are updated or added
MTC - {EB704011-1402-11D3-8E-77-00-A0-C9-69-72-3B}
```

Variable Stress Test

Argument “-cvus” : **Check Variable Usage situation by Shutdown.**

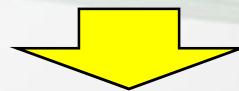
- Type command “H2OUVE -cvus [-count]”
- The argument -count used to specified the number of times.
- To detect that BIOS updates the variables during the POST time.
- There are 3 situations:
 - Variables are added or modified : list those variables
 - System do reclaim : show system do reclaim message
 - No change : pass the test

Variable Stress Test

Argument "-cvus" : **Check Variable Usage situation by Shutdown.**

```
fs0:\Insyde\Tool\H20UVE\Demo> H20UVE-Sx64.efi -cvus
```

```
Read BIOS data...
Start testing.....
1.
This is 1 times to run longrun
Variable usage condition check #0
NV used bytes: 0x2DF5
Elapsed time: 44784 (ms)
2013/01/01 00:23:08
System reboot !
```



```
Read BIOS data...
Start testing.....
Variable usage condition check #1
NV used bytes: 0x2E3D
Test Fail - variables are updated or added
MTC - {EB704011-1402-11D3-8E-77-00-A0-C9-69-72-3B}
```


Variable Stress Test

Argument “-cvrd” : **Check Variable Reclamation on Dxe.**

- Type command “H2OUVE -cvrd [-count]”
- The argument -count used to specified the number of times.
- This function is not supported in secure flash mode.
- Write a ‘dirty-byte’ to the variable storage remaining area.
- BIOS will do reclaim while detect there have any byte in remaining area is ‘dirty-byte’ (not 0xFF) during POST time.

```
fs0:\Insyde\Tool\H2OUVE\Demo> H2OUVE-Sx64.efi -cvrd
```

```
Read BIOS data...
Start testing.....
1.
This is 1 times to run longrun
Variable reclamation check
leftsize:C8B to write
Tool modify free space's data
Elapsed time: 30525 (ms)
2014/05/09 19:43:36
System reboot !
```

Variable Stress Test

Argument “-cvrs” : **Check Variable Reclamation on Smm.**

- Type command “H2OUVE -cvrs [-count]”
- The argument -count used to specified the number of times.
- This function is not supported in secure flash mode.
- Create variables until remaining size reaches the reclaim threshold and trigger reclamation event.

```
fs0:\Insyde\Tool\H2OUVE\Demo> H2OUVE-Sx64.efi -cvrs
```

```
Read BIOS data...
Start testing.....
1.
This is 1 times to run longrun
Variable reclamation check
leftsize:5B3 to write
Do recliam 500 times...start .
Do recliam 1 times.
Do recliam 2 times.
Do recliam 3 times.
Do recliam 4 times.
```

Variable Stress Test

Argument "-cvssf" : **Check Variable fill Setting (fill 0xFF).**

- Type command "H2OUE -cvssf [-count] [-bc]"
- The argument -count used to specified the number of times.
- The argument -bc used to specified the Partial Erase Size Base on Block Size = 0x1000. (Number * BlockSize)
- Fill 0xFF to NV Variable Area and reboot, the system should rebuild the NV Variable Area after reboot.
- This function is not supported in secure flash mode.

```
fs0:\Insyde\Tool\H2OUE\Demo> H2OUE-Sx64.efi -cvssf
```

```
Read BIOS data...
```

```
Start testing.....
```

```
1.
```

```
This is 1 times to run longrun
```

```
NVStorage rebuild testing (fill 0xFF)
```

```
Clean NVStorage's variable area already
```

```
Elapsed time: 33202 (ms)
```

```
2014/06/05 10:15:48
```

Variable Stress Test

Argument "-cvs0" : **Check Variable fill Setting (fill 0x00).**

- Type command "H2OUE -cvs0 [-count] [-bc]"
- The argument -count used to specified the number of times.
- The argument -bc used to specified the Partial Erase Size Base on Block Size = 0x1000. (Number * BlockSize)
- Fill 0x00 to NV Variable Area and reboot, the system should rebuild the NV Variable Area after reboot.
- This function is not supported in secure flash mode.

```
fs0:\Insyde\Tool\H2OUE\Demo> H2OUE-Sx64.efi -cvs0
```

```
Read BIOS data...
```

```
Start testing.....
```

```
1.
```

```
This is 1 times to run longrun
```

```
NVStorage rebuild testing (fill 0x0)
```

```
Clean NVStorage's variable area already
```

```
Elapsed time: 33295 (ms)
```

```
2014/06/05 10:19:29
```

Variable Stress Test

Argument "-cgnv" : **Check GetNextVariableName on RT.**

- Type command "H2OUE -cgnv"
- Verify Runtime GetNextVariableName function.
- Compare the variable between parsed from BIOS ROM and Runtime GetNextVariableName.

```
fs0:\Insyde\Tool\H2OUE\Demo> H2OUE-Sx64.efi -cgnv
```

```
Read BIOS data...
```

```
Start testing.....
```

```
Ready to reclaim
```

```
GetNextVariableName PASS
```

Variable Stress Test

Argument "-cssr" : **Check Synchronization between SMM and RT SetVariable.**

- Type command "H2OUE -cssr"
- Verify synchronization between SMM and RT SetVariable.
- Behavior
 - Write new variable in the sequence: SMM, RT, SMM, RT.
 - Then read variable data in the sequence: SMM, RT, SMM, RT and check if variable corrupted.

```
fs0:\Insyde\Tool\H2OUE\Demo> H2OUE-Sx64.efi -cssr
```

```
Start testing.....
SMM SetVariable
RT SetVariable
SMM SetVariable
RT SetVariable
SMM GetVariable
RT GetVariable
SMM GetVariable
RT GetVariable
SMM/RT SetVariable Sync Test PASS
```

Variable Stress Test

Argument "-cvsd" : **Check Variable is existing while changing variable State to Deleted_transition in Smm variable services.**

- Type command "H2OUVE -cvsd"
- Create a variable with state Added and change the variable state from Added to Deleted_Transition, then read the variable to ensure the variable is valid.

```
fs0:\Insyde\Tool\H2OUVE\Demo> H2OUVE-Sx64.efi -cvsd  
  
Read BIOS data...  
Start testing.....  
Testing item that check variable state in Smm Variable Service.  
Result:PASS!  
End testing.
```


Variable Stress Test

Argument "-sipi" : **Send IPI**.

- Type command "H2OUVE -sipi"
- Send IPI interrupt message to CPU and all except self processors.
- If processors cannot resume normally, the system may crush.

```
fs0:\Insyde\Tool\H2OUVE\Demo> H2OUVE-Sx64.efi -sipi
```

```
Start Send IPI testing.....  
Send SMI successfully!  
Send IPI testing end!
```

Variable Stress Test

Argument “-cvax” : **A group test.**

- Type command “H2OUVE -cvax”
- This test is a group test to test –cvhc, -cnvs, -dvfi, -vorv, -cvur, -cvus, -cvsf, -cvs0 one by one
- Before run this test, make sure system can be login automatically and run H2OUVE with administrator permission without input password

```
kanos@kanos-u1510-64:~$ sudo h2ouve-lx64 -cvax
```

```
* * * Check Variable Long-Run Test * * *  
Run Variable Long-Run Test Round [1/1]  
Test [cvhc] Function, Check Step [1/1]  
Checking Variable Header...
```



Graphical Mode

Graphical Mode

Introduce graphical mode

- Both Windows and Linux version support graphical mode.
 - Linux version supports Linux kernel 2.6.18 and later.
- Running H2OUE and `-g` argument will open graphical mode.
- When graphical mode initialized, it opens with empty window.
- Users can load runtime settings or image settings by clicking runtime & image buttons.
- When loading BIOS image, it loads all settings in that image.

- Main Frame

- Menu Bar

- Tool Bar

- Setup Editor

- String Editor

- Variable Editor

- Multi-Config Editor

- Setup

- Hotkey

- Logo

- SMBIOS



■ Main Frame

- Menu Bar
- Tool Bar
- Setup Editor
- String Editor
- Variable Editor
- Multi-Config Editor
 - Setup
 - Hotkey
 - Logo
 - SMBIOS

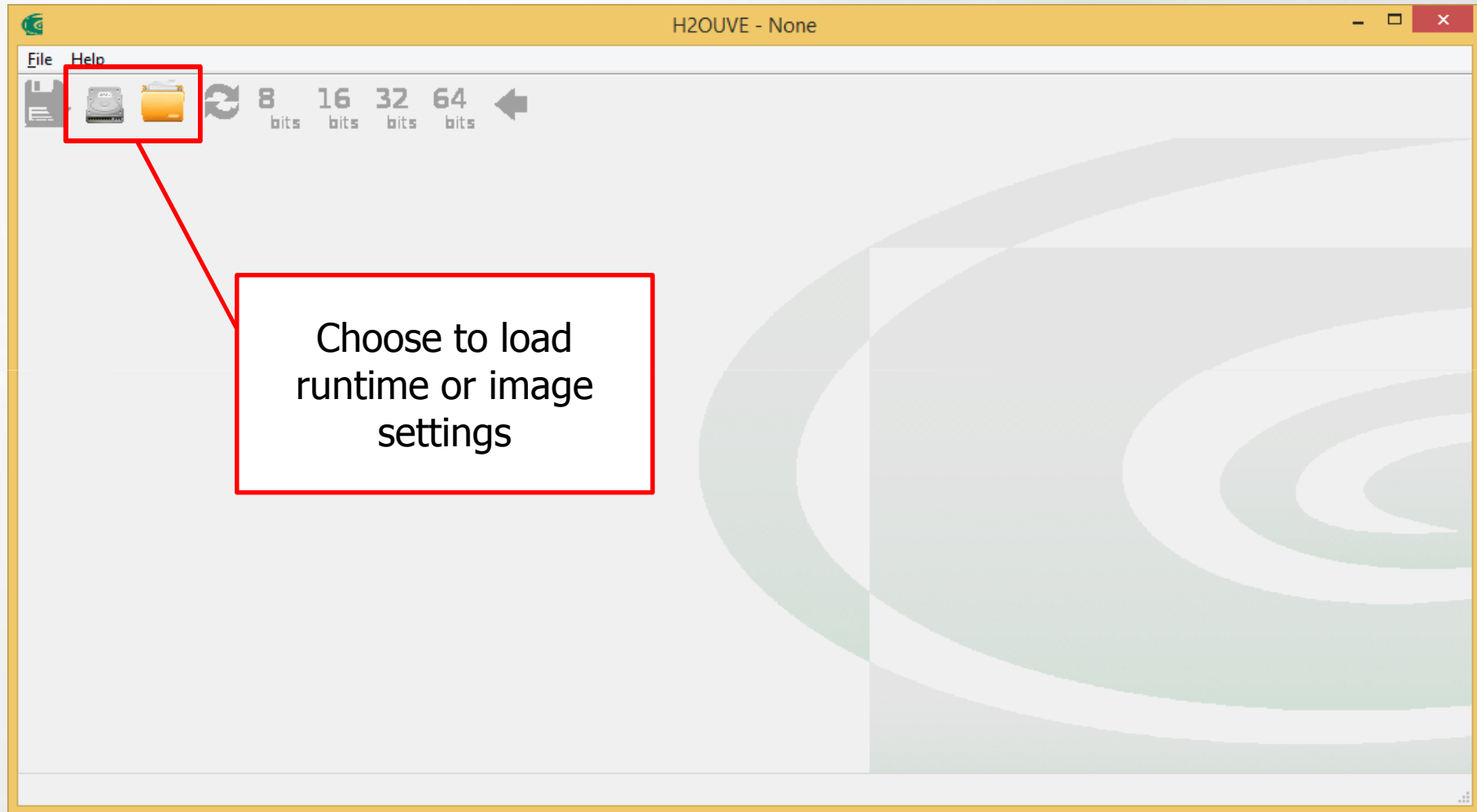
Main Frame

Editors of graphical H2OUVE

- There are 4 editors in H2OUVE
 - Setup editor: run setup web page to modify BIOS settings.
 - String editor: load setup strings and BIOS browser strings if any.
 - Variable editor: load all variables.
 - Multi-Config editor: load Multi-Config region and there exists some handlers for different type config data.

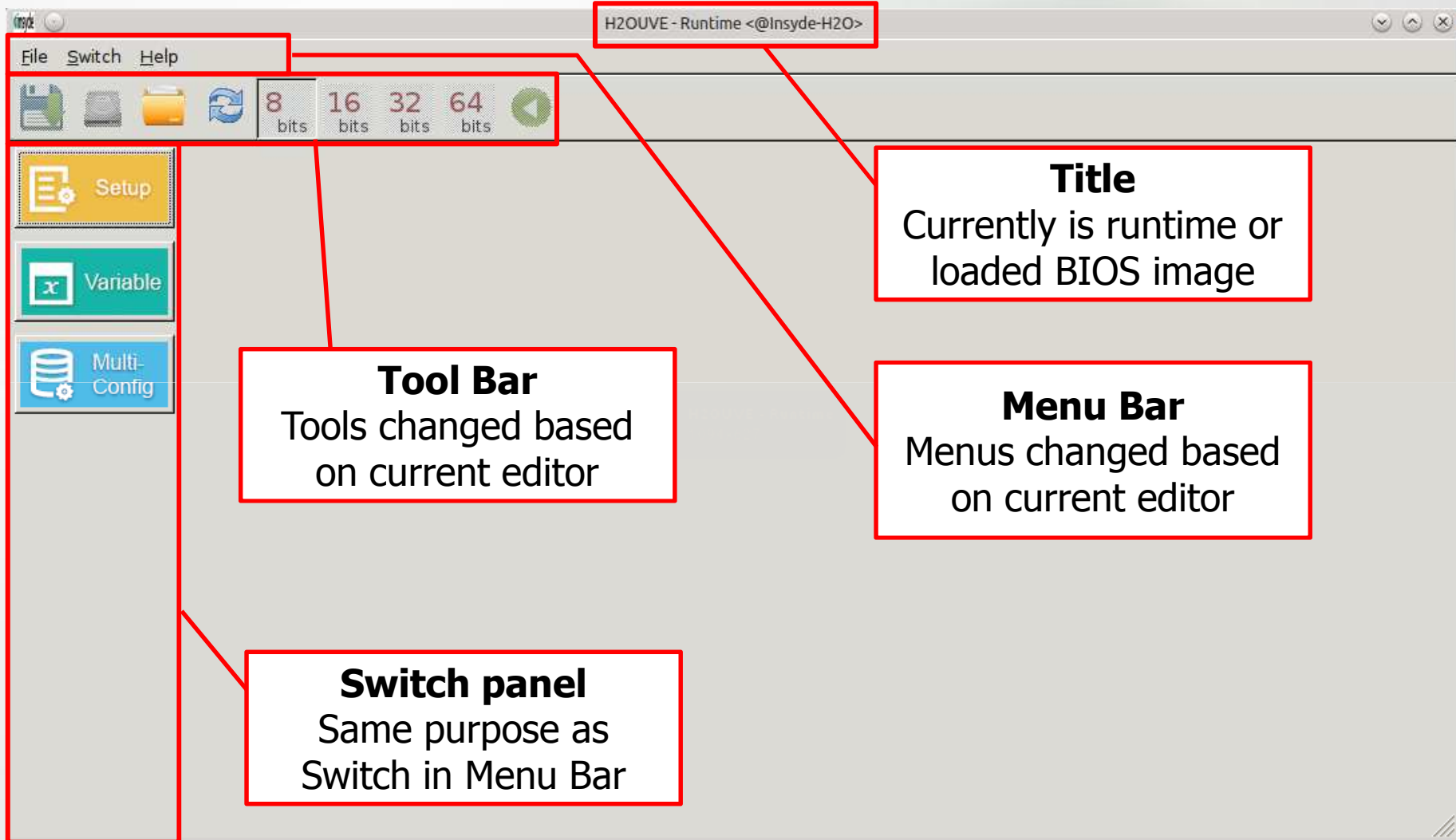
Main Frame

Initial H2OUVE opens nothing



Main Frame

Introduce the interface of graphical H2OUVE



- Main Frame

- Menu Bar

- Tool Bar

- Setup Editor

- String Editor

- Variable Editor

- Multi-Config Editor

- Setup

- Hotkey

- Logo

- SMBIOS

Menu Bar

File menu

Functions in file menu enabled by activated editor.



Switch runtime or load a BIOS image

- Save: Save settings to runtime or loaded BIOS image
- Save as: It's available when BIOS image loaded

Dump BIOS region data into file

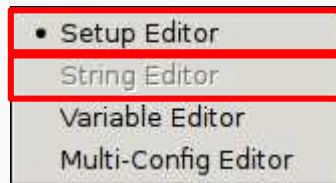
These 2 functions enabled when switch to some editor such like Setup Editor and Variable Editor

Exit H2OUVE

Menu Bar

Switch menu

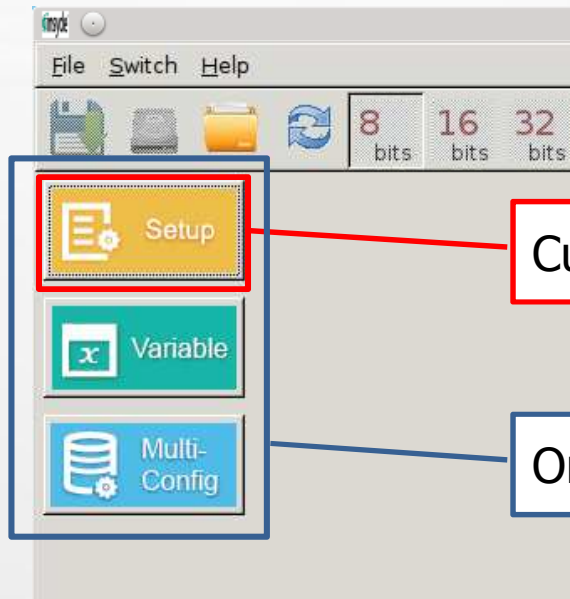
- It's a editor switch menu used to switch each editor.



Current editor with a dot at front

Disabled editor means it's not supported

- It's same as the left panel of content window.



Current editor with a dot line around it

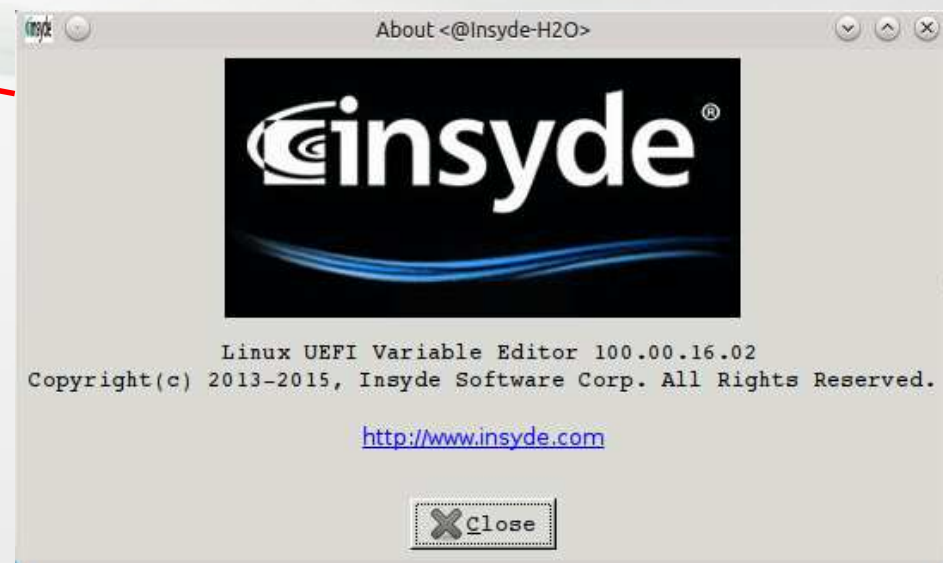
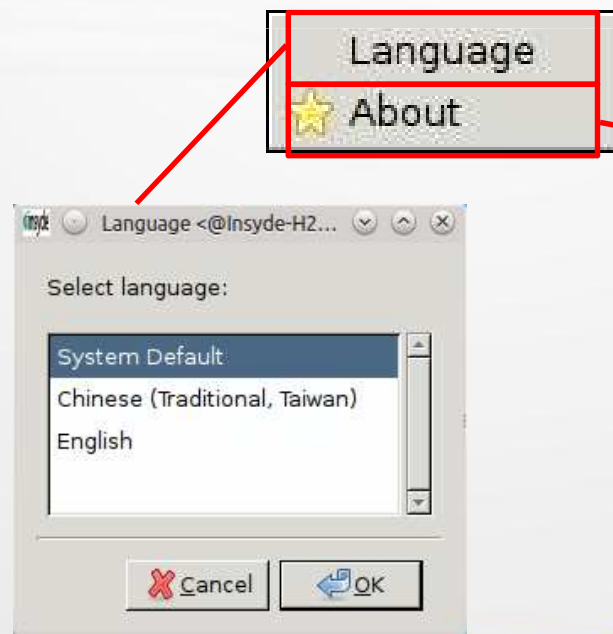
Only show supported editors

Menu Bar

Help menu

There are 2 functions under Help menu.

- Language
 - H2OUVE supports multi-language, but there are 2 language now.
 - There will be more language in the future.
- About
 - Show current version of H2OUVE.
 - Show copyright of H2OIVE.



- Main Frame

- Menu Bar

- Tool Bar

- Setup Editor

- String Editor

- Variable Editor

- Multi-Config Editor

- Setup

- Hotkey

- Logo

- SMBIOS

Tool Bar

Tool buttons changed by current mode and activated editor.

Save Button

Same as File>Save to save current settings

BIOS image Button

Same as File>Image to switch to load BIOS image

Display Bit Button

Display raw data in 8/16/32/64 bits and they are available when switch to Variable editor



Back to Previous Button

Used to go back to the up level page

Runtime Button

Same as File->Runtime to switch to runtime mode

Refresh Button

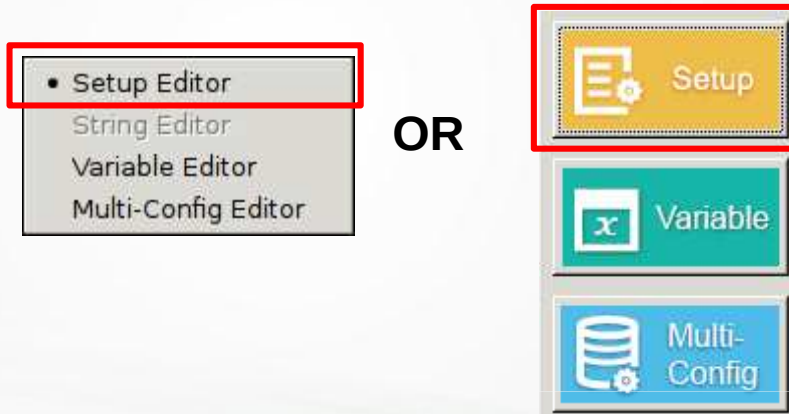
Drop all changes and load current settings

- 
- Main Frame
 - Menu Bar
 - Tool Bar
 - Setup Editor
 - String Editor
 - Variable Editor
 - Multi-Config Editor
 - Setup
 - Hotkey
 - Logo
 - SMBIOS

Setup editor

Modify setup settings

- Click on "Setup Editor" to switch to Setup editor

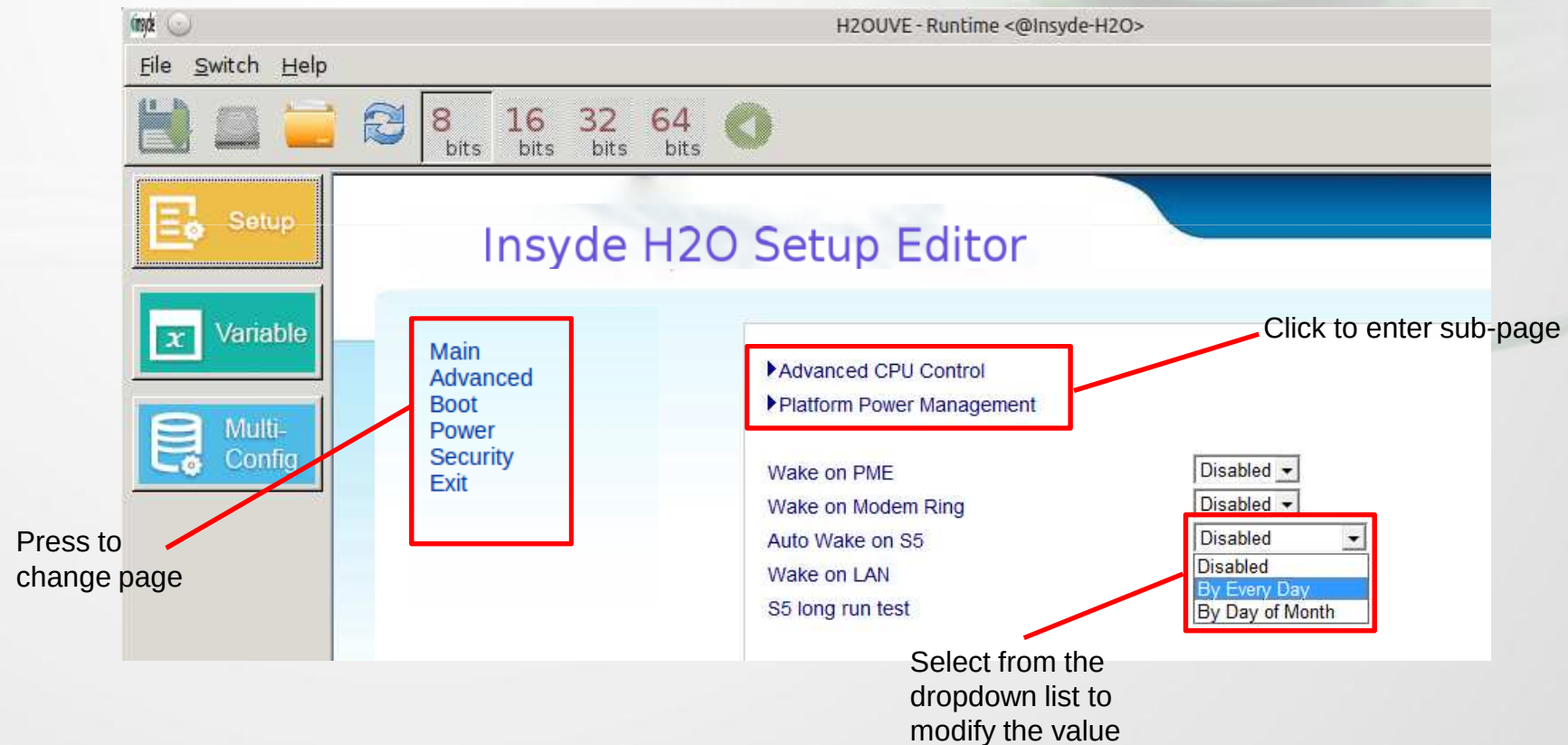


- In different mode, Setup Editor loads different current settings
 - Runtime mode: Load currently settings that system is running
 - Image mode: Load default settings of BIOS image
- Different save operation
 - Runtime mode: Save changes when leaving with "Save and Exit"
 - Image mode: Save changes when leaving with "Save and Exit" then click "Save button"

Setup editor

Introduce interface of setup browser

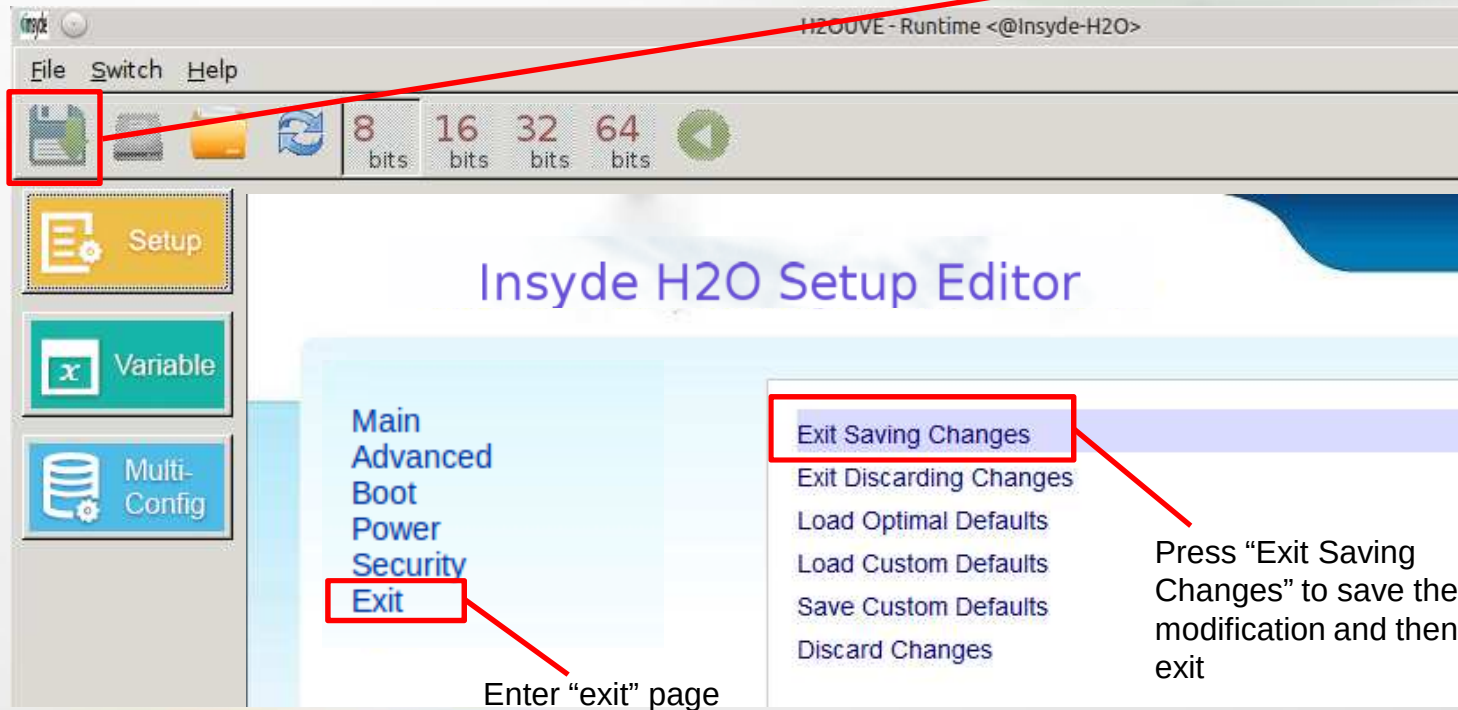
- The left panel of setup browser is main menu
- There are sub-page, oneof and numeric items



Setup editor

Save settings

- Enter "Exit" page to save the changes.
- In runtime mode, every changes effect when run the Exit->Save Changes
- In image mode, please remember to click the **Save button** to save chages to image or changes may lost when H2OUVE closed.



Setup editor

Different banner of setup page for different source

- The banner shows the type of setup page, there are 3 types now
 - Runtime Setup Editor
 - Default Setup Editor of one BIOS image
 - Default Setup Editor of Multi-Config

Runtime

Insyde H2O Setup Editor

Current User: Admin

Main
Advanced
Boot

System Date

[03/09/2015]

Image

Insyde H2O Setup Editor (Default)

Current User: Admin

Main
Advanced
Boot

System Date

[03/09/2015]

Multi-Config

Multi-Config Setup Editor

Current User: Admin

ConfigId: 0x1 Name:

Main
Advanced
Boot

System Date

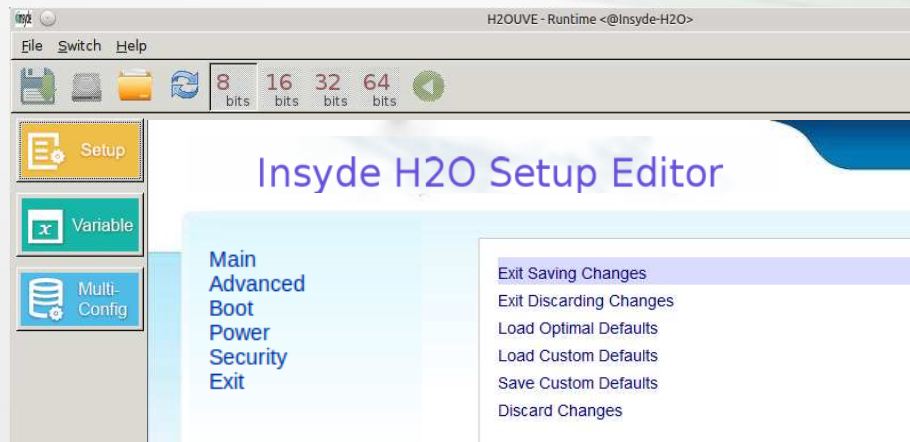
[03/09/2015]

Setup editor

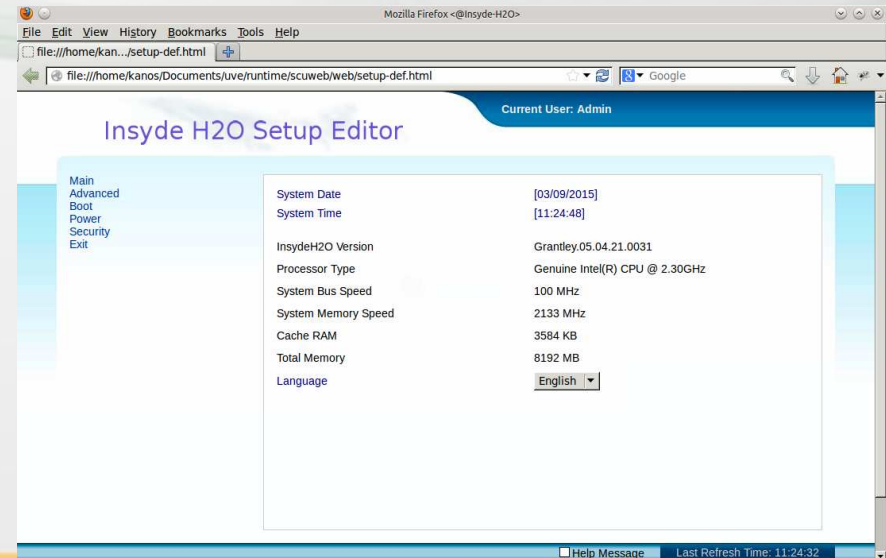
Difference between Windows and Linux

- It's different operation when switch to Setup Editor
 - Windows: Show setup content in right panel
 - Linux: Pop-up a default browser to show setup content
 - The setup browser must be closed to continue other operations like switch to another editor or save all changes

Windows



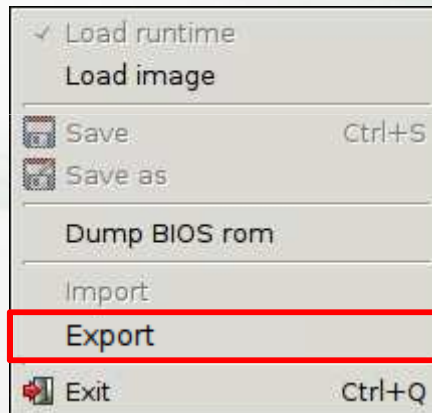
Linux



Setup editor

Export function

- There is **Export** enabled under **File menu**
- Export setup settings into a setup formatted file
- If it's Linux version, please close the setup browser then run this function



- Main Frame

- Menu Bar
- Tool Bar

- Setup Editor

- String Editor

- Variable Editor

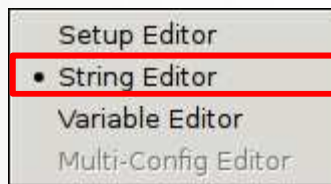
- Multi-Config Editor

- Setup
- Hotkey
- Logo
- SMBIOS

String editor

An editor used to modify strings show in BIOS

- String Editor supports in **image mode only**
- Press "String Editor" to start variable editor



OR



String editor

Introduce interface of String Editor

- Please remember that there is no String Editor when it's runtime mode, it shows when it's image mode.

The screenshot shows the String Editor interface with the following annotations:

- Tab to switch to string list:** Points to the 'String' tab in the left sidebar.
- Select string packages, it shows remain chars between ():** Points to the 'Package' dropdown menu.
- Select language:** Points to the 'Lang' dropdown menu.
- Remaining space:** Points to the 'Remained Chars: 0' label.
- String list:** Points to the list of string packages in the main table.
- Adjust size of column by moving this edge line:** Points to the vertical line between the 'Prompt' and 'Description' columns.

	Prompt	
1	Commit Changes and Exit	
2	Discard Changes and Exit	
3	Load Option Reconnect	
4	Boot Options	Modify system boot options
5	Driver Options	Modify boot driver options
6	Console Options	Modify system console options
7	Boot From File	Boot system from a file or device
8	Set Boot Next Value	Modify next boot behavior
9	Set Time Out Value	Modify automatic boot time-out value
10	Reset System	
11	Go Back To Main Page	
12	Add Boot Option	Add EFI Application or Removable Fs as Boot Option
13	Delete Boot Option	Will be valid immediately
14	Change Boot Order	
15	Add Driver Option	Add .EFI Driver as Driver Option
16	Delete Driver Option	Will be valid on next boot

String editor

Modify strings

- No matter in runtime mode or image mode, it needs to click "Save button" to save changes

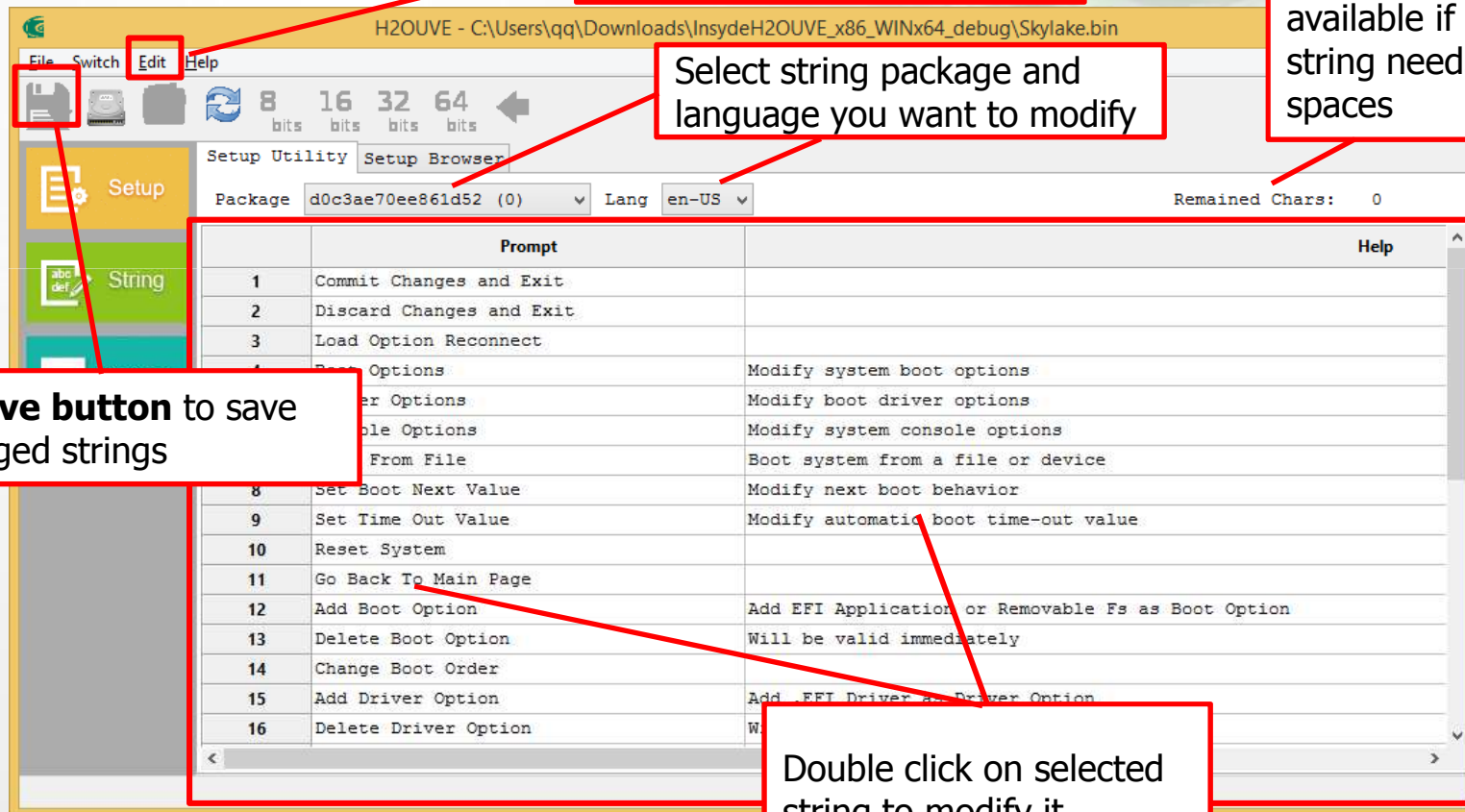
A new Edit menu used to find specific string

Make sure that remaining space is available if new string needs more spaces

Select string package and language you want to modify

Click **Save button** to save all changed strings

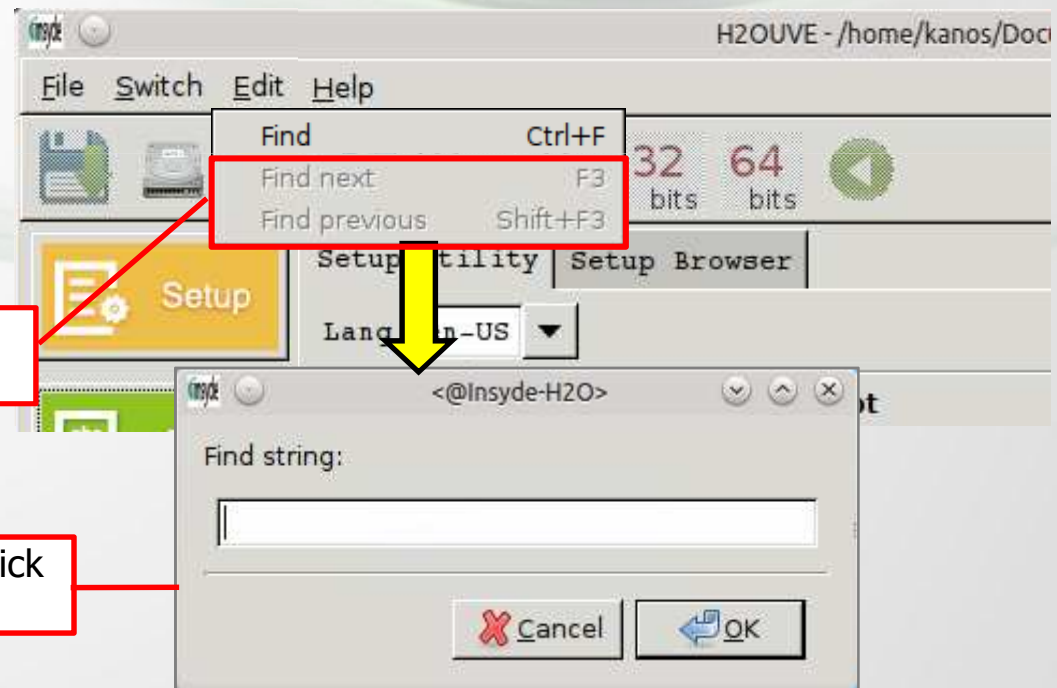
Double click on selected string to modify it



String editor

Find specific string

- Find string by click functions under Edit menu
- Find string by hotkeys
 - Find: Ctrl + F
 - Find next: F3
 - Find previous: Shift+F3



Those functions enabled after Find used

Pop-up this dialog when click Find or hotkey Ctrl+F

- Main Frame

- Menu Bar
- Tool Bar

- Setup Editor

- String Editor

- Variable Editor

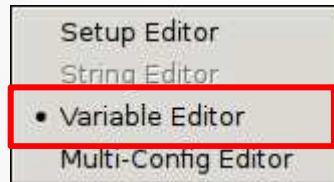
- Multi-Config Editor

- Setup
- Hotkey
- Logo
- SMBIOS

Variable Editor

To edit variables of current platform or a ROM image

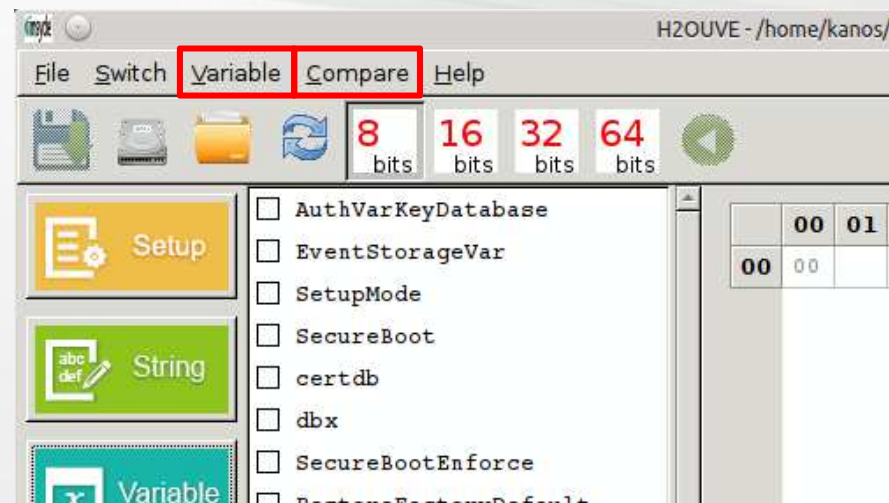
- Press "Variable Editor" to start variable editor



OR



- There are new menus named [Variable](#) and [Compare](#)



Variable Editor

Introduce interface of Variable Editor

The screenshot shows the Variable Editor interface with several annotations:

- Change the width of each column to display**: Points to the column width controls (8, 16, 32, 64 bits) at the top.
- Variable List**: Points to the list of variables on the left side.
- Raw Data**: Points to the header of the data table.
- Raw Data displayed in ASCII**: Points to the ASCII view of the data.
- This is adjustable for the 2 table**: Points to a vertical slider control between the two data tables.
- Click right button here to pup-up a [context menu](#)**: Points to a right-click button on the variable list.

The interface includes a menu bar (File, Switch, Variable, Compare, Help), a toolbar with icons for file operations and bit widths, a left sidebar with category buttons (Setup, String, Variable, Multi-Config), a variable list, and a main area with two data tables.

00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
00	00															0	.														

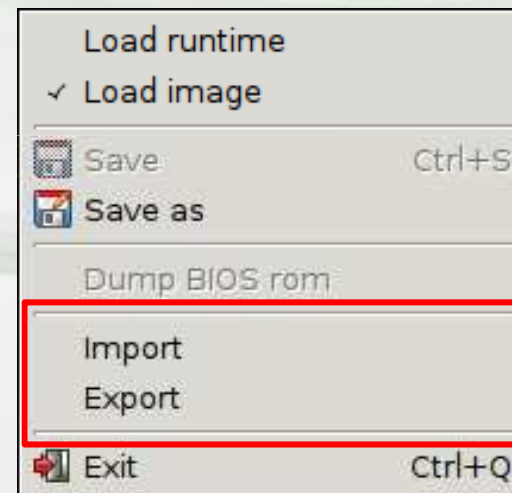
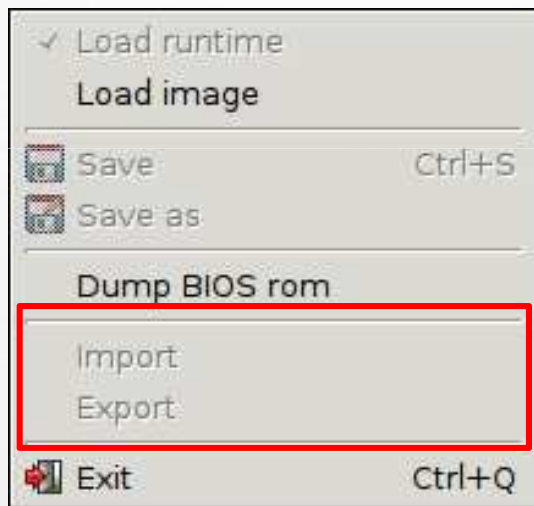
Context menu options:

- Add
- Import
- Select
- Show By
- List items
- Data Format

Variable Editor

Menu "File"

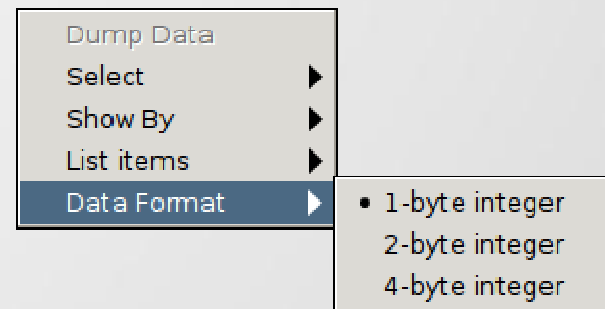
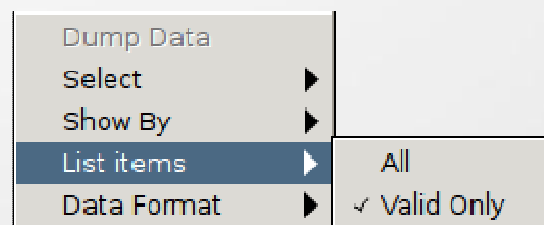
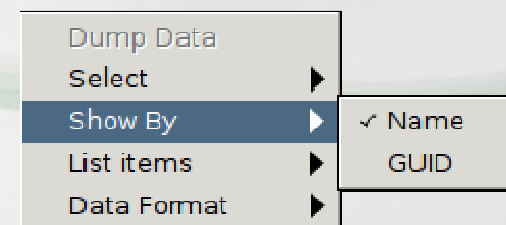
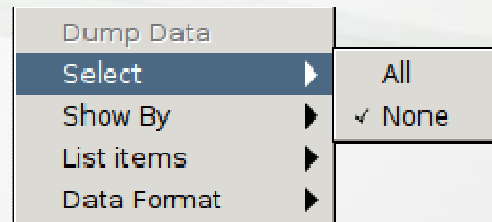
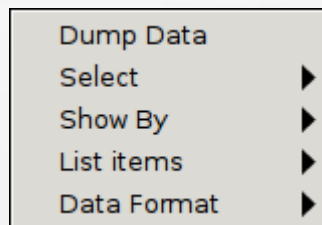
- The Import & Export function enabled.
 - Import : Load variable from a file
 - Export : Write variable to a file



Variable Editor

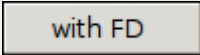
Menu "Variable"

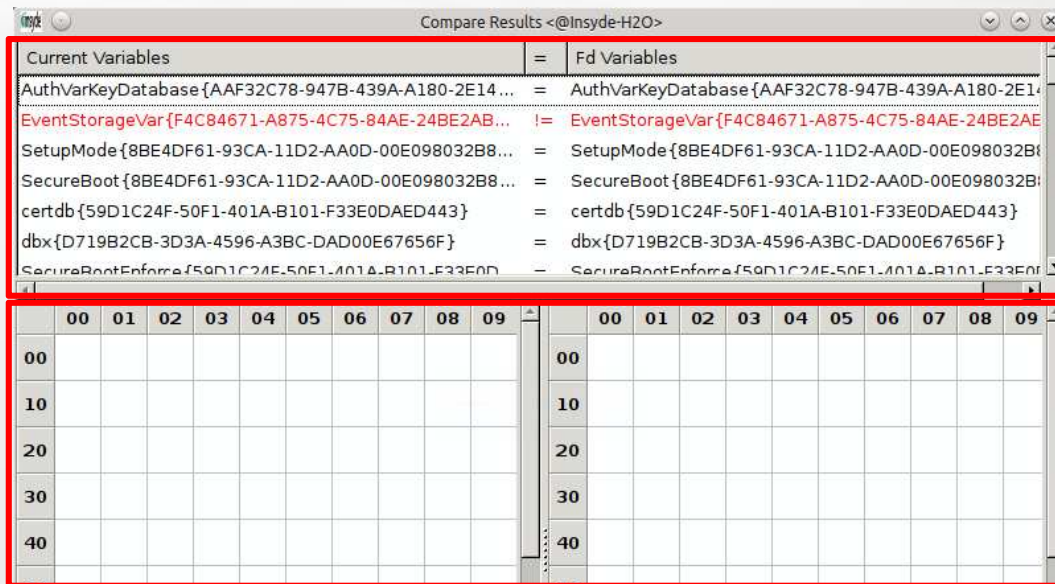
- Dump Data: Dump current variable data into variable file
- Select: Select all variable or none
- Show By: Display Name or GUID in the variable list
- List items: List all variables or valid variable only
- Data Format: Display raw data in selected width



Variable Editor

Menu "Compare"

- Under menu Compare->with FD 
- Compare all current variables with selected BIOS image file called FD.
- The different variable show in **RED** color in list. Current variables on left side and Fd variables on right side.
- Double click variable to show their raw data



Compare Results -> @Insyde-H2O>

Current Variables	=	Fd Variables
AuthVarKeyDatabase {AAF32C78-947B-439A-A180-2E14 ...}	=	AuthVarKeyDatabase {AAF32C78-947B-439A-A180-2E14 ...}
EventStorageVar {F4C84671-A875-4C75-84AE-24BE2AB ...}	!=	EventStorageVar {F4C84671-A875-4C75-84AE-24BE2AE ...}
SetupMode {8BE4DF61-93CA-11D2-AA0D-00E098032B8 ...}	=	SetupMode {8BE4DF61-93CA-11D2-AA0D-00E098032B8 ...}
SecureBoot {8BE4DF61-93CA-11D2-AA0D-00E098032B8 ...}	=	SecureBoot {8BE4DF61-93CA-11D2-AA0D-00E098032B8 ...}
certdb {59D1C24F-50F1-401A-B101-F33E0DAED443}	=	certdb {59D1C24F-50F1-401A-B101-F33E0DAED443}
dbx {D719B2CB-3D3A-4596-A3BC-DAD00E67656F}	=	dbx {D719B2CB-3D3A-4596-A3BC-DAD00E67656F}
SecureBootEnforce {59D1C24F-50F1-401A-B101-F33E0DAED443}	=	SecureBootEnforce {59D1C24F-50F1-401A-B101-F33E0DAED443}

	00	01	02	03	04	05	06	07	08	09
00										
10										
20										
30										
40										

	00	01	02	03	04	05	06	07	08	09
00										
10										
20										
30										
40										

Variable List

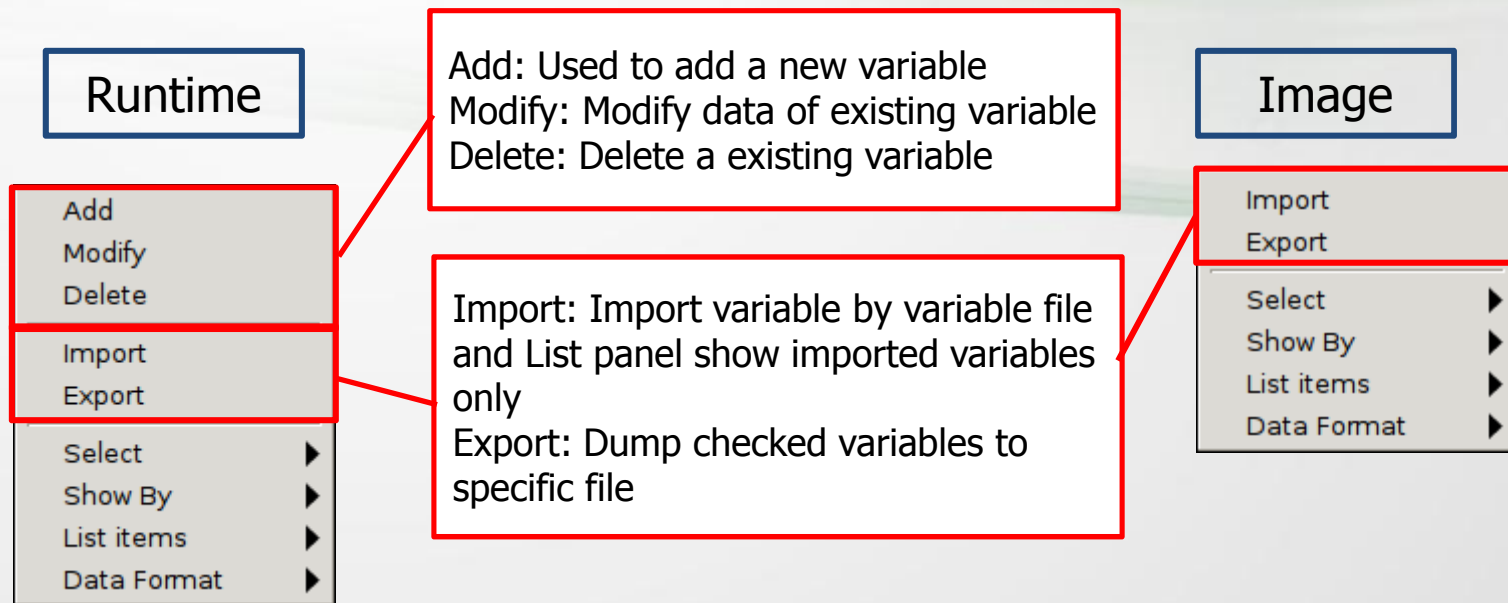
RED color means
different variable

Raw Data

Variable Editor

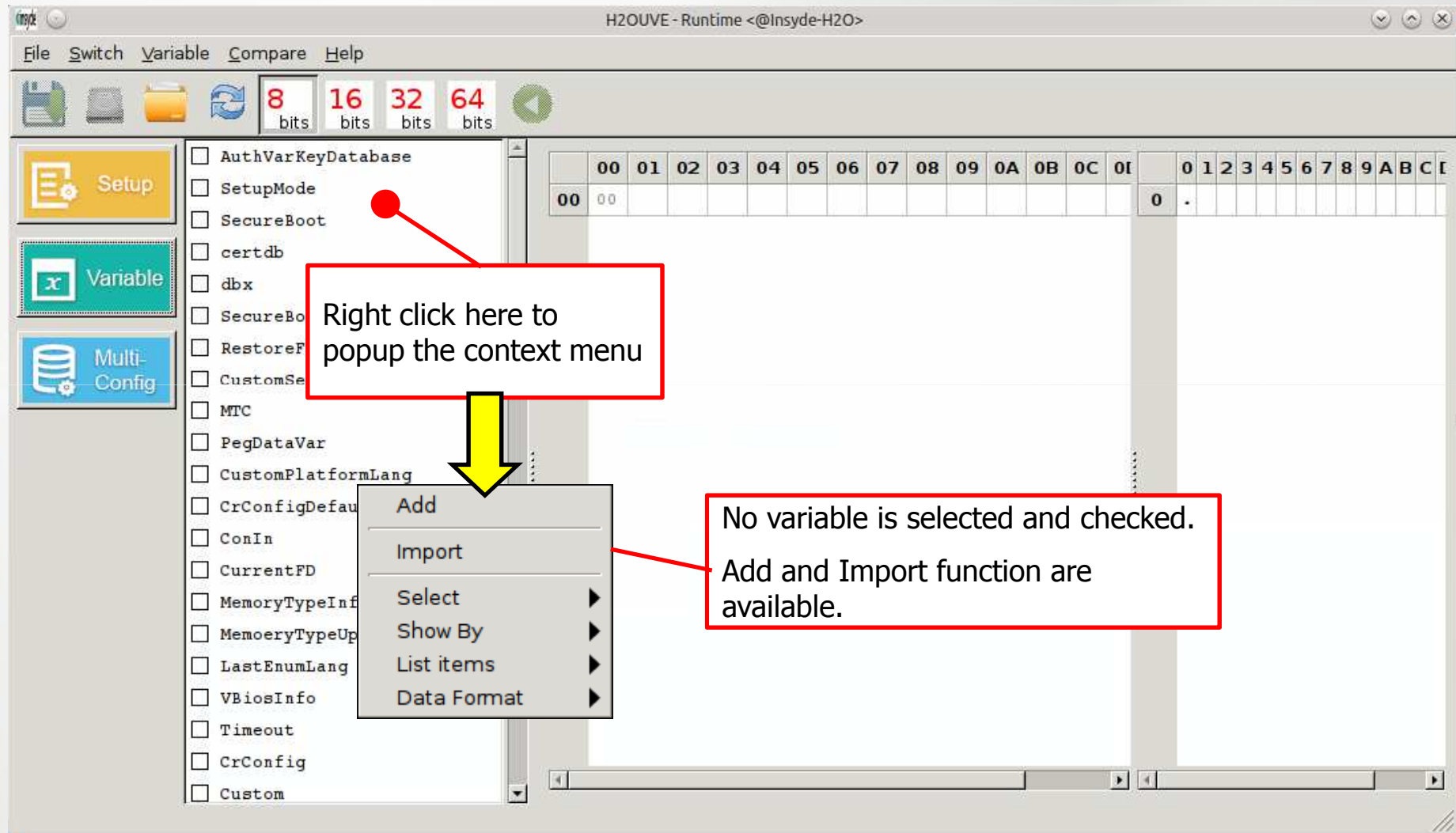
Context menu of variable list panel

- There are a little different for context menu shows in runtime and image mode.
- Add/Modify/Delete functions exists in runtime mode only.
- Other functions are all the same in either runtime and image mode.



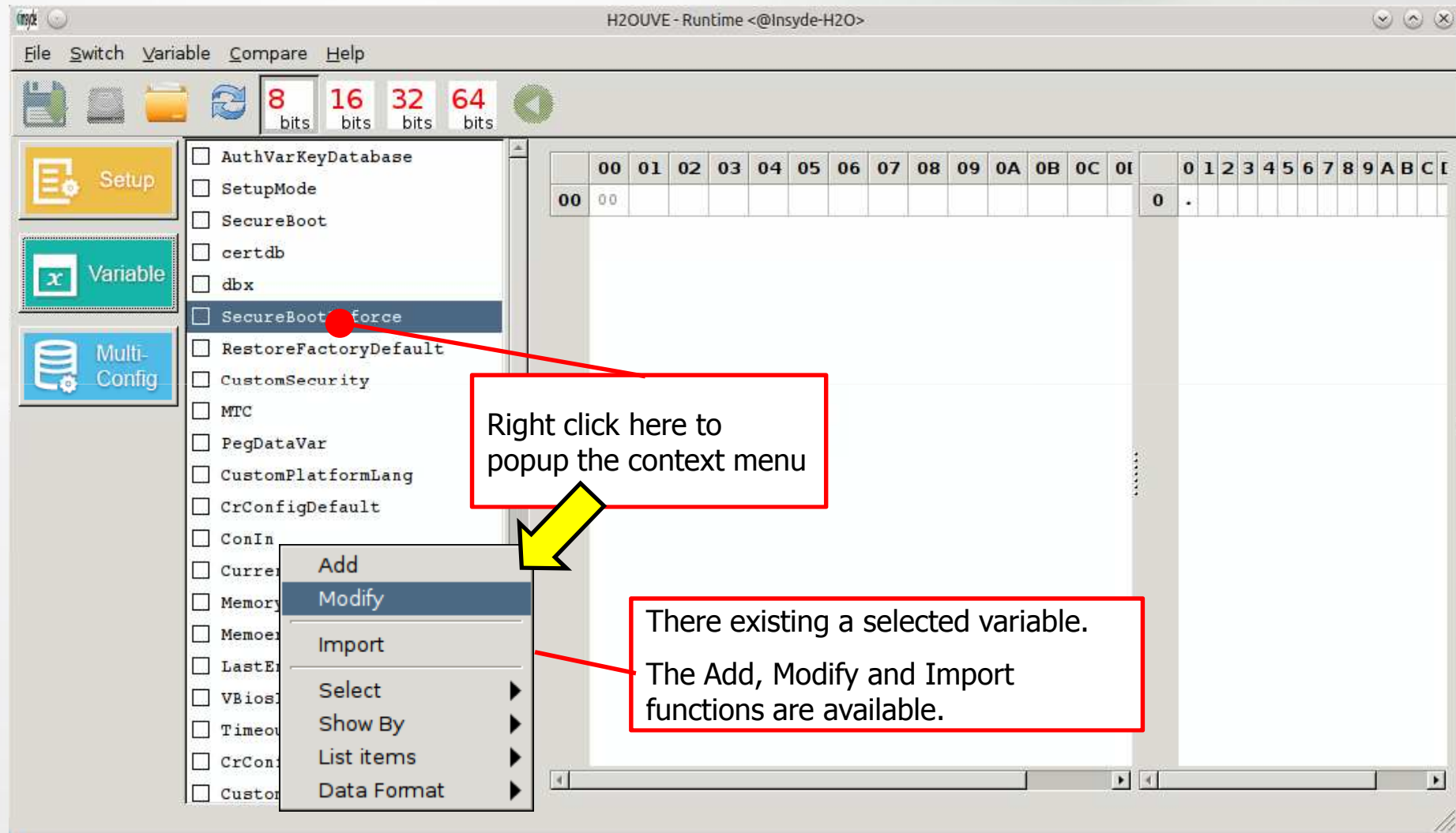
Variable Editor

Click right button without selected or checked items in runtime mode



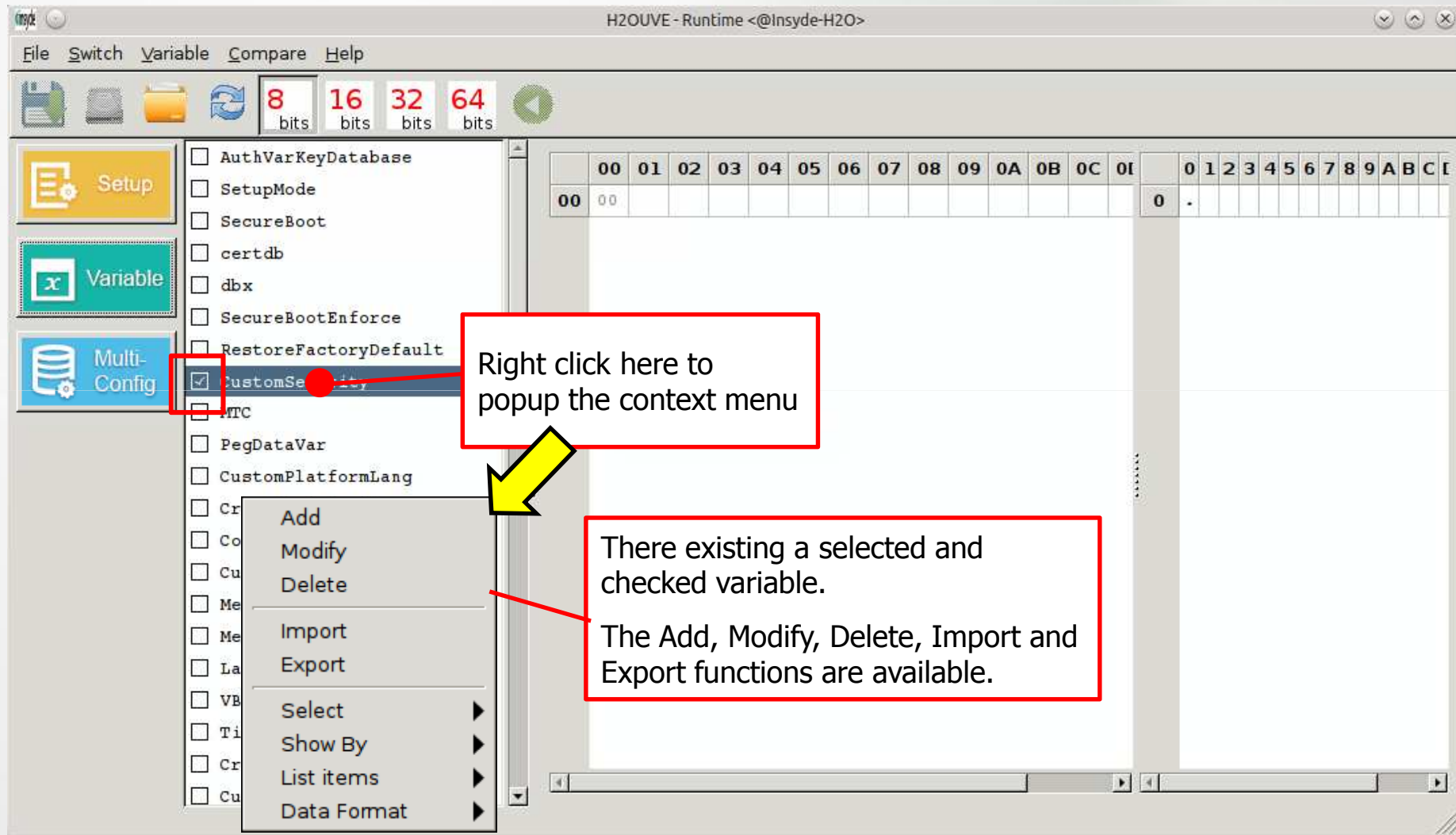
Variable Editor

Click right mouse button with selected item in runtime mode



Variable Editor

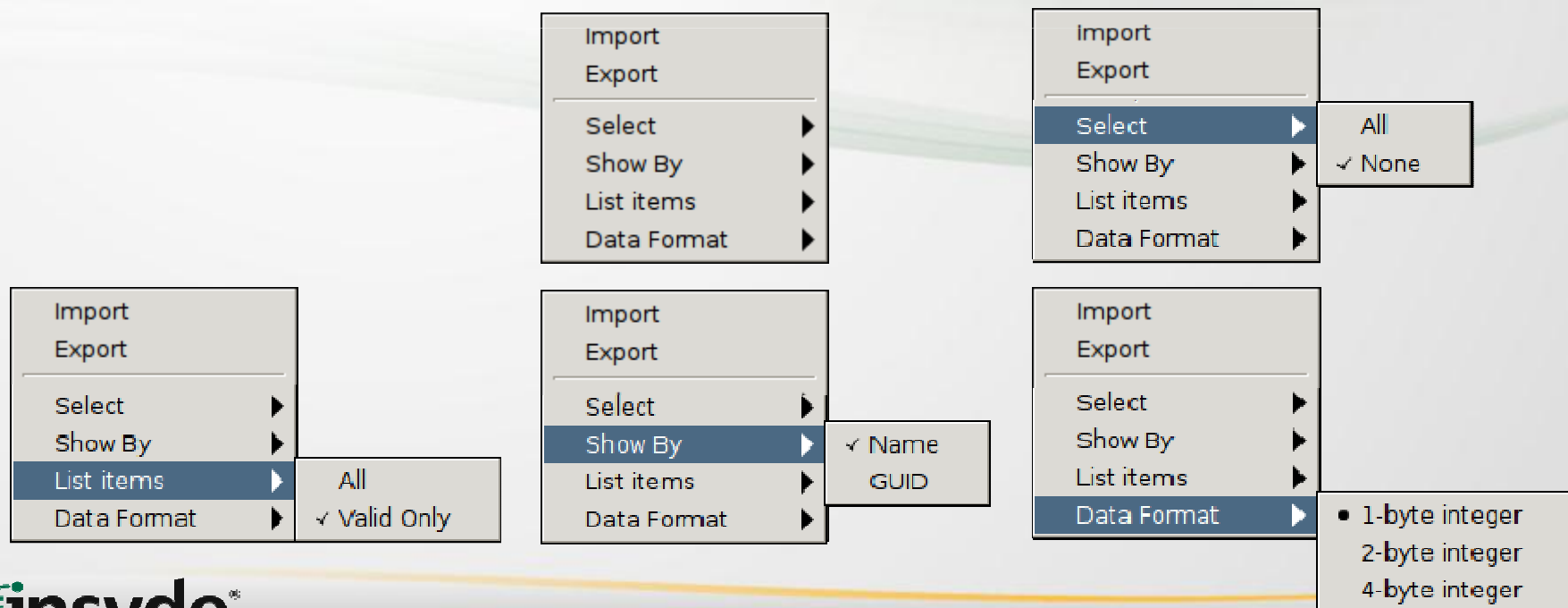
Click right mouse button with selected and checked item in runtime mode



Variable Editor

Common functions in context menu

- The common functions in context menu pop-up on the variable list has same functions include Import and Export under Variable menu of menubar
- Take a look on [Variable menu](#)
- Take a look on [Import & Export](#)



Variable Editor

Modify raw data of variable

- No matter in runtime mode or image mode, it needs to click "Save button" to save changes

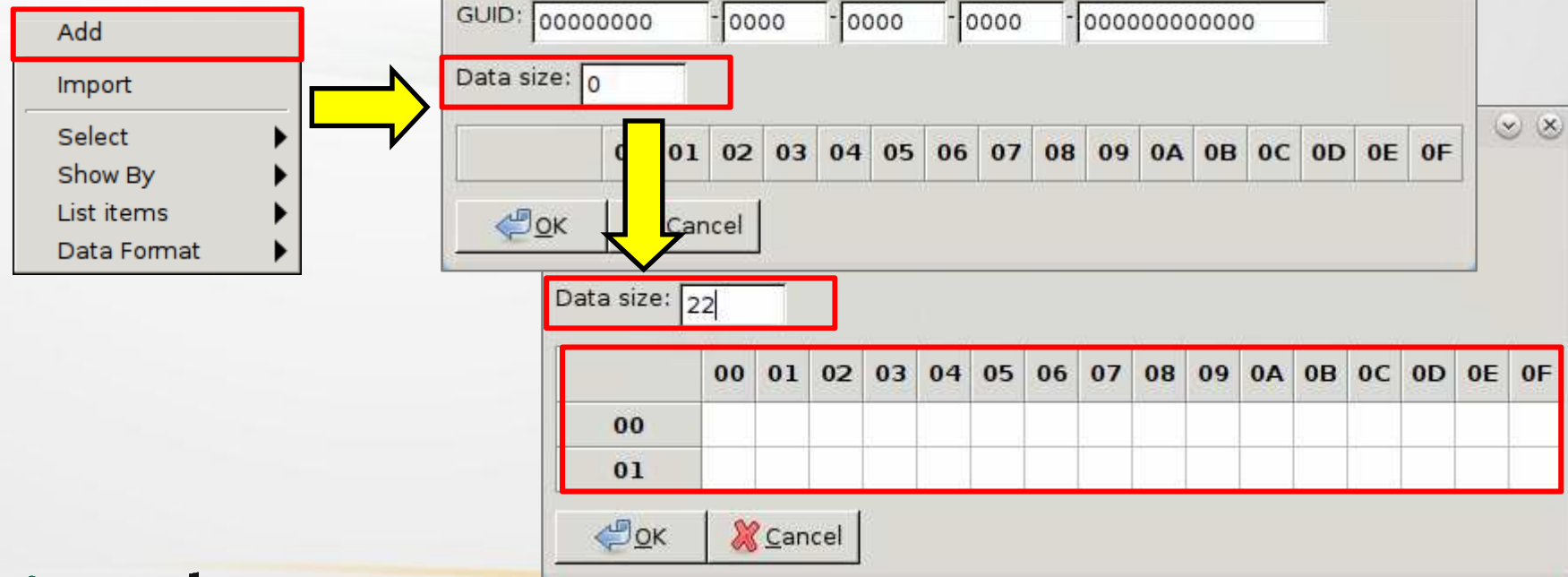
The screenshot shows the Variable Editor application window. The title bar reads "H2OUVE -". The menu bar includes "File", "Switch", "Variable", "Compare", and "Help". The toolbar contains icons for saving, opening, and other functions. A list of variables is shown on the left, including "AuthVarKeyDatabase", "SetupMode", "SecureBootEnforce", "RestoreFactoryDefault", "CustomSecurity", "MTC", "PegDataVar", "CustomPlatformLang", "CrConfigDefault", "ConIn", "CurrentFD", "MemoryTypeInfo", "MemoryTypeUpdateInfo", "LastEnumLang", "VBiosInfo", "Timeout", "CrConfig", and "Custom". The right panel displays raw data in hexadecimal and ASCII formats. Annotations with red boxes and arrows point to specific features:

- A red box around the "Save" icon in the toolbar with the text: "Click **Save button** to save all variables".
- A red box around the variable list with the text: "Double click variable item to display its raw data in the right panel".
- A red box around the ASCII column header with the text: "Show ASCII code here".
- A red box around the main data area with the text: "Modify raw data here".

Variable Editor

Add variable in runtime mode

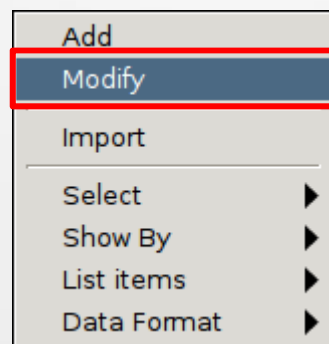
- Click mouse right button on list to show context menu
- Choose *Add* function to pop-up a Variable dialog
- Fill Name, GUID and change data size to add raw data



Variable Editor

Modify variable in runtime mode

- Select variable listed in List Panel
- Click mouse right button selected variable to show context menu
- Choose *Modify* function to pop-up a Variable dialog with existing name, GUID, data size and raw data
- The Name and GUID fields are read-only
- Users can only modify the data size and raw data



The Variable dialog box is titled 'Variable <@Insyde-H2O>'. It contains the following fields:

- Name: PegDataVar (read-only)
- GUID: 28CF6D27-0694-45D0-A06A-15266F1E8346 (read-only)
- Data size: 32 (modifiable)
- Raw data table (modifiable)

At the bottom are OK and Cancel buttons.

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
1	FF	FF	FF	FF	00	00	00	00	00	00	00	00	00	00	00	00
2	00	00	00	00	00	00	00	00	00	00	00	00	01	FF	00	00

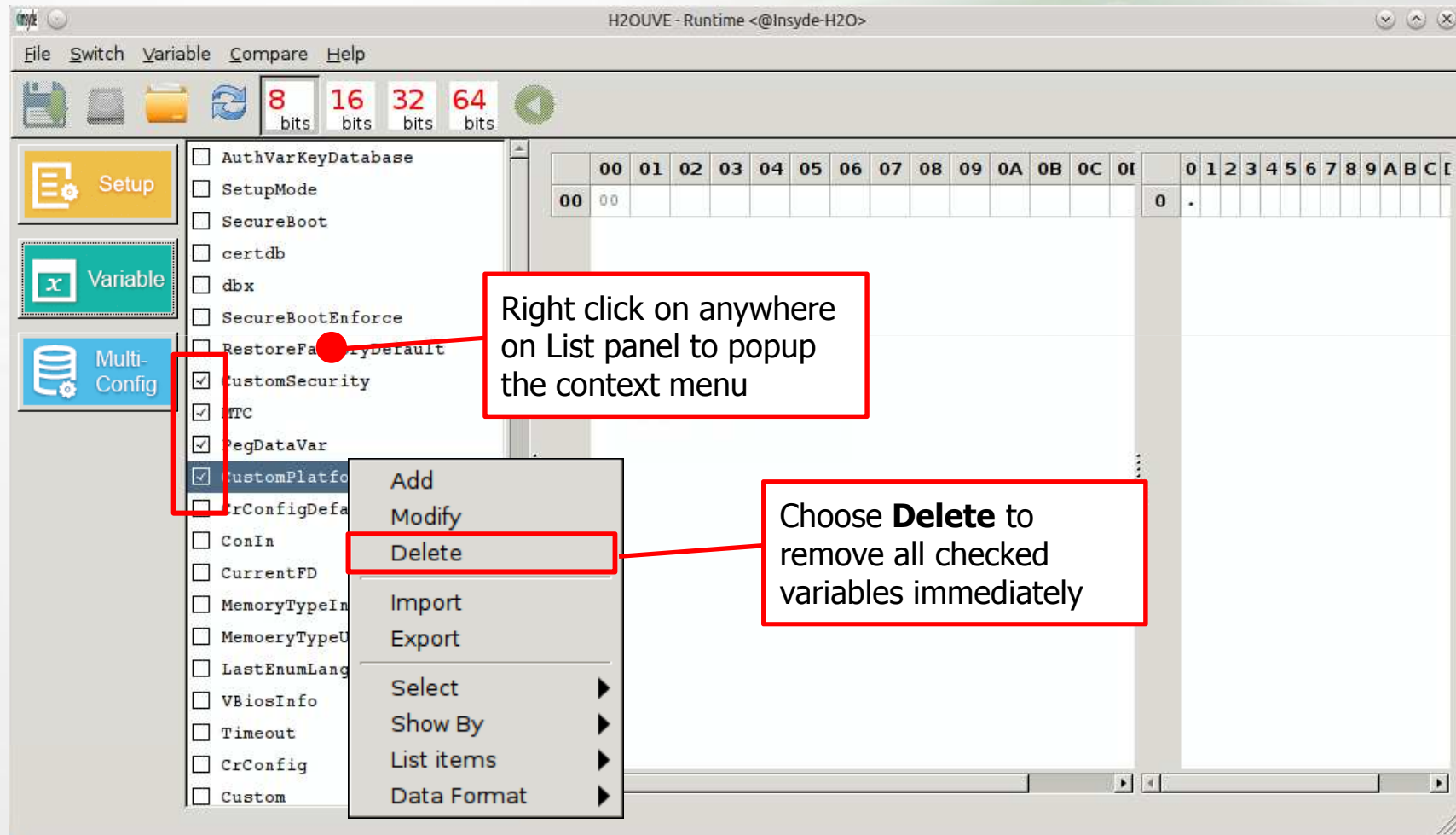
Read-Only

Modifiable

Variable Editor

Delete variables in runtime mode

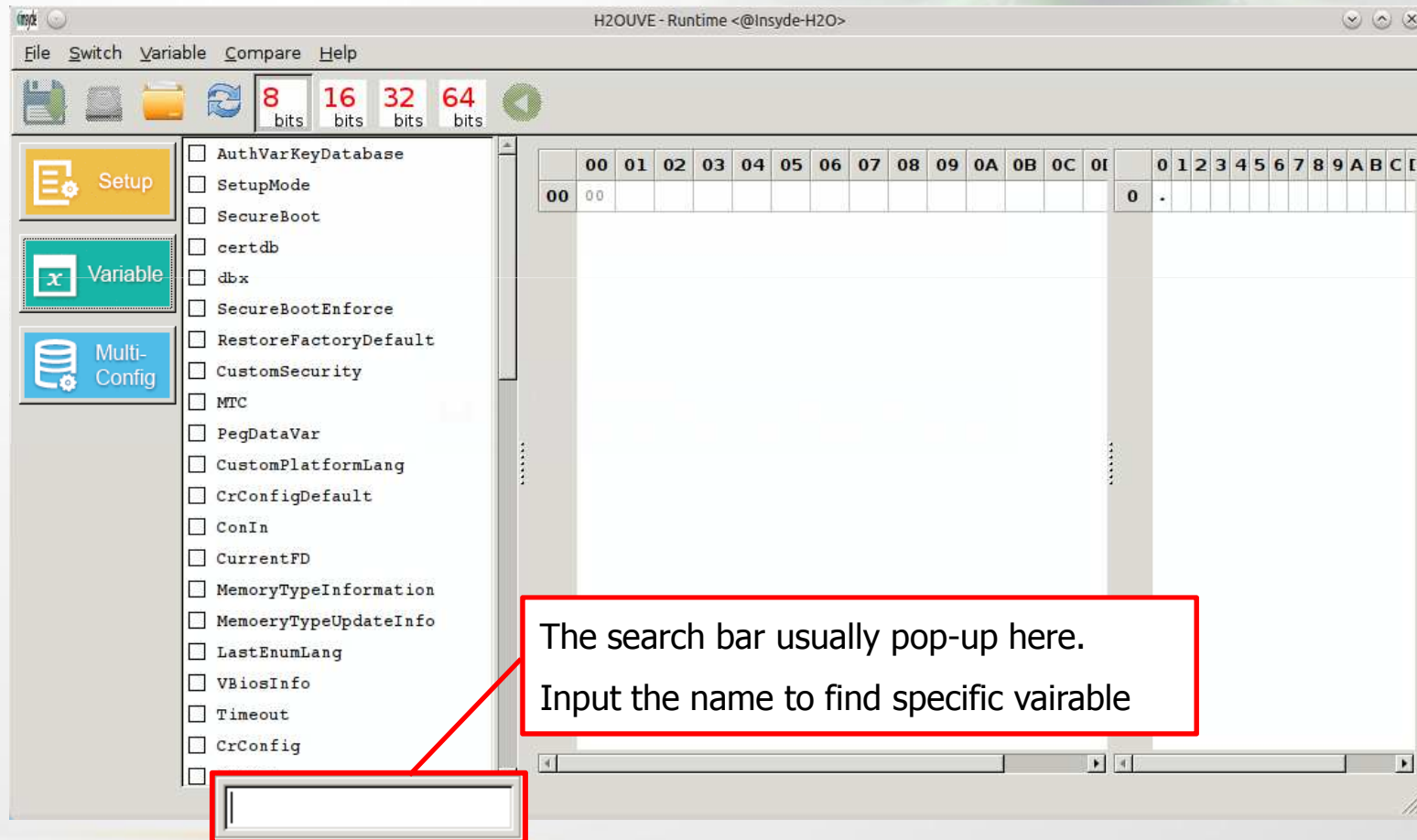
- Checked some variables which are going to be deleted



Variable Editor

Find specific variable

- Click anywhere on List Panel
- Use hotkey, ctrl+f to find specific variable



- Main Frame

- Menu Bar
- Tool Bar

- Setup Editor

- String Editor

- Variable Editor

- Multi-Config Editor

- Setup
- Hotkey
- Logo
- SMBIOS

Multi-Config Editor

Important information

- **This feature is not available after 100.00.16.06.**
- **Kernel 5.1 doesn't support Multi-Config.**

Multi-Config Editor

Introduce interface of Multi-Config Editor

Menu bar

H2OUVE - /home/kanos/Documents/uve/thisga.fd <@Insyde-H2O>

File Switch Help

8 bits 16 bits 32 bits 64 bits

Indicate runtime mode or BIOS image mode

Tool bar: quick functions in Menu bar

No.	Name	Type	Config ID	Active	Edit	Delete
0h		Setup	0h	☐	Edit	Delete
1h			0h	☐	Edit	Delete
2h			5h	☐	Edit	Delete
3h			6h	☐	Edit	Delete
4h		Setup	9h	☐	Edit	Delete

Editors: Responsible for all kinds of different config data in Multi-Config

Total number of Config Data: 5

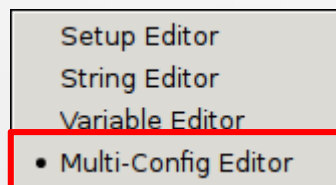
Load default setup from: BIOS

Remaining memory size (bytes): 0x3d92

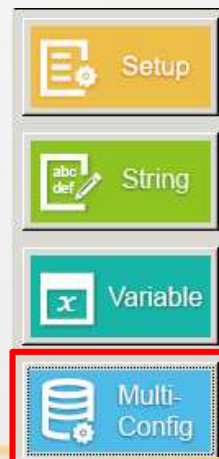
Multi-Config Editor

To edit settings in Multi-Config region

- There are 4 types data supported by Multi-Config Editor, but only 2 are ready
 - Setup type: Default setup data
 - Hotkey type: Hot key settings in post menu
 - SMBIOS type: Not ready
 - Logo type: Not ready
- Press "Multi-Config Editor" to start Mutli-Config Editor



OR



Multi-Config Editor

Introduce interface of Multi-Config Editor

Setting List

No checkbox no support
Checked: Enabled
Unchecked: Disabled

No.	Name	Type	Config ID	Valid	Mode	Post	Active	Edit	Delete
0h		Setup	0h	Valid	R/W	☐	☐	Edit	Delete
1h		HotKey	0h	Valid	R/W		☐	Edit	Delete
2h		Setup	5h	Valid	R/W	☐	☐	Edit	Delete
3h		Setup	6h	Invalid	RO			Edit	Delete
		Setup	9h	Invalid	R/W			Edit	Delete

Click right button here on item to add/delete/modify/duplicate

Click right button here on empty space to add item

All editable

Click once or double click on item to pop up more detail

Click once to delete item

Information of Multi-Config region

Total number of Config Data: 5
Load default setup from: BIOS
Remaining memory size (bytes): 0x3d92

Multi-Config Editor

Describe how to add one setting in Multi-Config region

1. Click right button on item list or empty area of item list area
2. Choose "Add"
3. Select the type to add

The screenshot shows the Multi-Config Editor interface. The main window has a menu bar (File, Switch, Help) and a toolbar with icons for file operations and bit widths (8, 16, 32, 64 bits). Below the toolbar is a table with columns: No., Name, Type, Config ID, Valid, and Mode. The table contains five rows of data. A context menu is open over the table, showing options: Add, Dump, Edit, and Delete. A red box highlights the 'Add' option. A yellow arrow points from the 'Add' option to the 'Multi-Config Type' dialog. The dialog has a title bar 'Multi-Config Type <...>' and a list of types: Setup, Hotkey, Smbios, and Logo. The 'Setup' type is selected. The dialog has 'Cancel' and 'OK' buttons.

No.	Name	Type	Config ID	Valid	Mode
0h		Setup	0h	Valid	R/W
1h		HotKey	0h	Valid	R/W
2h		Setup	5h	Valid	R/W
3h		Setup	6h	Invalid	RO
4h			9h	Invalid	R/W

Click right button here on item list to show **Add** choice

Click right button on empty area to show **Add** choice

Multi-Config Type <...>

Choose type:

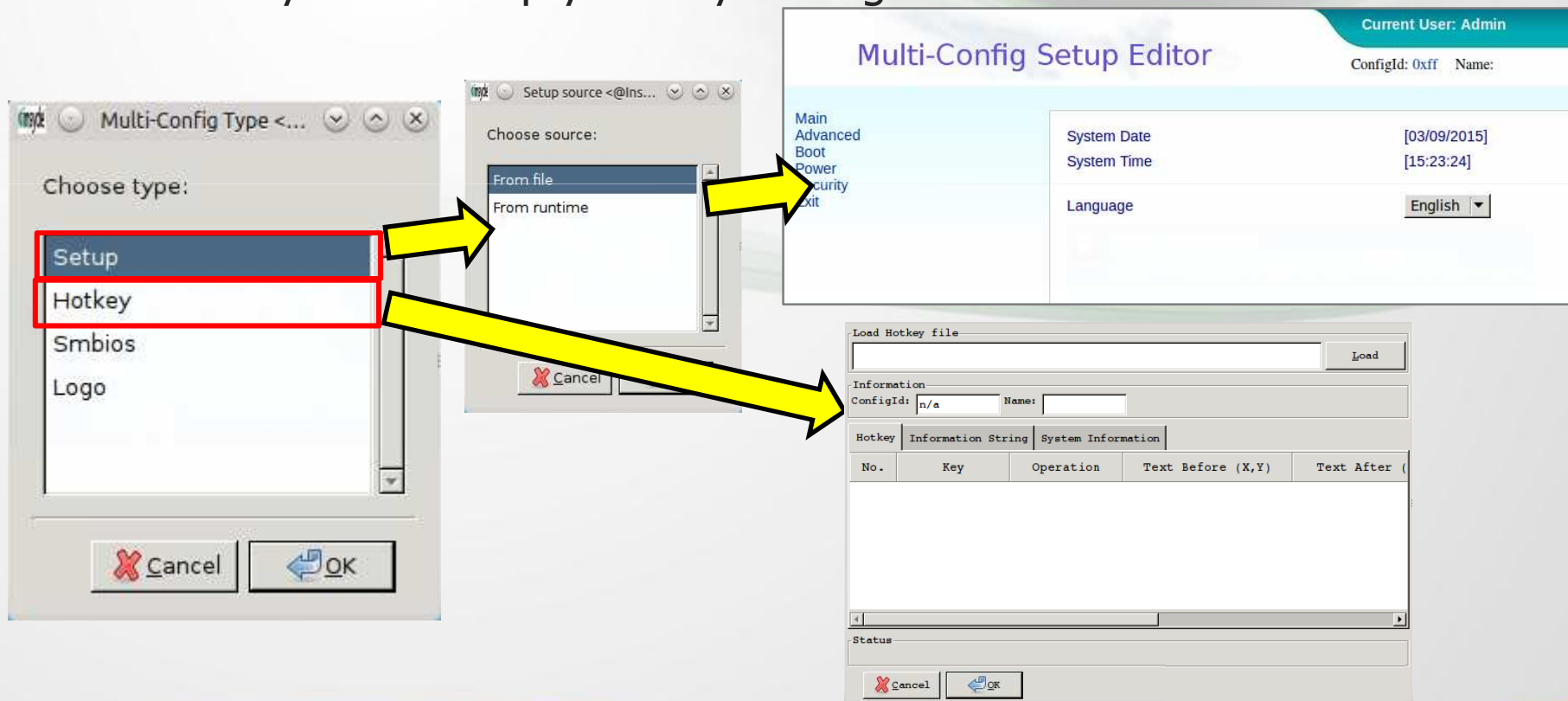
- Setup
- Hotkey
- Smbios
- Logo

Cancel OK

Multi-Config Editor

Describe how to add one setting in Multi-Config region

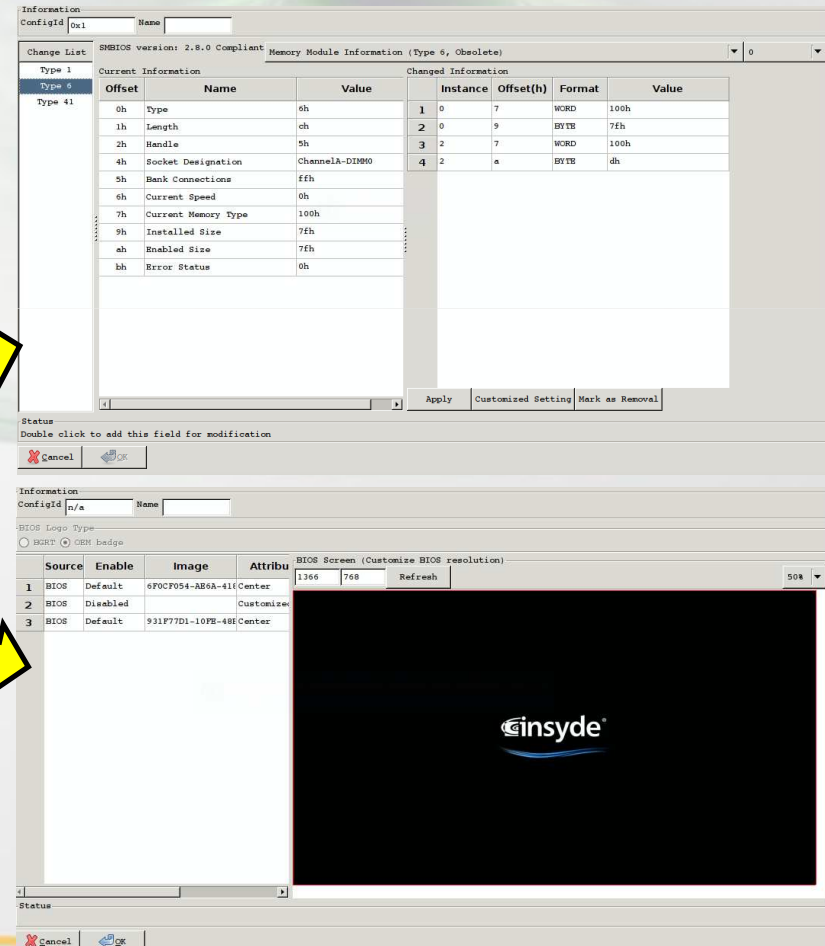
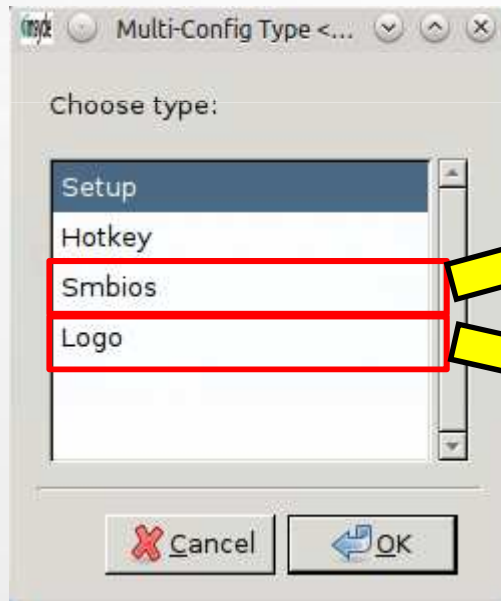
- Based on the chosen type, there will be different result
 - Setup: Choice about the source to add
 - Hotkey: Show empty hotkey setting



Multi-Config Editor

Describe how to add one setting in Multi-Config region

- Smbios: Customized SMBIOS settings
- Logo: Modify BIOS logo settings

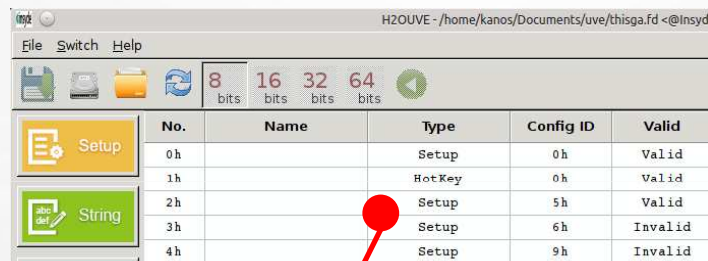


Multi-Config Editor

Describe how to dump one setting in Multi-Config region

1. Click right button on item list
2. Choose "Dump"
3. Input the file name to the dumped settings

Show chosen type

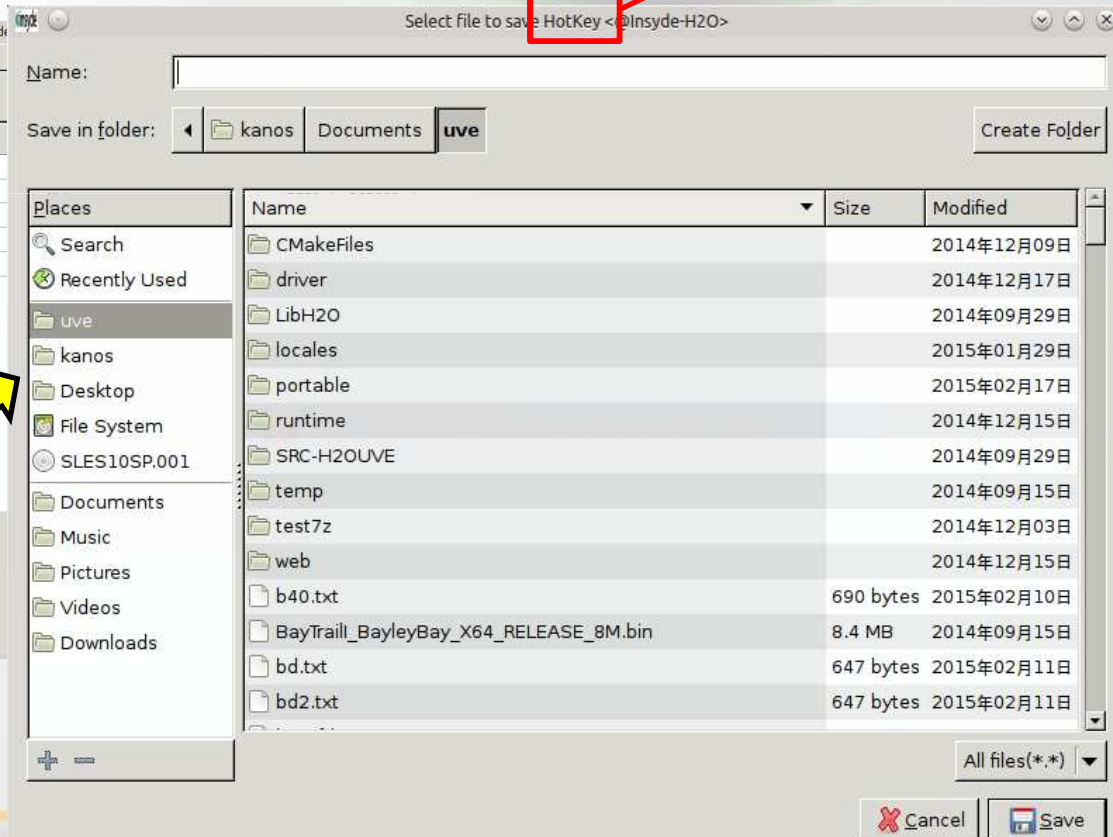


The screenshot shows the Multi-Config Editor interface. On the left, there are buttons for Setup, String, Variable, and Multi-Config. The main area displays a table with columns: No., Name, Type, Config ID, and Valid. The table contains four rows of settings. A red dot is placed on the right side of the table, indicating where to click the right button.

No.	Name	Type	Config ID	Valid
0h		Setup	0h	Valid
1h		HotKey	0h	Valid
2h		Setup	5h	Valid
3h		Setup	6h	Invalid
4h		Setup	9h	Invalid

Click right button on item list to show **Dump** choice

Add
Dump
Edit
Delete



Multi-Config Editor

Describe how to modify one setting in Multi-Config region

1. Click right button on item list or press **Edit button**
2. Choose "Edit"
3. The detail editor based on selected item

The screenshot displays the Multi-Config Editor interface. On the left, a sidebar contains icons for Setup, String, Variable, and Multi-Config. The main window shows a table of configuration items. A red circle highlights the right-click context menu, and a red box highlights the 'Edit' button. A yellow arrow points from the 'Edit' button to the 'Multi-Config Setup Editor' dialog box. Another yellow arrow points from the 'Edit' button to the 'Edit' button in the context menu.

Press **Edit button**

Click right button on item list to show **Edit** choice

No.	Name	Type	Config ID	Valid	Mode	Post	Active	Edit	Delete
0h		Setup	0h	Valid	R/W	☐	☐	Edit	Delete
1h		HotKey	0h	Valid	R/W	☐	☐	Edit	Delete
2h		Setup	5h	Valid	R/W	☐	☐	Edit	Delete
3h		Setup	6h	Invalid	RO	☐	☐	Edit	Delete
4h		Setup	9h	Invalid	R/W	☐	☐	Edit	Delete

Multi-Config Setup Editor

Current User: Admin

ConfigId: 0x1 Name:

Main
Advanced
Boot
Power
Security

System Date [03/09/2015]
System Time [11:12:34]

Information
ConfigId: 0x0 Name:

No.	Key	Operation	Text Before (X,Y)
0	F1	Device Manager	
1	F2	Setup	
2	Delete	Boot Manager	
3	F10	Boot Maintenance	
4	ESC	Front Page	"Press Esc for boot options"\n+Location:(m

Status

Cancel OK

Multi-Config Editor

Describe how to delete setting in Multi-Config region

1. Click right button on item list or press **Delete** button
2. Choose "Delete"

The screenshot shows the Multi-Config Editor window. The title bar reads "H2OUIVE - /home/kanos/Documents/uve/thisga.fd <@Insyde-H2O>". The menu bar includes "File", "Switch", and "Help". Below the menu bar is a toolbar with icons for file operations and a row of bit width selectors: "8 bits", "16 bits", "32 bits", and "64 bits". On the left is a sidebar with buttons for "Setup", "String", "Variable", and "Multi-Config". The main area contains a table with the following columns: "No.", "Name", "Type", "Config ID", "Valid", "Mode", "Post", "Active", and two action buttons: "Edit" and "Delete".

No.	Name	Type	Config ID	Valid	Mode	Post	Active	Edit	Delete
0h		Setup	0h	Valid	R/W	☐	☐	Edit	Delete
1h		HotKey	0h	Valid	R/W		☐	Edit	Delete
2h		Setup	5h	Valid	R/W	☐	☐	Edit	Delete
3h		Setup	6h	Invalid	RO			Edit	Delete
4h		Setup	9h	Invalid	R/W			Edit	Delete

A red dot is placed on the "Delete" button of the first row (0h). A red box with the text "Click right button on item list to show **Delete** choice" has a yellow arrow pointing to a context menu. The context menu contains the options "Add", "Dump", "Edit", and "Delete", with "Delete" highlighted by a red box. Another red box with the text "Press **Delete** button" has a red arrow pointing to the "Delete" button in the table's action column.

Total number of Config Data: 5

Multi-Config Editor

Read-Only item

- RO in Mode column means that it's a read-only item.
- Most columns which is read-only and can't be modified.
- Users can still open its detail by pressing Edit button or double clicking on one item.

H2OUVE - /home/kanos/Documents/uve/thisga.fd <@Insyde-H2O>

File Switch Help

8 bits 16 bits 32 bits 64 bits

No.	Name	Type	Config ID	Valid	Mode	Post	Active	Edit	Delete
0h		Setup	0h	Valid	R/W	☐	☐	Edit	Delete
1h		HotKey	0h	Valid	R/W		☐	Edit	Delete
2h		Setup	5h	Valid	R/W	☐	☐	Edit	Delete
3h		Setup	6h	Valid	RO	☐	☐	Edit	Delete
4h		Setup	9h	Invalid	R/W			Edit	Delete

Those columns are not editable when **Mode** column is read-only (RO)

Those 2 column are still editable when **Mode** column is read-only (RO)

Multi-Config Editor

Invalid item

- Invalid in Valid column means that item can't set Post and Active.
- Post and Active are not controlled by its Mode.

H2OUVE - /home/kanos/Documents/uve/thisga.fd <@Insyde-H2O>

File Switch Help

8 bits 16 bits 32 bits 64 bits

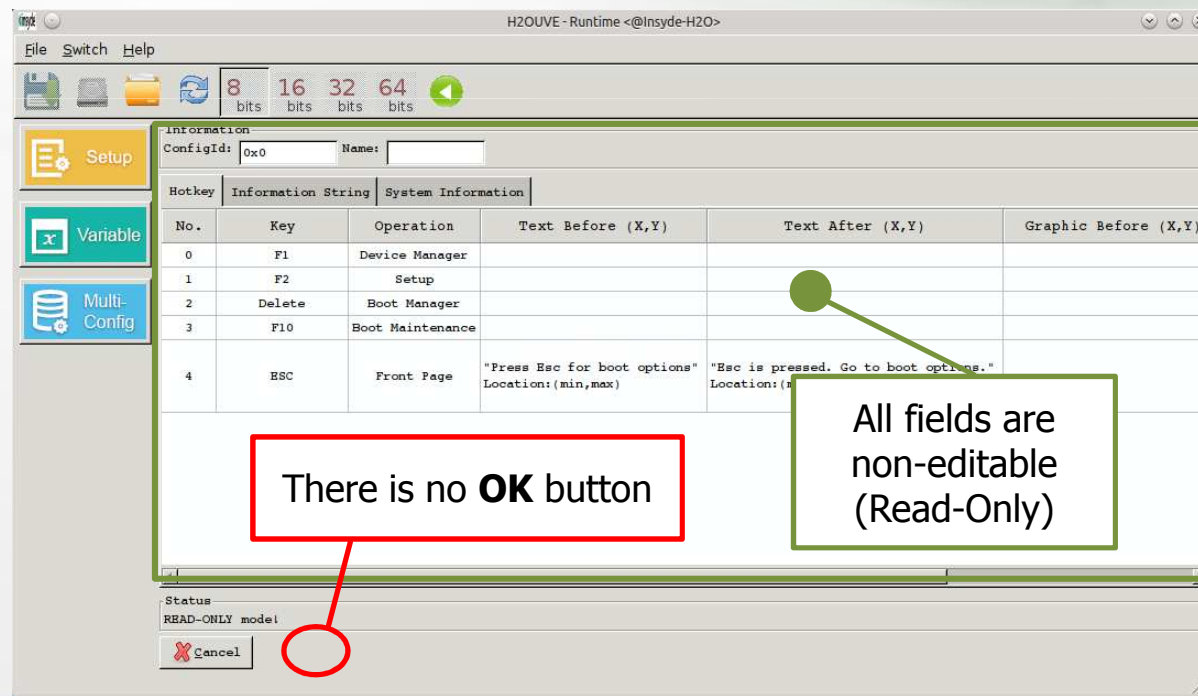
No.	Name	Type	Config ID	Valid	Mode	Post	Active		
0h		Setup	0h	Valid	R/W	☐	☐	Edit	Delete
1h		HotKey	0h	Valid	R/W		☐	Edit	Delete
2h		Setup	5h	Valid	R/W	☐	☐	Edit	Delete
3h		Setup	6h	Valid	RO	☐	☐	Edit	Delete
4h		Setup	9h	Invalid	R/W			Edit	Delete

Those 2 column are **NOT** available when **Valid** column is invalid

Multi-Config Editor

Edit Read-Only item

- All fields can't be modified.
- H2OUIVE can't stop user to change options in setup data, but it won't be modified.
- Other types in Multi-Config will show its detail without OK button.



- Main Frame

- Menu Bar
- Tool Bar

- Setup Editor

- String Editor

- Variable Editor

- Multi-Config Editor

- Setup

- Hotkey

- Logo

- SMBIOS

Multi-Config Editor

Setup information

- It's same page with [Setup Editor](#) with different banner
- There are config id and name list on top of content section

The banner shows that it's a Multi-Config Setup Editor

Config id and name are here

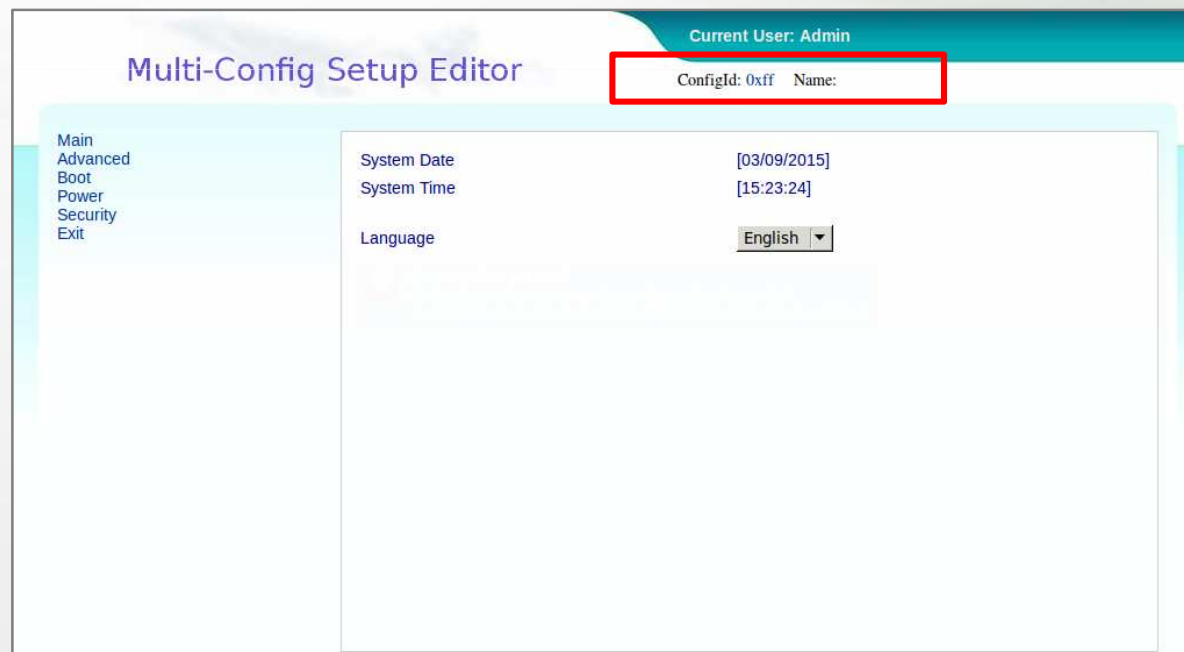
The screenshot shows the 'Multi-Config Setup Editor' interface. At the top, a teal header bar displays 'Current User: Admin'. Below this, the title 'Multi-Config Setup Editor' is centered. To the right, there is a section for 'ConfigId: 0x1' and 'Name:'. The main content area is divided into two columns. The left column contains a menu with options: 'Main', 'Advanced', 'Boot', 'Power', 'Security', and 'Exit'. The right column displays system information: 'System Date' as '[03/09/2015]' and 'System Time' as '[11:12:34]'. Red boxes and lines are used to highlight and label specific parts of the interface: the title, the config ID and name fields, and the main content area.

Same operation with Setup Editor

Multi-Config Editor

Add/Edit setup setting

- The config id is 0xff with empty name. Config Id 0xff won't be Id of any existing item
- Same operation with [Setup Editor](#)
- The new item will be added after [Exit->Save and Exit](#) runs successfully



- Main Frame

- Menu Bar
- Tool Bar

- Setup Editor

- String Editor

- Variable Editor

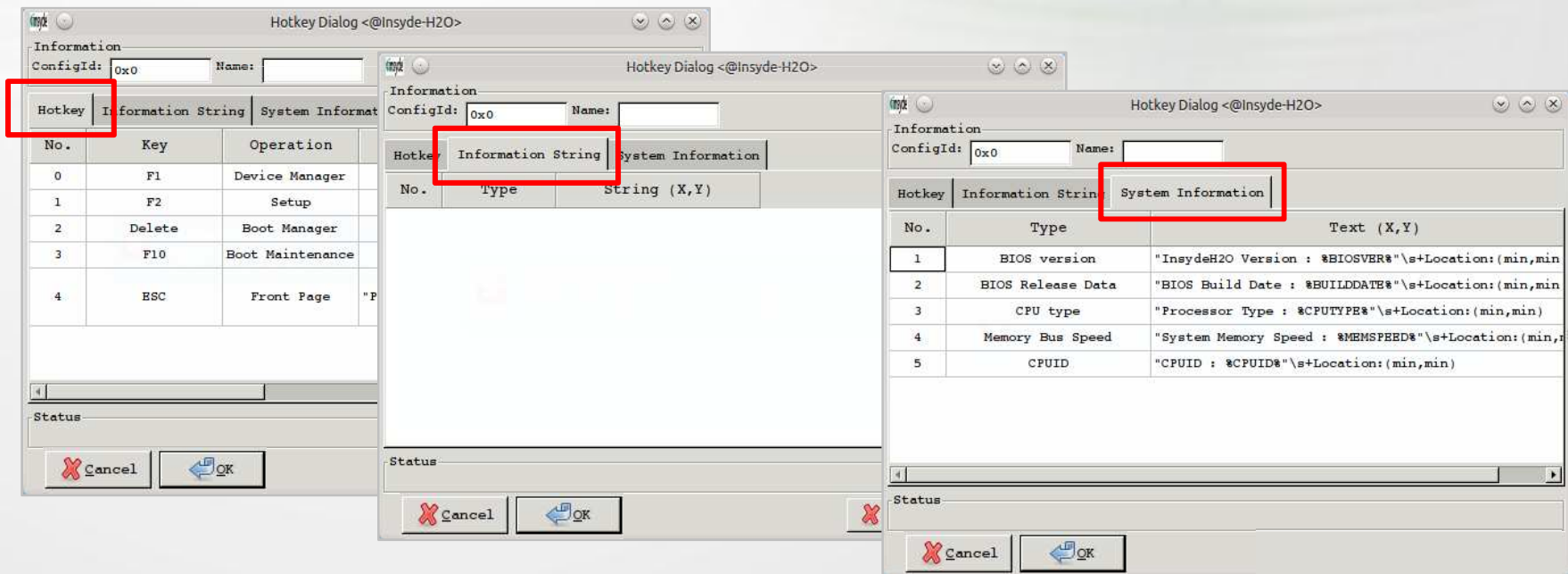
- Multi-Config Editor

- Setup
- Hotkey
- Logo
- SMBIOS

Multi-Config Editor

Hotkey information

- There are 3 different parts in Hotkey section
 - Hokey: Set hotkey for different operation
 - Information String: Set text or image to show additional information
 - System Information: Assign information for each system information



Multi-Config Editor

Introduce the interface of Hotkey handler

Config id and name are here

Information

ConfigId: Name:

Hotkey | Information String | System Information

No.	Key	Operation	Text Before (X,Y)
0	F1	Device Manager	
1	F2	Setup	
2	Delete	Boot Manager	
3	F10	Boot Maintenance	
4	ESC	Front Page	"Press Esc for boot options"\s+Loca

Status

Tabs to separate all types

Item list

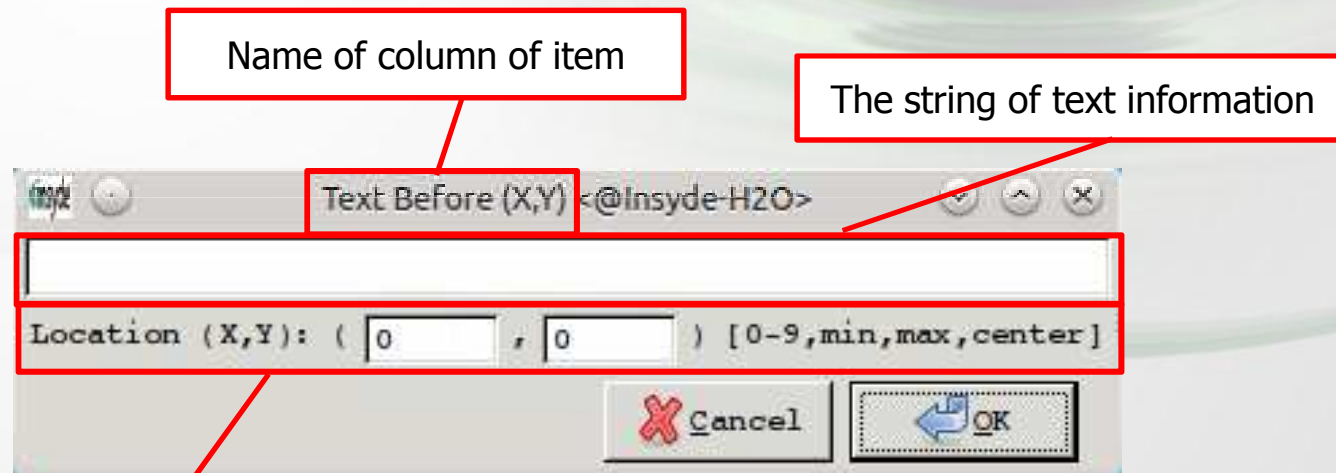
Double click on **text** or **image** column to show more details of each item

Show result of operation

Multi-Config Editor

Detail of selected item of Hotkey settings

- Double click on text or image column of selected item will pop-up a dialog to show its details and it's capable to edit it



Multi-Config Editor

Add hotkey setting

- This new Hotkey setting can be loaded from file
- The other parts are same as [Hotkey dialog](#)
- The new item will be added by click **OK** button

The screenshot shows the 'Multi-Config Editor' dialog box. At the top, there is a section titled 'Load Hotkey file' which contains a text input field and a 'Load' button. Below this is the 'Information' section with fields for 'ConfigId' (containing 'n/a') and 'Name'. Underneath is a tabbed interface with three tabs: 'Hotkey', 'Information String', and 'System Information'. The 'Hotkey' tab is active, displaying a table with columns: 'No.', 'Key', 'Operation', 'Text Before (X,Y)', and 'Text'. The table is currently empty. At the bottom of the dialog are 'Cancel' and 'OK' buttons. Red boxes and arrows highlight the 'Load' button, the 'OK' button, and the table area, with corresponding text labels: 'Load hotkey file', 'Press OK to add item', and 'Load hotkey file'.

No.	Key	Operation	Text Before (X,Y)	Text
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- Main Frame

- Menu Bar
- Tool Bar

- Setup Editor

- String Editor

- Variable Editor

- Multi-Config Editor

- Setup
- Hotkey
- Logo
- SMBIOS

Multi-Config Editor

BIOS logo information

- There are 2 mode for BIOS logo presentation.
 - BGRT: Indicating only one logo can be presented at center of screen
 - OEM BADG: One or more logos can exist at different position on screen
- There are 2 types of BIOS logo.
 - BIOS default settings: Original logo settings exist in BIOS, and the default option is used for those original logo
 - Additional settings: User added images
- 3 options for each logo setting.
 - Enabled: User added image to logo setting, how many available options depends on the BIOS logo mode.
 - Default: Take BIOS default logo setting to run, and its most options are read-only.
 - Disabled: Make logo setting disabled, its all options are read-only.
- Accept BMP, JPG, GIF, PCX, TGA images for logo setting.

Multi-Config Editor

Introduce interface of BIOS Logo handler

Indicate current type of BIOS logo setting
BGRT: Only one logo exist and always at center
OEM BADG: Accept multiple logos with customized position

Simulating BIOS Logo screen with user input screen resolution and presentation ratio.
Please make sure your resolution is correct or it may cause different result than your expectation.
The ratio is used for user to view all simulating area.

ConfigId: n/a Name:

BIOS Logo Type
☐ BGRT ☒ OEM badge

	Source	Enable	Image	Attribu
1	BIOS	Default	6F0CF054-AB6A-418	Center
2	BIOS	Disabled		Customize
3	BIOS	Default	931F77D1-10FE-48E	Center

BIOS Screen (Customize BIOS resolution)
1366 768 Refresh 50%

Click right button of mouse to add a row or delete a selected row

Logos on this simulating may be moved by holding left mouse button while it's OEM BADF mode

Status
Cancel OK

Additional message or warnings

Multi-Config Editor

Add/Edit logo setting in BGRT mode

- There is only one logo can exist in BGRT mode.
- The logo is always presented at center.

The screenshot shows the 'Multi-Config Editor' interface in BGRT mode. It includes a table for logo settings, a preview window, and various control buttons. Red boxes and arrows highlight specific features and instructions.

Information
ConfigId: n/a Name:

BIOS Logo Type
☒ BGRT ☐ OEM badge

	Source	Enable	Image
1	BIOS	Disabled	
2	BIOS	Default	931F77D1-10FE-48E
3	BIOS	Disabled	

BIOS Screen (Customize BIOS resolution)
1366 768 Refresh 50%

Status
Every logo should be at center in BGRT mode

Buttons: Cancel OK

Annotations:

- Change solution and ratio to show how logo present in BIOS screen (points to resolution fields)
- Only one Enabled or Default logo setting exists (points to the 'Default' row in the table)
- Press **OK** to add item (points to the OK button)
- Logo is read-only (points to the Insyde logo in the preview window)

Multi-Config Editor

Add/Edit logo setting in OEM BADG mode

- There can exist more than 1 logos in OEM BADG mode.
- Logos can be put on any location on screen.

The screenshot shows the Multi-Config Editor interface. At the top, there's an 'Information' section with 'ConfigId' set to 'n/a' and a 'Name' field. Below it, the 'BIOS Logo Type' section has two radio buttons: 'BGRT' (unselected) and 'OEM badge' (selected). A table lists four logos, with the first three highlighted by a red box. To the right of the table, there's a 'BIOS Screen (Customize BIOS resolution)' section with input fields for '1366' and '768', and a 'Refresh' button. A large preview window shows the 'insyde' logo on a black background. A red box highlights a 'JPG' icon in the bottom right corner of the preview window. A red box highlights the 'Add' and 'Delete' buttons in the bottom left corner of the interface.

	Image	Attribute	Offset X	Offset Y
1	6F0CF054-AB6A-41f	Center	0	0
2		Customized	0.0%	0.
3	931F77D1-10FE-48E	Center	0	0
4	/home/kanos/Docu	CenterRight	1	0

BIOS Screen (Customize BIOS resolution)
1366 768 Refresh

Enabled logo setting can put image on anywhere of screen

There are most available options to be edited if the logo setting is Enabled, otherwise are read-only

Click right button on setting list to **Add** a new setting or **Delete** selected setting

- Main Frame

- Menu Bar
- Tool Bar

- Setup Editor

- String Editor

- Variable Editor

- Multi-Config Editor

- Setup
- Hotkey
- Logo
- SMBIOS

Multi-Config Editor

Introduce interface of SMBIOS handler

Change List

Show compliant
SMBIOS version

SMBIOS Type List

Instance List exists
in runtime mode

Information
ConfigId 0x1 Name

Change List
Type 1
Type 6
Type 41

SMBIOS version: 2.8.0 Compliant Memory Module Information (Type 6, Obsolete) 0

Current Information			Changed Information				
Offset	Name	Value		Instance	Offset(h)	Format	Value
0h	Type	6h	1	0	7	WORD	100h
1h	Length	ch	2	0	9	BYTE	7fh
2h	Handle	5h	3	2	7	WORD	100h
4h	Socket Designation	ChannelA-DIMM0	4	2	a	BYTE	dh
5h	Bank Connections	ffh					
6h	Current Speed	0h					
7h	Current Memory Type	100h					
9h	Installed Size	7fh					
ah	Enabled Size	7fh					
bh	Error Status	0h					

Changed Information:
Show modified elements.
Click right mouse button to Add/Delete row.

Apply: accept this change to Change List
Customized Setting: add a row for customized change
Mark as Removal: mark this instance to be removed

Apply Customized Setting Mark as Removal

Status
Double click to add this field for modification

Cancel OK

Current Information:
Show current elements

Show usage message or warnings.

Multi-Config Editor

Add/Edit SMBIOS setting

Step 6: After **Apply** the **Changed Information**, the modified **Type** is added here

Step 1: Choose target **Type**

Step 2: Choose target **Instance**, or change it in first column of **Changed Information**. Instance option doesn't exist while it's BIOS image mode

ConfigId: 0x1 Name:

SMBIOS version: 2.8.0 Compliant Memory Module Information (Type 6, Obsolete) 0

Change List

- Type 1
- Type 6
- Type 41

Current Information

Offset	Name	Value
0h	Type	6h
1h	Length	ch
2h	Handle	5h
4h	Socket Designation	ChannelA-DIMM0
5h	Bank Connections	ffh
6h	Current Speed	0h
7h	Current Memory Type	100h
9h	Installed Size	7fh
ah	Enabled Size	7fh

Changed Information

Instance	Offset(h)	Format	Value	
1	0	7	WORD	100h
2	0	9	BYTE	7fh
3	2	7	WORD	100h
4	2	a	BYTE	dh

Apply Customized Setting Mark as Removal

Status: Double click to add this field for modification

Cancel OK

Step 3: Double click one element for modification in **Changed Information**. The **Value** column is not available while it's BIOS image

Step 4: Select **Instance** and assign new **Value**. Change **Offset** and **Format** while it's customized setting. The **Instance** column is selection element when it's runtime mode instead of input column in BIOS image mode.

Step 5: Click **Apply** to add above modified elements to **Change List**

Step 7: Click **OK** button to create a config data for Logo setting

Thanks!

